



**Technical
Institute of America**

Project Management Processes PMBOK 8th Edition



Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Governance	Initiate Project or Phase	- Integrate and Align Project Plans - Plan Sourcing Strategy	- Manage Project Execution - Manage Quality Assurance - Manage Project Knowledge	- Monitor and Control Project Performance - Assess and Implement Changes	Close Project or Phase
Scope		- Plan Scope Management - Elicit and Analyze Requirements - Define Scope - Develop Scope Structure		- Monitor and Control Scope - Validate Scope	
Schedule		- Plan Schedule Management - Develop Schedule		Monitor and Control Schedule	
Finance		- Plan Financial Management - Estimate Costs - Develop Budget		Monitor and Control Finances	
Stakeholders	Identify Stakeholders	- Plan Stakeholder Engagement - Plan Communications Management	- Manage Stakeholder Engagement - Manage Communications	- Monitor Stakeholder Engagement - Monitor Communications	
Resources		- Plan Resource Management - Estimate Resources	- Acquire Resources - Lead the Team	Monitor and Control Resourcing	
Risk		- Plan Risk Management - Identify Risks - Perform Risk Analysis - Plan Risk Responses	Implement Risk Responses	Monitor Risks	

Initiate Project or Phase

- The process of developing a document to formally authorize a project or a phase
- Outlines the project objectives
- Defines the authority of the project manager
- Provides the project manager with the authority to put the resources together to project activities
- The approved project charter formally initiates the project

Initiate Project or Phase

Inputs

- Business documents
 - Business case
 - Benefits management plan
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Brainstorming
 - Focus groups
 - Interviews
- Interpersonal and team skills
 - Conflict management
 - Facilitation
 - Meeting management
- Meetings
- Responsibility assignment matrix
- Project canvas
- Etc.

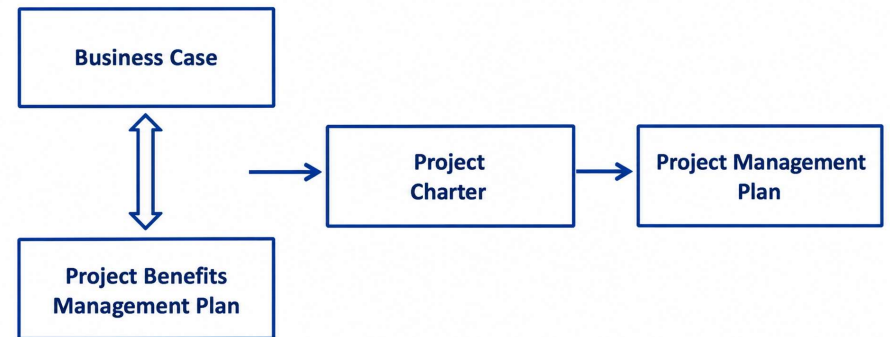
Outputs

- Project charter
- Assumption log
- Etc.

Initiate Project or Phase Inputs

- Business Documents - Contain specific information as to why a project should be initiated.
- There are two main documents the business case and the benefits management plan.
 - **Business Case** - Necessary information that determines whether or not the project is worth the required investment
 - Market Demand, Customer Request, Organizational Need, Legal requirement
 - **Project Benefits Management Plan**
 - Describes the main benefits that the project will produce once it is completed and how to measure the benefits. The project benefit could be the product, service, or result.
 - It maybe created by doing a cost-benefit analysis a project.

Initiate Project or Phase Inputs



Initiate Project or Phase Inputs

- Agreements
 - Service Level Agreements (SLA)
 - Letters of intent
 - Contract between internal and external customer
 - Work required to be performed for Payment
- Enterprise Environmental Factors
- Organizational Processes Assets

Initiate Project or Phase Tools

- Expert judgment
- Data gathering
 - Brainstorming
 - Focus groups
 - Interviews
- Interpersonal and team skills
 - Conflict management
 - Facilitation
 - Meeting management
- Meetings

Initiate Project or Phase Tools

- Responsibility assignment matrix
 - A **Responsibility Assignment Matrix (RAM)** shows which resources are assigned to specific work packages.
 - A common example of a RAM is a **RACI chart**.
 - **RACI** stands for **Responsible, Accountable, Consulted, and Informed**.
 - **Responsible** means the person doing the work.
 - **Accountable** means the person who makes sure the work gets done.
 - **Consulted** means the expert or stakeholder who provides input, while **Informed** means the person who needs updates.
 - There should be only one accountable person for each work package to avoid confusion and blame.

Initiate Project or Phase Tools

Activity	Sponsor	Project Manager	Business Analyst	Developer
Define project vision	A	R	C	I
Define high-level scope	A	R	R	C
Identify stakeholders	A	R	C	I
Draft & approve charter	A	R	C	I

Simple RACI Chart

Initiate Project or Phase Tools

- Project canvas
 - A **project canvas** is a visual tool used to outline and plan the key parts of a project.
 - It gives the project manager and stakeholders a quick, high-level view of the project.
 - Early in the project, the canvas may be simple and high level.
 - As the project progresses, the canvas can become more detailed.

Initiate Project or Phase Tools

Purpose - Build a new e-commerce website	Objective - Increase conversion rate from 5% → 15% - Improve customer engagement	Project Approval Requirements - Fully functional e-commerce website (desktop + mobile) - Page load time under 3 seconds
Scope: - 20 Page e-commerce site - Responsive website design - Product catalog + inventory system	Deliverables - Home page - Contact us page - 15 product pages - Shopping Cart + Checkout - About page	Assumptions/Constraints - Current web server will be allowed to handle the new site - Time constraints of 3 months for development.
Milestone - Design acceptance - MVP Launch - Customers' orders	Stakeholders - Project Manager - Project Team - Sale manager (sponsor)	Risk - Web server failure - Gateway integration failure
Project Phases - Collecting Requirements - Design - Coding and testing - Implementation	Team - Business Analyst - Web Designer - Web programmer - Tester	Lesson Learned - Create mock layout early - Test each product page multiple times.

Sample Project Canvas for a web development project

Initiate Project or Phase Output

- **Output**
 - **Project Charter**
 - Formally authorizes the existence of the project and it assigns the Project Manager and their Authority Level
 - Signed by the organization Senior Management
 - High Level requirements & risks
 - Preliminary Project Budget and Schedule
 - Project Purpose or justification
 - **Assumption Log**
 - A list of things that you perceive to be true (assumptions) and things that might constrain the project.

Identify Stakeholders

- Identifies the people, groups, or organizations that have a stake in the project.
- It includes analyzing stakeholder interests, involvement, influence, interdependencies, and impact on project success.
- The key benefit is helping the project team decide how to engage each stakeholder or stakeholder group.
- Stakeholder identification should happen regularly, not just once at the beginning.
- Continuous stakeholder identification can support risk management as the project environment changes.
- This process is performed periodically throughout the project as needed.

Identify Stakeholders

ITTO

Inputs

- Project charter
- Business documents
 - Business case
 - Benefits management plan
- Project management plan
 - Communications management plan
 - Stakeholder engagement plan
- Project documents
 - Change log
 - Issue log
 - Requirements documentation
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Questionnaires and surveys
 - Brainstorming
- Data analysis
 - Stakeholder analysis
 - Document analysis
- Data representation
 - Stakeholder mapping/representation
- Meetings
- Etc.

Outputs

- Stakeholder register
- Change requests
- Project management plan updates
 - Requirements management plan
 - Communications management plan
 - Risk management plan
 - Stakeholder engagement plan
- Project document updates
 - Assumption log
 - Issue log
 - Risk register
- Etc.

Identify Stakeholders Inputs

- **Project Charter**
 - Identifies the key stakeholder list, sponsor, and customer as a starting point for further identification.
- **Business Documents**
 - Business case - identifies the project objectives and stakeholders who care about the business value.
 - Benefits management plan - identifies stakeholders who will receive the benefits and who are responsible for tracking them.
- **Project Management Plan**
 - Communications management plan - if it exists, lists previously identified stakeholders and their communication needs.
 - Stakeholder engagement plan - documents currently identified stakeholders and engagement strategies.

Identify Stakeholders Inputs

- **Project Documents**

- Change log - changes may introduce new stakeholders or alter existing ones' roles.
- Issue log - issues often surface affected stakeholders previously overlooked.
- Requirements documentation - identifies stakeholders with requirements that must be met.

- **Agreements**

- Vendor contracts and partnership agreements identify external stakeholders with formal roles.

- **EEFs and OPAs**

- Organizational culture, government regulations (which may create regulatory stakeholders), stakeholder registers from prior projects, and lessons learned.

Identify Stakeholders Tools

- **Expert Judgment**
 - Specialists with knowledge of the political environment, industry, technology, and prior project stakeholder lists.
- **Data Gathering**
 - Questionnaires and surveys - efficient way to gather views from a large or dispersed group of potential stakeholders.
 - Brainstorming - team-based ideation to identify any stakeholder who could affect or be affected by the project.
- **Data Analysis**
 - Stakeholder analysis - examines who the stakeholders are and how they feel about the project: their role (sponsor, team member, customer, etc.), how the project affects them (positively or negatively), whether they are active or passive, and their power and authority.
 - Document analysis - reviews existing documents (charter, agreements, lessons learned) to identify stakeholders.

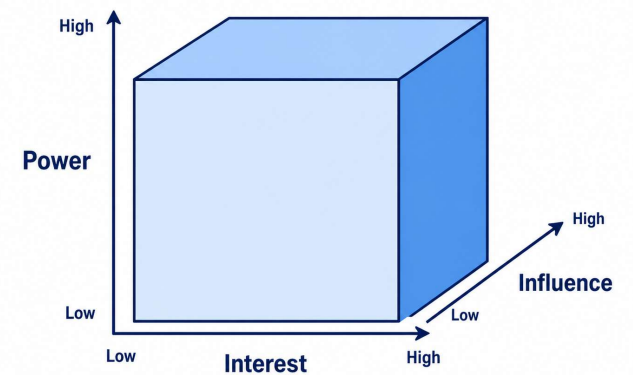
Identify Stakeholders Tools

- Data Representation
 - Stakeholder Mapping/Representation
 - Power/interest grid, power/influence grid, or impact/influence grid



Identify Stakeholders Tools

- Data Representation
 - Stakeholder Mapping/Representation
 - Stakeholder cube
 - A three-dimensional methodology to support the mapping of a stakeholder's interest, power, and influence



Identify Stakeholders Tools

- Data Representation
 - Method to categorize stakeholders.
 - Salience model:
 - Power: Level of authority
 - Urgency: Immediate attention
 - Legitimacy: How appropriate is their involvement
 - Directions of Influence:
 - Upward: Senior management
 - Downward: Team members
 - Outward: Vendors, government, public, end-users
 - Sideward: peers such as other project managers
 - Prioritization

Identify Stakeholders Output

- Stakeholder Register
 - stakeholder register is a project document that stores information about project stakeholders.
 - It includes details used to identify, assess, and classify stakeholders.
 - Identification information includes name, role, organizational position, location, and contact details.
 - Assessment information includes stakeholder requirements, expectations, influence, and impact.
 - It may also show which project phase the stakeholder has the most influence or impact.
 - Stakeholder classification may include internal/external, power/interest, impact/influence, or other models chosen by the project manager.

Identify Stakeholders Output

- **Change Requests**
 - Identification of new stakeholders (or changes to previously identified ones) may trigger change requests to management plans or project documents.
- **Project Management Plan Updates**
 - Requirements management plan - new stakeholders may have new requirements.
 - Communications management plan - new stakeholders need tailored communication.
 - Risk management plan - new stakeholders may introduce or affect risks.
 - Stakeholder engagement plan - updated to include the new stakeholders and their engagement strategies.
- **Project Document Updates**
 - Assumption log - assumptions about stakeholders documented.
 - Issue log - issues raised by or about stakeholders.
 - Risk register - risks involving stakeholders identified during analysis.

Planning Focus Area

- 19 Processes across all 7 domains
- Creates the Project Management plan and project documents used to execute, monitor and control and close the project

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Integrate and Align Project Plans

- The process of bringing together all subsidiary management plans, baselines, and supporting project documents into a single, integrated Project Management Plan
- Defines how the project will be executed, monitored, controlled, and closed
- The approved Project Management Plan becomes the baseline against which performance is measured throughout the project

Integrate and Align Project Plans - ITTO

Inputs

- Project charter
- Outputs from other processes
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Brainstorming
 - Checklists
 - Focus groups
 - Interviews
- Interpersonal and team skills
 - Conflict management
 - Facilitation
 - Meeting management
- Meetings
- Project canvas
- Etc.

Outputs

- Project management plan
- Etc.

Integrate and Align Project Plans - Inputs

- **Project Charter**
 - Provides the high-level project description, key deliverables, and assumptions used as the starting point for the PM Plan
 - Defines the authority of the project manager and the success criteria
- **Outputs from Other Planning Processes**
 - All subsidiary management plans
 - All baselines
- **Enterprise Environmental Factors**
 - Organizational culture, infrastructure, regulations, and PMIS
- **Organizational Process Assets**
 - Standard templates, lessons learned, configuration management knowledge bases

Integrate and Align Project Plans - Tools

- **Expert Judgment**

- Input from people with specialized knowledge in the relevant management domains (scope, schedule, cost, risk, procurement, etc.)

- **Data Gathering**

- Brainstorming - to elicit ideas and approaches for integrating plans
- Checklists - to confirm all required components of the PM Plan are addressed
- Focus groups - to gather feedback on the proposed approach
- Interviews - one-on-one with subject matter experts and stakeholders

Integrate and Align Project Plans - Tools

- **Interpersonal and Team Skills**
 - Conflict management - to resolve competing demands among management plans
 - Facilitation - to drive consensus across plan owners
 - Meeting management - to keep planning meetings focused and productive
- **Meetings**
 - Planning workshops to integrate the subsidiary plans
- **Project canvas**
 - A visual tool used to outline and plan the key parts of a project.

Integrate and Align Project Plans - Output

- **Project Management Plan**

- The integrated, approved master document that defines how the project is executed, monitored, controlled, and closed
- Includes development approaches (predictive, iterative, hybrid) for each major deliverable
- References all subsidiary management plans:
 - Scope, Schedule, Cost, Quality, Resource, Communications, Risk, Procurement, Stakeholder Engagement
- Approved by the sponsor before execution begins
- Predictive Projects: Once plan is approved, any changes to any component will require an approved change request.

Integrate and Align Project Plans - Output

Common Project Management Plan Components
Scope Management Plan
Requirement Management Plan
Schedule Management Plan
Financial Management Plan
Quality Management Plan
Resource Management Plan
Communication Management Plan
Risk Management Plan
Procurement Management Plan
Stakeholder Engagement Plan
Change Management Plan
Configuration Management Plan
Scope Baseline
Schedule Baseline
Cost Baseline
Performance Measurement Baseline
Sourcing strategy plan
Project Life Cycle Description
Development Approach

Plan Sourcing Strategy

- The process of determining how the project will obtain the goods, services, and resources needed to deliver project objectives
- Decides what work will be performed in-house and what will be outsourced to external sellers
- Establishes the source-selection criteria the project will use to evaluate and choose vendors
- Documents the overall sourcing approach in a sourcing strategy plan that informs subsequent procurement planning
- Performed early in planning so that schedule, cost, and resource plans can reflect make-vs-buy decisions

Procurement Process

1. Identify procurement needs

- Determine what goods, services, or results the project needs from outside sources
- Perform make-or-buy analysis
 - Decide whether the project team should do the work internally or buy it from an outside seller

2. Create procurement strategy

- Decide the best delivery method, contract type, and procurement phases

3. Prepare procurement documents

- Create documents such as:
 - **SOW** — Statement of Work
 - **RFI** — Request for Information
 - **RFP** — Request for Proposal
 - **RFQ** — Request for Quote

• Simple example

- A company needs a new training website and decides whether to build it internally or hire a vendor

Procurement Process

4. Advertise the opportunity

- Let qualified sellers know about the work

5. Hold a bidder conference

- Answer vendor questions and make sure everyone receives the same information

6. Receive proposals or quotes

- Sellers submit their responses based on the bid documents

7. Evaluate sellers

- Compare vendors based on cost, quality, experience, technical ability, delivery dates, and other criteria

8. Select seller and negotiate contract

- Choose the best vendor and agree on cost, schedule, scope, payment terms, and responsibilities

9. Monitor contract performance

- Make sure the seller delivers the work on time, within budget, and according to the contract

10. Close procurement

- Confirm all work is completed, obligations are fulfilled, and there are no outstanding issues

Simple example

- After reviewing vendor proposals, the company chooses the best website developer, signs the contract, tracks progress, and closes the contract when the website is delivered

Contracts

- Contracts explains how goods, services, or results are acquired from vendors
- They affect project risk
 - Different contracts shift risk between the buyer and the seller
- They affect cost and payment
 - Contracts decide how the seller will be paid and who pays for overruns
- Project managers must understand both sides
- The goal is a win-win agreement
 - Contracts should protect both parties
 - Good contracts encourage fairness, performance, and project success

Contract Types

- **Fixed-Price Contract** Price is agreed upfront for a clearly defined scope
 - Buyer gets budget certainty
 - Seller has more cost risk
 - Best when the work is well understood
 - **Example:** A contractor agrees to build a training room for **\$50,000**
- **Cost-Reimbursable Contract** Buyer pays the seller's actual costs plus a fee/profit
 - Good when scope is unclear or high risk
 - Buyer has more cost risk
 - Seller has less financial risk
 - **Example:** A company hires a research team and agrees to pay all approved costs plus a fixed fee

Contract Types

- **Time and Materials Contract** Buyer pays for labor hours and materials used
 - Mix of fixed-price and cost-reimbursable
 - Flexible, but costs must be monitored closely
 - Common for consulting, IT support, and maintenance work
 - **Example:** A consultant charges **\$150 per hour** plus software expenses
- **Target-Cost Contract** Buyer and seller agree on a target cost
 - Savings or overruns are shared based on an agreement
 - Encourages cost control and efficiency
 - Useful for large or complex projects
 - **Example:** If the target cost is **\$1 million** and the project finishes under budget, both buyer and seller share the savings

Plan Sourcing Strategy

ITTO

Inputs

- Project charter
- Project management plan
 - Scope management plan
 - Quality management plan
 - Scope baseline
 - Schedule management plan
 - Financial management plan
 - Resource management plan
- Project documents
 - Milestone list
 - Requirements documentation
 - Requirements traceability matrix
 - Quality metrics
 - Resource requirements
 - Project team assignments
 - Risk register
 - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Market research
- Make-or-buy analysis
- Source selection analysis
- Document analysis
- Etc.

Outputs

- Sourcing strategy plan
 - Insourcing or outsourcing decisions
 - Source selection criteria
- Etc.

Plan Sourcing Strategy - Inputs

- **Project Charter**
 - Provides the high-level project description, key deliverables, and constraints that influence sourcing decisions.
 - Identifies the project sponsor and the PM's authority to commit to vendor agreements.
- **Project Management Plan**
 - Scope management plan - defines how scope will be managed.
 - Quality management plan - sets quality standards a vendor must meet.
 - Scope baseline - define the work to be procured.
 - Schedule management plan - drives delivery dates and milestones the vendor must hit.
 - Financial management plan - cost controls applied to procurements.
 - Resource management plan - identifies internal resource availability, which informs make-vs-buy decisions.

Plan Sourcing Strategy - Inputs

- **Project Documents**
 - Milestone list - key dates the sourcing plan must support.
 - Requirements documentation - the detailed requirements that the procured product or service must satisfy.
 - Requirements traceability matrix - links each requirement back to a deliverable so vendor work can be traced.
 - Quality metrics - measurable criteria a deliverable must meet, used in source selection criteria.
 - Resource requirements - skills, materials, and equipment needed; gaps drive sourcing.
 - Project team assignments - who is already assigned, highlighting where external help is needed.
 - Risk register - risks that influence sourcing strategy (e.g., vendor concentration risk, single-source risk).
 - Stakeholder register - identifies stakeholders with influence over sourcing decisions (e.g., procurement, legal, finance).

Plan Sourcing Strategy - Inputs

- **Enterprise Environmental Factors (EEFs)**
 - Marketplace conditions, supplier performance data, regulatory requirements, and contracting policies that constrain or enable sourcing options.
- **Organizational Process Assets (OPAs)**
 - Pre-approved vendor lists, prior contracts, lessons learned from prior procurements, and standard templates used to build the sourcing plan.

Plan Sourcing Strategy - Tools

- **Expert Judgment**
 - Input from individuals with specialized knowledge in procurement, contracting, the relevant industry, and regulatory requirements.
- **Market Research**
 - Examination of industry and specific vendor capabilities to determine what is available, at what cost, and under what terms.
 - Includes reviewing trade publications, supplier conferences, and online reviews to understand market trends.

Plan Sourcing Strategy - Tools

- **Make-or-Buy Analysis**
 - A technique used to determine whether particular work can best be accomplished by the project team (make) or purchased from outside sources (buy).
 - Considers cost, capability, capacity, schedule, risk, and strategic value of keeping the work in-house.
- **Source Selection Analysis**
 - Reviews the methods used to evaluate and select sellers, such as least cost, qualifications only, quality-based, fixed budget, or best value.
- **Document Analysis**
 - Reviews existing documents (charter, requirements, prior contracts) to extract information that informs the sourcing approach.

Plan Sourcing Strategy - Outputs

- **Procurement Management plan**
 - Component of the PM plan that describes how the team will acquire goods and services from outside the performing organization, including the type of bidding (international, national, local)
- **Sourcing strategy plan**
 - Component of the PM plan that defines what work will be insourced vs outsourced, with rationale tied to value, schedule, risk, and cost
 - Tailored to the project: can be formal or informal, detailed or high-level, and may or may not include procurement specifics
 - Also include determining the type of contract to be used on the project
 - Two key components: insourcing/outsourcing decisions and source selection criteria

Plan Scope Management

- The process of creating a scope management plan that documents how the project and product scope will be defined, validated, and controlled
- Establishes how the requirements will be elicited, analyzed, and documented through the requirements management plan
- Defines how the WBS will be created
- Specifies how formal acceptance of completed deliverables will be obtained and how scope changes will be processed
- Provides guidance and direction on how scope will be managed throughout the project life cycle

Plan Scope Management

ITTO

Inputs

- Project charter
- Project management plan
- Project documents
 - Requirements documentation
 - Risk register
 - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Interviews
 - Focus groups
 - Questionnaires and surveys
- Data analysis
- Test and inspection planning
- Etc.

Outputs

- Project management plan updates
 - Scope management plan
 - Requirements management plan
- Etc.

Plan Scope Management - Inputs

- **Project Charter**
 - Documents the project purpose, high-level project description, assumptions, constraints, and high-level requirements that the scope management plan must support.
- **Project Management Plan**
 - Provides components such as the quality management plan, project life cycle, and development approach that influence how scope will be planned, defined, and controlled.
- **Project Documents**
 - Requirements documentation - existing documented requirements that inform how requirements will be managed going forward.
 - Risk register - identified risks that may affect scope, such as scope creep or unclear requirements, that the scope management plan must address.
 - Stakeholder register - identifies stakeholders who will help elicit requirements and approve scope, and their level of engagement.

Plan Scope Management - Inputs

- **Enterprise Environmental Factors (EEFs)**
 - Organizational culture, infrastructure, and personnel administration that shape how scope can be planned and managed.
- **Organizational Process Assets (OPAs)**
 - Policies and procedures, historical information, lessons learned repository, and templates that support building the scope and requirements management plans.

Plan Scope Management - Tools

- **Expert Judgment**
 - Input from individuals with specialized knowledge or experience in similar projects, prior requirements, or scope management.
- **Data Gathering**
 - Interviews - one-on-one discussions with stakeholders to understand expectations, requirements, and concerns about scope.
 - Focus groups - bringing prequalified stakeholders and subject matter experts together to learn about expectations and attitudes regarding scope.
 - Questionnaires and surveys - written sets of questions used to gather information from a large number of respondents quickly when stakeholders are geographically dispersed.

Plan Scope Management - Tools

- **Data Analysis**
 - Techniques used to analyze the project context and identify the best approach to define, validate, and control scope.
- **Test and Inspection Planning**
 - Determines how the product, deliverable, or service will be tested or inspected to verify it meets the stakeholder's needs and acceptance criteria.

Plan Scope Management - Output

- **Project Management Plan Updates**
- **Scope Management Plan**
 - A component of the project management plan that describes how the scope will be defined, developed, monitored, controlled, and validated.
 - Documents how the scope statement will be prepared, how the WBS will be created from the scope statement, and how the scope baseline will be approved and maintained.
 - Specifies how formal acceptance of completed deliverables will be obtained.
- **Requirements Management Plan**
 - A component of the project management plan that describes how project and product requirements will be analyzed, documented, and managed.
 - Defines how requirements activities will be planned, tracked, and reported.

Elicit and Analyze Requirements

- The process of determining, documenting, and managing stakeholder needs and requirements to meet project objectives
- Provides the basis for defining and managing the project scope, including product scope
- Captures requirements at multiple levels: business, stakeholder, solution (functional and non-functional), transition, project, and quality
- Engages stakeholders directly through interviews, focus groups, workshops, and other techniques
- Produces requirements documentation that is traceable, complete, consistent, and acceptable to key stakeholders

Elicit and Analyze Requirements - ITTO

Inputs

- Project charter
- Agreements
- Business case
- Project documents
 - Assumption log
 - Lessons learned register
 - Stakeholder register
- Project management plan
 - Requirements management plan
 - Scope management plan
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Decision-making
- Data gathering
 - Benchmarking
 - Brainstorming
 - Focus groups
 - Interviews
 - Questionnaires and surveys
- Data analysis
 - Document analysis
- Data representation
- Interpersonal and team skills
 - Nominal group
- Design thinking
- Prioritization/ranking
- Meetings
- Etc.

Outputs

- Requirements documentation
- Etc.

Elicit and Analyze Requirements - Inputs

- **Project Charter**
 - Documents the high-level project description, success criteria, and stakeholder list - the starting point for elicitation.
- **Agreements**
 - Contracts and SLAs that contain stated and implied requirements the project must satisfy.
- **Business Case**
 - Documents the business need that drove the project; helps validate that elicited requirements truly support the value proposition.

Elicit and Analyze Requirements - Inputs

- **Project Documents**
 - Assumption log - assumptions and constraints affecting requirements.
 - Lessons learned register - prior project insights about elicitation techniques that worked or failed.
 - Stakeholder register - identifies who provides which requirements and their level of influence.
- **Project Management Plan**
 - Requirements management plan - defines how requirements will be elicited, analyzed, and managed.
 - Scope management plan - defines how scope (and therefore requirements) will be defined and validated.
- **EEFs and OPAs**
 - Industry standards, regulations, organizational templates, and lessons learned that shape elicitation approach.

Elicit and Analyze Requirements - Tools

- **Expert Judgment**
 - Specialists in business analysis, requirements engineering, and the relevant subject matter help shape and validate requirements.
- **Decision-Making**
 - Techniques such as voting, autocratic, or multicriteria analysis used to converge on a final set of requirements.
- **Data Gathering**
 - Benchmarking - comparing actual or planned products to those of comparable organizations to identify best practices.
 - Brainstorming - rapid generation of ideas in a group setting to surface candidate requirements.
 - Focus groups - facilitated sessions with prequalified stakeholders to learn about expectations.
 - Interviews - one-on-one or small-group discussions to elicit detailed requirements.
 - Questionnaires and surveys - written queries used when stakeholders are dispersed or numerous.

Elicit and Analyze Requirements - Tools

- **Data Analysis & Representation**
 - Document analysis reviews existing artifacts (contracts, policies, prior requirements) to extract requirements.
 - Data representation techniques such as affinity diagrams and mind maps.
- **Interpersonal and Team Skills**
 - Nominal group technique - structured method for group brainstorming that encourages contributions from everyone and facilitates quick agreement.
- **Design Thinking, Prioritization, Meetings**
 - Design thinking puts users at the center to understand needs deeply before defining solutions.
 - Prioritization/ranking (e.g., MoSCoW, weighted scoring) orders requirements by value and constraints.
 - Workshops and meetings bring stakeholders together to elicit, refine, and approve requirements.

Elicit and Analyze Requirements - Output

- **Requirements Documentation**
 - Requirements may start high level and become more detailed over time.
 - Before baselining, requirements should be clear, measurable, testable, traceable, complete, consistent, and accepted by key stakeholders.
 - The acceptance criteria is a set of conditions that are met before deliverables are accepted.
 - Could include:
 - **Business requirements:** strategic objectives and high-level organizational needs.
 - **Stakeholder requirements:** needs of stakeholders or stakeholder groups.
 - **Solution requirements:** features, functions, and characteristics of the product, service, or result.
 - **Functional requirements:** what the product must do.
 - **Nonfunctional requirements:** how the product must perform, such as security, reliability, safety, and performance.
 - **Transition/readiness requirements:** temporary needs such as training or data conversion.
 - **Project requirements:** milestones, constraints, and contractual obligations.
 - **Quality requirements:** tests, certifications, validations, and acceptance criteria.

Elicit and Analyze Requirements - Output

- **Requirements Traceability Matrix**
 - Links product requirements to the deliverables that satisfy them.
 - It helps ensure each requirement supports business value and project objectives.
 - It helps confirm that approved requirements are completed by the end of the project.
 - The matrix may link requirements to business needs, project objectives, WBS deliverables, design, development, and test cases.
 - Common attributes include requirement ID, description, owner, source, priority, status, and status date.
 - Additional attributes may include stability, complexity, and acceptance criteria.

Define Scope

- The objective is to describe the project, product, and expected value to be delivered.
- The description may be detailed or high level depending on the project approach.
- In predictive projects, scope is usually defined early and structured in the WBS.
- In adaptive or hybrid projects, scope is progressively defined through the backlog.
- This process also identifies quality requirements and quality standards for deliverables.
- It determines how the project will show that quality requirements have been met.

Define Scope

ITTO

Inputs

- Project charter
- Assumption log
- Project management plan
- Requirements documentation
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Decision-making
- Data analysis
- Decomposition
- Interpersonal and team skills
 - Facilitation
- Product analysis
- Etc.

Outputs

- Project documents
 - Project scope statement
 - Requirements documentation
- Etc.

Define Scope - Inputs

- **Project Charter**
 - Provides the high-level description of the project and the product, the project objectives, and the success criteria - the starting point for the detailed scope statement.
- **Assumption Log**
 - Lists assumptions and constraints that shape the project scope and any decisions made about what is in or out of scope.
- **Project Management Plan**
 - The scope management plan within the PM Plan documents the approach for defining scope and the level of detail needed in the scope statement.
- **Requirements Documentation**
 - The full set of analyzed requirements from which the team selects the final project requirements to include in the scope.
- **EEFs and OPAs**
 - Organizational culture, infrastructure, personnel administration, marketplace conditions, policies, procedures, templates, and lessons learned from previous projects.

Define Scope - Tools

- **Expert Judgment**
 - Input from those experienced in similar projects to ensure the scope is realistic, complete, and feasible.
- **Decision-Making**
 - Multicriteria decision analysis and voting to select the final project requirements from the broader requirements documentation.
- **Data Analysis**
 - Alternatives analysis to evaluate ways to fulfill the requirements and objectives identified in the charter.

Define Scope - Tools

- **Decomposition**
 - Breaking down the project deliverables into smaller, more manageable components, lays groundwork for the WBS.
- **Interpersonal and Team Skills**
 - Facilitation - keeps key stakeholders aligned on what is in scope and what is excluded; resolves differences.
- **Product Analysis**
 - Tools such as product breakdown, requirements analysis, systems analysis, and value engineering used to translate high-level descriptions into tangible deliverables.

Define Scope - Output

- **Project Documents**
- **Project Scope Statement**
 - The project scope statement describes the project scope, product scope, major deliverables, assumptions, and constraints.
 - It helps determine whether change requests are inside or outside the project boundaries.
 - Key components include:
 - **Project scope description:** details the product, service, or result.
 - **Deliverables:** unique and verifiable outputs produced by the project.
 - **Acceptance criteria:** conditions that must be met before deliverables are accepted.
 - **Project exclusions:** items that are specifically out of scope.
 - The project charter is high level, while the project scope statement provides a more detailed description of scope.

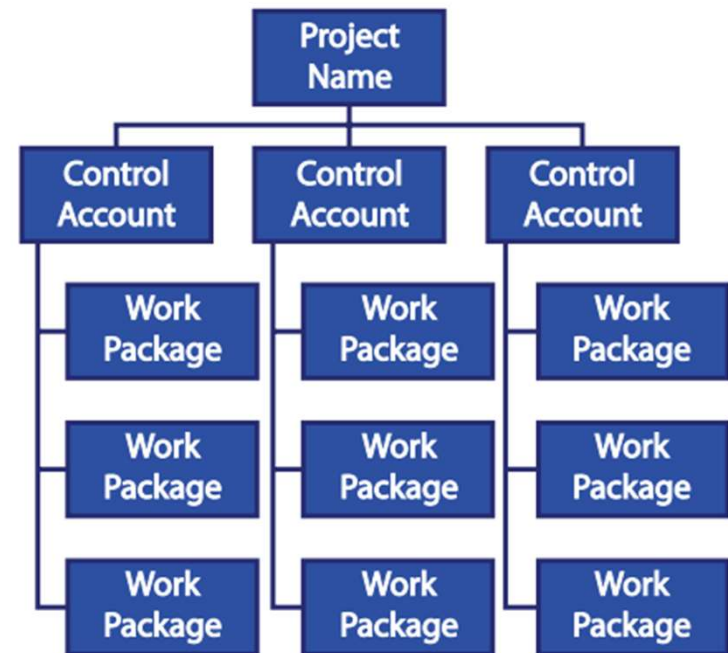
Define Scope - Output

- **Project Documents**
- **Requirements Documentation (updated)**
 - May be updated as a result of decisions made during scope definition (e.g., requirements removed, refined, or reprioritized).
- **Quality Management Plan**
 - Describes how quality policies, procedures, and guidelines will be applied to the project.
 - It explains how the project team will achieve the project's quality objectives.
 - Early quality planning can reduce rework, lower costs, and prevent schedule delays.
 - The plan may include:
 - Quality standards used by the project
 - Quality roles and responsibilities
 - Deliverables and processes subject to quality review
 - Quality control and quality management activities
 - Quality tools, methods, and procedures
 - Nonconformance and continuous improvement processes

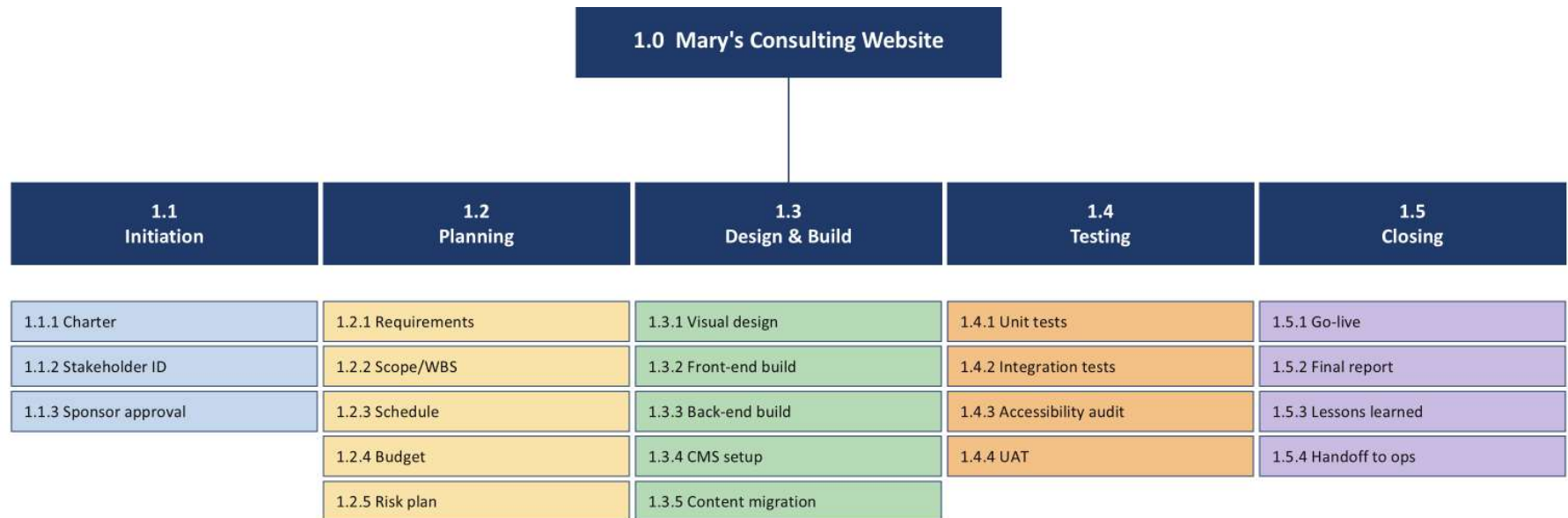
Develop Scope Structure

- In predictive projects, the WBS breaks the total project scope into smaller, manageable work packages.
- The WBS helps define project deliverables and supports planning, tracking, and control.
- Work packages make it easier to assign ownership, measure progress, and manage accountability.
- For complex projects, a WBS dictionary may provide extra details for each WBS component.
- Agile:
 - In Agile projects, the WBS is represented by the product backlog.
 - The backlog breaks work into epics, features, and user stories.

Develop Scope Structure Work Breakdown Structure (WBS)



Develop Scope Structure Work Breakdown Structure (WBS)



Develop Scope Structure

ITTO

Inputs

- Project management plan
- Project documents
 - Project scope statement
 - Requirements documentation
- Approved changes
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Brainstorming
- Decomposition
- Etc.

Outputs

- Scope baseline
- Work breakdown structure (WBS)
- WBS dictionary
- User stories
- Product backlog
- Etc.



Develop Scope Structure

- Inputs

- **Project Management Plan**
 - The scope management plan documents how the WBS will be created, maintained, and approved.
- **Project Documents**
 - Project scope statement - describes the deliverables, acceptance criteria, and project boundaries that the scope structure must reflect.
 - Requirements documentation - the detailed requirements that the WBS work packages must satisfy.
- **Approved Changes**
 - Approved change requests that affect scope must be incorporated into the WBS.
- **EEFs and OPAs**
 - Industry standards (e.g., government regulations, construction standards) and organizational templates, policies, and lessons learned from prior WBS development.

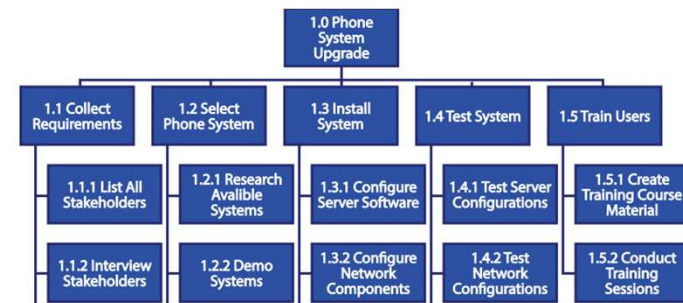
Develop Scope Structure

- Tools

- **Expert Judgment**
 - Input from individuals who have experience developing WBSs for similar projects.
- **Brainstorming**
 - Group technique used to generate the components of the scope structure when working from the project scope statement and requirements.
- **Decomposition**
 - The technique of dividing and subdividing the project scope and deliverables into smaller, more manageable parts.
 - Activities for decomposition: identify deliverables; structure and organize the WBS; decompose upper levels into lower-level components; assign identification codes; verify the decomposition is correct.
 - Each descending level represents an increasingly detailed definition of the project work.

Develop Scope Structure - Output

- **Scope Baseline**
 - The approved version of the
 - Scope statement
 - WBS
 - WBS dictionary
 - Used as the basis for comparison.
 - Can be changed only through formal change-control procedures.
- **Work Breakdown Structure (WBS)**
 - Hierarchical decomposition of the total scope of work to be carried out by the project team.
 - Lowest-level components are called work packages.



Develop Scope Structure - Output

- **WBS Dictionary**
- Document that provides detailed information about each WBS component (work description, owner, schedule, cost, acceptance criteria, etc.).

Project Name: Phone System Upgrade		WorkPackageID:1.3.1	
Work Package Name: Configure Server Software			
Work Package Description: Install a new virtual server. Install the phone server software. Configure the software to support 100 phone and voice mails to e-mail. Ensure all updates are applied to the operating system before installing the phone system software.			
Assigned to:	Bob Peterson	Duration:	5 Days
Date Assigned:	12/30/2017	Due Date:	1/30/2018
Estimated Cost:	\$5,000	Account Code:	PSU-882.3

Develop Scope Structure - Output

- **User Stories**
 - In adaptive projects, short descriptions of features written from the user's perspective.
- **Product Backlog**
 - An ordered list of features, requirements, and user stories that represents the planned work for the product.
 - Reordered as priorities and understanding evolve.

Plan Schedule Management

- Establishes how the project schedule will be planned, developed, managed, performed, and maintained.
- It provides guidance for managing the schedule throughout the project.

Plan Schedule Management

ITTO

Inputs

- Project charter
- Project management plan
 - Scope management plan
- Development approach
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data analysis
 - Alternative analysis
- Meetings
- Etc.

Outputs

- Project management plan updates
 - Schedule management plan
- Etc.

Plan Schedule Management

- Inputs

- **Project Charter**
 - Defines the summary milestone schedule, project approval requirements, and high-level constraints that influence scheduling decisions.
- **Project Management Plan**
 - Scope management plan - documents how the scope will be defined, developed, and validated, which directly impacts schedule planning.
- **Development Approach**
 - Predictive, iterative, incremental, agile, or hybrid - dictates how the project schedule is built and managed.
- **Enterprise Environmental Factors**
 - Organizational culture, marketplace conditions, scheduling tools, and resource availability that constrain or enable the schedule approach.
- **Organizational Process Assets**
 - Schedule templates, scheduling methodology, lessons learned from prior projects, and historical schedule data.

Plan Schedule Management

- Tools

- **Expert Judgment**
 - Specialists with experience in scheduling, scheduling software, the specific industry, and similar previous projects.
- **Data Analysis**
 - Alternative analysis - evaluates different methods of executing and accomplishing the work in a given level of detail, time, and budget.
 - Reviews multiple scheduling methodologies (CPM, critical chain, iterative) to determine the best fit for the project.
- **Meetings**
 - Planning meetings to develop the schedule management plan, involving the PM, sponsor, team members, and other relevant stakeholders.

Plan Schedule Management - Output

Project Management Plan Updates:

- **Schedule Management Plan**
 - Defines how the project schedule will be developed, monitored, and controlled.
 - It may be formal or informal, detailed or high level, depending on the project needs.
 - It establishes the scheduling method, tools, and procedures the team will use.
 - It may define release, wave, or iteration lengths for adaptive projects.
 - It identifies the level of accuracy for duration estimates and any contingency allowances.
 - It defines units of measure, such as hours, days, weeks, or quantity-based measures.
 - It explains how the schedule model will be updated and maintained during execution.
 - It sets control thresholds for acceptable schedule variance before action is needed.
 - It defines performance measurement rules, such as percent complete
 - It identifies the format and frequency of schedule reports.

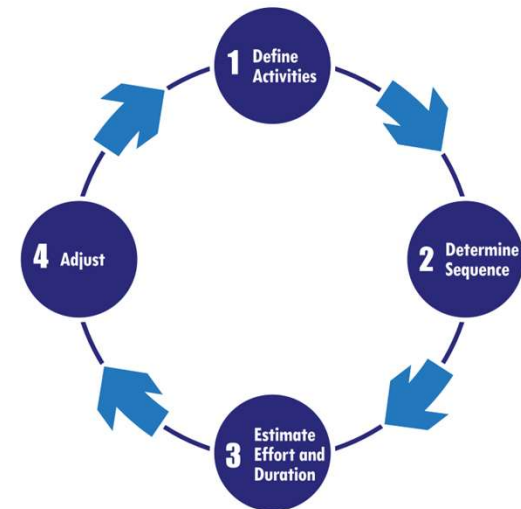
Develop Schedule

- Creates the schedule model used to execute, monitor, and control the project.
- It analyzes activity sequences, durations, resource needs, and schedule constraints.
- The schedule identifies planned start and finish dates for activities and milestones.
- Developing the schedule is an iterative process.
- The team may need to review and revise duration estimates, resource estimates, and schedule reserves.
- The goal is to create an approved project schedule that can be used as the schedule baseline.
- The schedule baseline is used to track progress and measure performance.

Develop Schedule

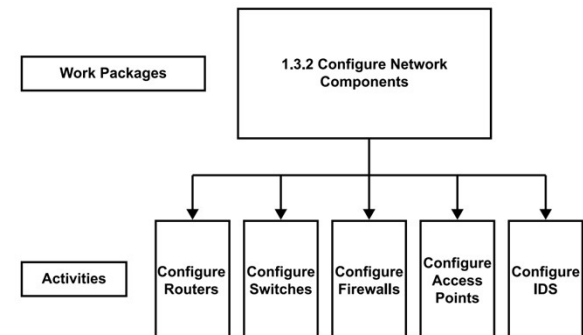
Four Steps to Develop the Schedule

1. **Define Activities:** identify the specific work activities needed to produce the project deliverables.
2. **Determine Sequence:** arrange activities in the correct order based on dependencies and relationships.
3. **Estimate Effort and Duration:** estimate the work effort, resources, and time needed to complete each activity.
4. **Adjust:** review and refine the schedule based on constraints, resources, risks, and project priorities.



Develop Schedule

- **Step 1: Define Activities**
 - Define Activities identifies the specific work actions needed to produce the project deliverables.
 - Work packages are broken down into schedule activities.
 - These activities become the basis for estimating, scheduling, executing, monitoring, and controlling project work.



Develop Schedule

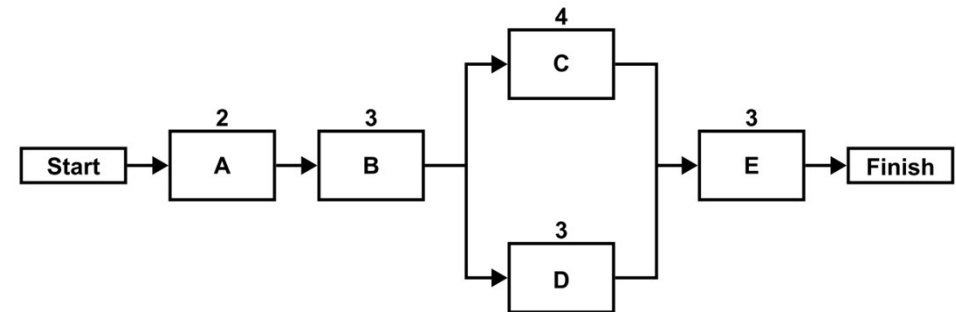
- **Step 1: Define Activities**
 - Key artifacts (output) may include:
 - **Activity List:** documented list of schedule activities needed to complete project work.
 - Includes activity ID, description, and enough detail so the team understands the work.
 - **Activity Attributes:** extra details about each activity, such as WBS ID, predecessors, successors, relationships, leads/lags, resources, constraints, and assumptions.
 - Attributes help with sequencing, scheduling, reporting, and organizing work.
 - **Milestone List:** identifies key project events or checkpoints.
 - Milestones have zero duration.
 - Milestones may be mandatory, such as contract requirements, or optional, such as internal target dates.

Develop Schedule

Step 2: Determine Sequence

- Determine Sequence identifies the logical order in which project activities should be performed.
- It defines dependencies and relationships between schedule activities.
 - Common relationships include:
 - Finish-to-start
 - Start-to-start
 - Finish-to-finish
 - Start-to-finish
- Leads and lags may be used to create a realistic and achievable schedule.
- Uses Precedence Diagramming Method (PDM),
 - A technique used for constructing a schedule model in which activities are represented by nodes and graphically linked by one or more logical relationships.
 - Outputs a Project Schedule Network Diagram

Develop Schedule



Develop Schedule

Step 2: Determine Sequence

- **Activity Dependency Relationships**
 - Activity relationships show how schedule activities depend on each other.
 - The four common dependency relationships are:
 - **Finish-to-Start (FS):** Activity A must finish before Activity B can start.
 - Most common relationship.
 - Example: The first coat of paint must dry before applying the second coat.
 - **Finish-to-Finish (FF):** Activity B cannot finish until Activity A finishes.
 - Example: Editing an article can start before writing is done, but editing cannot finish until writing is complete.
 - **Start-to-Start (SS):** Activity A must start before Activity B can start.
 - Example: Check-in cannot start until the venue opens.
 - **Start-to-Finish (SF):** Activity B cannot finish until Activity A starts.
 - Least common relationship.
 - Example: The day guard cannot leave until the night guard arrives.

Develop Schedule

Step 2: Determine Sequence

- **Dependency Determination**

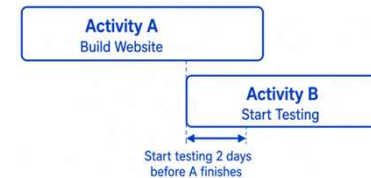
- Dependencies can also be classified by their nature and source.
- **Mandatory Dependencies:** required by the nature of the work.
 - Also called hard logic.
 - Example: Memory must be installed before the operating system can be installed.
- **Discretionary Dependencies:** based on preference, best practice, or experience.
 - Also called soft logic or preferred logic.
 - Example: Installing system updates before office software because it is easier to manage.
- **External Dependencies:** depend on something outside the project team's control.
 - Example: Waiting for permits, approvals, or supplier deliveries.
- **Internal Dependencies:** depend on work within the project team's control.
 - Example: Software must be installed before system testing can begin.

Develop Schedule

Step 2: Determine Sequence

- **Leads and Lags**

- Leads and lags are used to show overlap or delay between activities.
- Lead: Lead creates overlap between activities.
 - Example: Start taping fixtures after half the furniture has been removed.



- Lag: creates a delay between activities.
 - Example: Wait one day after painting before moving furniture back.



- Leads can help shorten the schedule when activities can safely overlap.
- Lags may be needed when work requires waiting time, curing time, approvals, or delays.

Develop Schedule

Step 3: Estimate Effort and Duration

- Estimate Effort and Duration determines how much work and time are needed to complete each activity.
- Estimates may be measured in hours, days, weeks, or other work periods.
- Estimating uses information such as scope, resource types, skill levels, resource quantities, and calendars.
- Common estimating techniques include:
 - Expert judgment
 - Delphi technique
 - Analogous estimating
 - Parametric estimating
 - PERT
 - Bottom-up estimating
- Estimates should be progressively elaborated as more information becomes available.
- Duration can be affected by resource availability, resource skill, constraints, risk, technology, and assumptions.

Develop Schedule

Step 4: Adjust

- Adjust means reviewing the draft schedule to determine whether it is realistic and acceptable.
- The draft schedule is reviewed against activity sequences, estimates, resources, constraints, and risks.
- If the schedule is not acceptable, the team revises it using alternative schedule options.
- Team members should review assigned activities to confirm dates are realistic and do not conflict with resource calendars.
- The schedule is analyzed for:
 - Logical relationship conflicts
 - Resource conflicts
 - Need for resource leveling
 - Schedule feasibility
- Once approved, the schedule becomes the baseline for tracking progress.

Develop Schedule

- **Common Estimation Techniques**
 - **Expert Judgment:** Uses people with experience to estimate the work. Accuracy depends on the expert's knowledge and the quality of information available.
 - **Analogous Estimating (Top-Down):** Uses similar past projects to estimate the current project. It is usually faster but less accurate because it is based on comparison.
 - **Bottom-Up Estimating:** Estimates smaller work components and adds them together. It is usually the most accurate but takes more time because it requires detailed information.
 - **Parametric Estimating:** Uses formulas, historical data, or statistical relationships. It can be highly accurate when the data and formula are reliable.
 - **Relative Size Estimation:** Compares work items by size or complexity, often used in Agile. It is useful for quick planning but is usually less precise than detailed estimating.

Develop Schedule

- **Common Estimation Techniques**
 - **Multipoint Estimating:** Uses multiple estimates, such as optimistic, most likely, and pessimistic. It is usually **more accurate than one-point estimating** because it considers uncertainty.
 - **Three-Point Estimating / PERT**
 - Three-point estimating uses three values to estimate activity duration or cost.
 - It is also known as **PERT**: Program Evaluation and Review Technique.
 - It helps account for uncertainty instead of using only one estimate.
 - The three estimates are:
 - **Optimistic (O):** best-case estimate; shortest possible time or lowest cost.
 - **Most Likely (M):** realistic estimate; what is most likely to happen.
 - **Pessimistic (P):** worst-case estimate; longest time or highest cost.

Develop Schedule

- PERT Formulas to Know
- Beta Distribution / Weighted Average
 - Formula: **$(O + 4M + P) / 6$**
 - Gives more weight to the most likely estimate.
 - Usually more realistic than a simple average.
- Triangular Distribution / Simple Average
 - Formula: **$(O + M + P) / 3$**
 - Gives equal weight to all three estimates.
- Standard Deviation
 - Formula: **$(P - O) / 6$**
 - Measures uncertainty or variability in the estimate.
- Example:
 - O = 2 days, M = 4 days, P = 6 days
 - Beta: **$(2 + 4(4) + 6) / 6 = 4$ days**
 - Triangular: **$(2 + 4 + 6) / 3 = 4$ days**
 - Standard Deviation: **$(6 - 2) / 6 = 0.67$ days**

Develop Schedule

Activity	Optimistic	Realistic	Pessimistic	PERT (Beta)	SD	Triangular
A	5	10	20	10.83	2.5	11.67
B	20	30	50	31.67	5	33.33
D	50	55	80	58.33	5	61.67
E	30	50	150	63.33	20	76.67

Develop Schedule

Inputs

- Project charter
- Project management plan
 - Scope management plan
- Development approach
- Project documents
 - Activity attributes
 - Activity list
 - Assumption log
 - Basis of estimates
 - Duration estimates
 - Lessons learned register
 - Milestone list
 - Project schedule network diagrams
 - Project team assignments
 - Resource calendars
 - Resource requirements
 - Risk register
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Decomposition
- Rolling wave planning
- Precedence diagramming method
- Logical relationship
- Leads and lags
- Dependency determination and integration
- Estimation techniques
- Reserve analysis
- Data analysis
 - What-if analysis
 - Simulation
 - Alternative analysis
- Voting
- Schedule network analysis
- Schedule compression
- Critical path method
- Critical chain method
- Resource optimization
 - Resource leveling
- Project management information system
- Agile release planning
- Etc.

Outputs

- Schedule baseline
- Project schedule
- Schedule data
- Project calendars
- Change requests
- Project management plan updates
 - Schedule management plan
- Project document updates
 - Activity list
 - Activity attributes
 - Assumption log
 - Basis of estimates
 - Duration estimates
 - Lessons learned register
 - Milestone list
 - Project schedule network diagrams
 - Resource requirements
 - Risk register
- Etc.

Develop Schedule - Inputs

- **Project Charter**
 - Defines summary milestone schedule and project approval requirements that influence schedule development.
- **Project Management Plan**
 - Scope management plan - drives how scope is decomposed and scheduled.
 - Schedule management plan – how to create the schedule
- **Development Approach**
 - Predictive vs adaptive vs hybrid - drives whether you use a Gantt-based schedule or a release plan with iterations.
- **Project Documents - Activities & Estimates**
 - Activity list - activities that need to be scheduled.
 - Activity attributes - successor/predecessor relationships, leads/lags, resource requirements per activity.
 - Assumption log & Basis of estimates - the assumptions and rationale behind duration estimates.
 - Duration estimates - time required for each activity.
 - Lessons learned register - prior schedule-development experience that should be applied here.

Develop Schedule - Inputs

- **Project Documents (cont.)**

- Milestone list - significant points or events whose dates anchor the schedule.
- Project schedule network diagrams - graphical representation of activities and their logical relationships.
- Project team assignments - who is performing each activity.
- Resource calendars - working days, shifts, and available resources by date.
- Resource requirements - skills, materials, equipment needed.
- Risk register - schedule risks and their planned responses.

- **Agreements**

- Vendor contracts that specify deliverable dates and milestones the schedule must accommodate.

- **EEFs and OPAs**

- Government/industry standards, communication channels, scheduling tools, scheduling methodology, calendar templates, and historical project information.

Develop Schedule - Tools

- **Expert Judgment**
 - Specialists in scheduling methodology, scheduling software, and similar prior projects.
- **Decomposition & Rolling Wave Planning**
 - Decomposition - breaking down work packages into activities.
 - Rolling wave planning - near-term work is planned in detail; future work is planned at a higher level (progressive elaboration).
- **Precedence Diagramming Method (PDM)**
 - A technique used for constructing a schedule model in which activities are represented by nodes and graphically linked by one or more logical relationships.

Develop Schedule - Tools

- **Leads and Lags**
 - Lead – Overlap of activities
 - Lag - delay between activities.
- **Dependency Determination & Integration**
 - Mandatory (hard logic), discretionary (preferred), external (outside the project), or internal dependencies.
- **Estimation, Reserves, Data Analysis**
 - Estimation techniques
 - Reserve analysis – adding additional time for risk.
 - What-if analysis - 'what if scenario X happens?' to test schedule resilience.
 - Simulation (Monte Carlo) - many possible schedules generated to assess probability of meeting target dates.

Develop Schedule - Tools

- **Critical Path & Critical Chain**
 - Critical Path Method (CPM) - longest path through the network diagram
 - Critical Chain Method - critical path approach that focuses on managing buffers and constrained resources.
- **Schedule Compression**
 - Crashing - add resources to shorten duration (typically increases cost).
 - Fast tracking - run activities in parallel that were originally sequential (typically increases risk).
- **Resource Optimization & Leveling**
 - Resource leveling adjusts start/finish dates to balance resource demand with availability; may extend the schedule.
- **PMIS, Agile Release Planning**
 - PMIS automates schedule modeling. Agile release planning decomposes the product roadmap into releases and iterations.

Develop Schedule - Output

- **Schedule Baseline**

- The approved version of the schedule model that can be changed only through formal change-control procedures.
- Used as the basis for comparison to actual results.

- **Project Schedule**

- The output of the schedule model that presents linked activities with planned dates, durations, milestones, and resources.
- Presented as bar charts, milestone charts, or network diagrams.

Develop Schedule - Output

- **Schedule Data**
 - Information for describing and controlling the schedule (milestones, activities, attributes, identified assumptions and constraints, resource histograms, alternative schedules).
- **Project Calendars**
 - Identifies working days and shifts available for scheduled activities.
- **Change Requests & Updates**
 - Change requests submitted via Integrated Change Control.
 - Schedule management plan, activity list, activity attributes, assumption log, basis of estimates, duration estimates, lessons learned register, milestone list, network diagrams, resource requirements, and risk register may all be updated.

Plan Financial Management

- Defines how project revenues and expenses will be estimated, budgeted, managed, monitored, and controlled.
- It provides guidance for managing project finances throughout the project.
- This process is usually performed once up front or at predefined points in the project.
- It helps ensure the project team understands how costs, budgets, funding, and financial performance will be handled.

Plan Financial Management

Inputs

- Project charter
- Project management plan
 - Schedule management plan
 - Risk management plan
- Project documents
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data analysis
 - Alternative analysis
- Meetings
- Etc.

Outputs

- Financial management plan
- Funding strategy
- Etc.

Plan Financial Management

- Inputs

- **Project Charter**
 - Defines the high-level financial constraints, pre-approved budget, and key sponsor approval requirements that frame the financial plan.
- **Project Management Plan**
 - Schedule management plan - drives the timing of when funds will be needed across the project life cycle.
 - Risk management plan - influences the size of contingency and management reserves built into the financial plan.
- **Project Documents**
 - Documents such as the stakeholder register and assumption log inform stakeholder funding expectations and assumptions underlying the financial approach.

Plan Financial Management

- Inputs

- **Enterprise Environmental Factors**
 - Marketplace conditions, currency exchange rates, published interest rates, and the organization's financial controls policies.
- **Organizational Process Assets**
 - Financial controls procedures, billing and accounting systems, financial reporting templates, and historical financial data from prior projects.

Plan Financial Management - Tools

- **Expert Judgment**

- Specialists in finance, accounting, treasury, risk, and the relevant industry help shape the financial approach.

- **Data Analysis**

- Alternative analysis - evaluates different funding options (self-funded, external loan, grant, milestone billing) and selects the most appropriate.

- **Meetings**

- Planning meetings with the sponsor, finance team, and PM to agree on the financial approach, controls, and reporting.

Plan Financial Management - Output

Financial Management Plan

- The financial management plan describes how project costs will be planned, structured, and controlled.
- It documents the financial management processes, tools, and techniques the project will use.
- It may define:
 - **Units of measure:** staff hours, days, weeks, or currency amounts.
 - **Level of precision:** how cost estimates are rounded.
 - **Level of accuracy:** acceptable estimate range, such as $\pm 10\%$.
 - **WBS/control accounts:** links costs to the WBS and accounting system.
 - **Control thresholds:** acceptable cost variance before action is required.
 - **Performance measurement rules:** EVM methods, control account measurement, and EAC forecasting.
 - **Reporting formats:** cost report format and frequency.
 - **Additional procedures:** funding choices, currency exchange handling, and cost recording.

Plan Financial Management - Output

- **Funding Strategy**
 - Defines how the project will secure the money needed to be successful.
 - Funding may come from internal or external sources.
 - Some projects may use one funding source, while others may combine multiple funding approaches.
 - Common funding strategies include:
 - **Fixed or reallocated internal budgets:** using existing organizational funds.
 - **Lump sum:** allocating the full project budget and reserves at once.
 - **Incremental disbursement:** releasing funds by phase, milestone, or need.
 - **External investment:** receiving capital from investors, often for ownership or future returns.
 - **Government or NGO grants:** funding from agencies, nonprofits, or philanthropists.
 - **Crowdfunding:** collecting small contributions from many supporters.
 - **Client contract:** funding provided through a customer or sponsor contract.

Estimate Costs

- The process of developing an approximation of the monetary resources needed to complete project work
- Determines the cost required to complete project work; estimates are based on the information known at a given point in time
- Cost trade-offs and risks must be considered (e.g., make vs buy, buy vs lease, sharing of resources)

Estimate Costs

Inputs

- Project management plan
 - Quality management plan
 - Scope baseline
- Project documents
 - Lessons learned register
 - Project schedule
 - Resource requirements
 - Risk register
- Make-or-buy decisions
- Work package estimation
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Analogous estimating
- Parametric estimating
- Bottom-up estimating
- Multipoint estimating
- Data analysis
 - Alternative analysis
 - Reserve analysis
 - Cost of quality
- Project management information system
- Decision-making
 - Voting
- Etc.

Outputs

- Cost estimates
- Basis of estimates
- Project document updates
 - Assumption log
 - Lessons learned register
 - Risk register
- Etc.



Estimate Costs - Inputs

- **Project Management Plan**
 - Quality management plan - identifies the level of quality required, which influences the cost of quality embedded in estimates.
 - Scope baseline - the project scope statement, WBS, and WBS dictionary define what work must be costed.
- **Project Documents**
 - Lessons learned register - applies prior estimating lessons to improve current estimates.
 - Project schedule - duration estimates and the timing of activities affect cost (e.g., resources costing more in peak season).
 - Resource requirements - the type, quantity, and characteristics of resources to be costed.
 - Risk register - identified risks may require contingency reserves built into the estimate.

Estimate Costs - Inputs

- **Make-or-Buy Decisions**
 - Decisions made in Plan Sourcing Strategy on what to build internally vs procure externally directly drive where costs land.
- **Work Package Estimation**
 - The granularity at which work has been decomposed; smaller work packages enable more accurate bottom-up estimates.
- **EEFs and OPAs**
 - Marketplace conditions, published commercial cost information, currency exchange rates, cost-estimating policies, templates, and historical cost data.

Estimate Costs - Tools

- **Expert Judgment**
 - Specialists with knowledge in similar prior projects, the industry, the discipline, and estimating methods.
- **Analogous Estimating**
 - Uses values from a similar previous project as the basis for estimating the current project.
 - Quick and inexpensive but generally less accurate; useful early when little information is known.
- **Parametric Estimating**
 - Uses a statistical relationship between historical data and other variables (e.g., \$ per page, \$ per square foot).
- **Bottom-Up Estimating**
 - Estimating component work and aggregating up. Most accurate but most time-consuming.
- **Multipoint Estimating (Three-Point)**
 - Uses optimistic (O), most likely (M), and pessimistic (P) estimates:
 - $\text{beta} = (O+4M+P)/6$.
 - $\text{triangular} = (O+M+P)/3$

Estimate Costs - Estimating Techniques

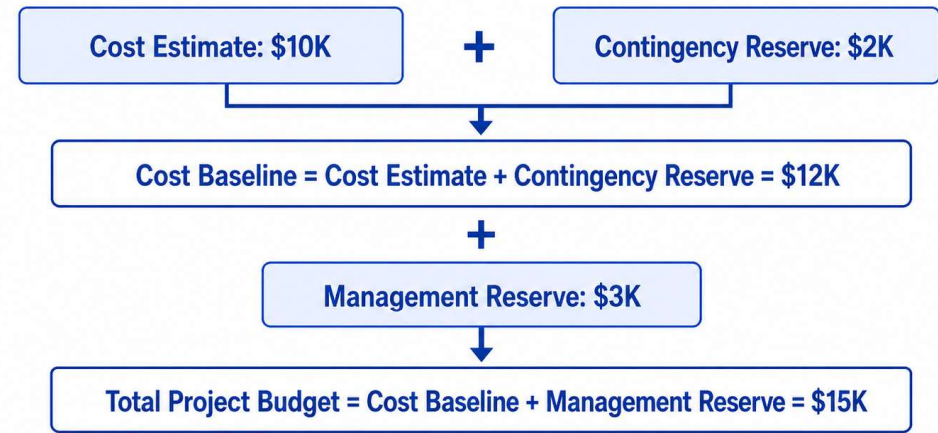
Analogous (Top-Down)	Parametric	Bottom-Up	Three-Point
Use actual cost from a similar past project	Use a statistical/unit-rate model (e.g. \$/page, \$/sqft)	Estimate each work package then aggregate up	Best, most likely, and worst case: $(O+4M+P)/6$
PROS / CONS Quick. Low cost. Less accurate.	PROS / CONS Quick. Defensible. Needs historical data.	PROS / CONS Most accurate. Most time-consuming.	PROS / CONS Accounts for uncertainty. Good for risky items.
TYPICAL ACCURACY -25% to +75%	TYPICAL ACCURACY -15% to +25%	TYPICAL ACCURACY -5% to +10%	TYPICAL ACCURACY -10% to +15%

Different techniques offer different trade-offs between speed, cost, and accuracy.

Estimate Costs - Tools

- **Data Analysis**
 - Alternative analysis - evaluates options to balance cost, schedule, scope, and quality.
 - Reserve analysis - contingency reserves for known risks; management reserves for unknown unknowns.
 - Cost of quality - includes cost of conformance (prevention + appraisal) and non-conformance (internal/external failure) in the estimate.
- **PMIS & Decision-Making**
 - PMIS aids in calculating, recording, and reporting estimates. Voting techniques converge multiple expert estimates.

Project Budget Buildup



Cost Estimate (\$10K)	Estimated cost of the work to complete the project.
Contingency Reserve (\$2K)	Set aside for known risks with active response plans; included in the cost baseline.
Management Reserve (\$3K)	Set aside for unknown-unknown risks; outside the cost baseline and controlled by management.
Total Project Budget (\$15K)	Total funds available = Cost Baseline (\$12K) + Management Reserve (\$3K) = \$15K.

Estimate Costs - Output

- **Cost Estimates**
 - Quantitative assessments of the probable costs required to complete project work.
 - May be presented as a single value or in ranges (e.g., \$10K +/- 10%).
 - Includes direct labor, materials, equipment, services, facilities, IT, contingency reserves, and indirect costs.
- **Basis of Estimates**
 - Supporting documentation that explains how each estimate was derived, including the assumptions, constraints, estimating method, range of possible estimates, and confidence level.
- **Project Document Updates**
 - Assumption log - updated with new assumptions made during estimating.
 - Lessons learned register - captures effective and ineffective estimating techniques used.
 - Risk register - updated with new risks identified during estimating, especially around uncertain costs.

Develop Budget

- The process of aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline
- Includes all funds authorized to execute the project
- Cost baseline is the approved version of the time-phased budget, excluding any management reserves
- Project funding requirements are derived from the cost baseline and may be incremental rather than continuous

Develop Budget ITTO

Inputs

- Project management plan
 - Financial management plan
 - Resource management plan
 - Scope baseline
- Project documents
 - Basis of estimates
 - Cost estimates
 - Project schedule
 - Risk register
- Business documents
 - Business case
 - Benefits management plan
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Cost aggregation
- Data analysis
 - Reserve analysis
- Historical information review
- Funding limit reconciliation
- Financing
- Etc.

Outputs

- Cost baseline
- Project funding requirements
- Project document updates
 - Cost estimates
 - Project schedule
 - Risk register
- Etc.

Develop Budget - Inputs

- **Project Management Plan**
 - Financial management plan - establishes the rules and controls under which the budget is built.
 - Resource management plan - identifies what resources are needed and at what cost.
 - Scope baseline - WBS and work packages provide the structure for aggregating costs.
- **Project Documents**
 - Basis of estimates - explains how each cost was estimated; needed to validate aggregations.
 - Cost estimates - the activity- or work-package-level figures that get aggregated into the budget.
 - Project schedule - drives the time-phasing of costs across the project life cycle.
 - Risk register - identified risks drive the contingency reserve included in the budget.

Develop Budget - Inputs

- **Business Documents**
 - Business case - documents the financial justification; the budget must support the expected benefits.
 - Benefits management plan - identifies the target benefits and timing the budget must enable.
- **Agreements**
 - Vendor contracts that fix portions of the budget at specific values and payment milestones.
- **EEFs and OPAs**
 - Currency exchange rates, market conditions, budgeting templates, organizational financial reporting standards, and historical budget data.

Develop Budget - Tools

- **Expert Judgment**
 - Specialists in financial planning, cost estimating, and the relevant industry refine and validate the aggregated budget.
- **Cost Aggregation**
 - Cost estimates are aggregated by work package per the WBS, then by control account, then to the project total.
- **Data Analysis**
 - Reserve analysis - establishes contingency reserves for known risks and management reserves for unknown risks; both included in the budget.

Develop Budget - Tools

- **Historical Information Review**

- Reviewing past similar projects to validate that the budget magnitude and pattern are reasonable.

- **Funding Limit Reconciliation**

- Reconciles planned expenditure with funding limits set by the organization (e.g., quarterly capital budgets) and may require schedule adjustments to align spending with available funds.

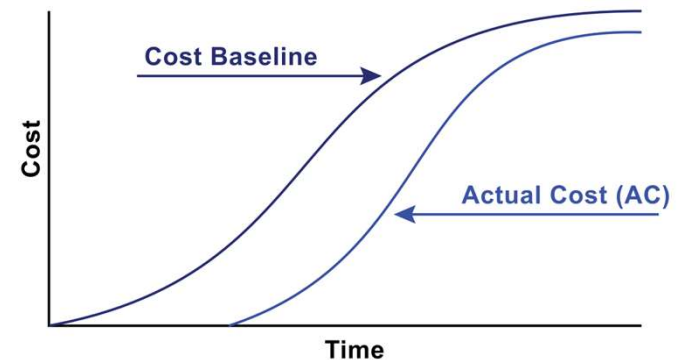
- **Financing**

- Acquiring funding for projects (loans, equity, customer advance payments) - especially relevant for long-duration or capital-intensive projects.

Develop Budget - Output

- **Cost Baseline**

- The approved version of the time-phased project budget, excluding any management reserves.
- Used as the basis for comparison to actual results; can only be changed through formal change-control procedures.
- Often visualized as an S-curve of cumulative cost over time.
- Equals the sum of activity cost estimates + contingency reserves; total project budget = cost baseline + management reserves.



Develop Budget - Output

- **Project Funding Requirements**
 - Total funding requirements + periodic funding requirements (monthly, quarterly, etc.) derived from the cost baseline.
 - Funding often occurs in incremental amounts that may not be evenly distributed; usually drawn at decision gates or milestone completion.
- **Project Document Updates**
 - Cost estimates, project schedule, and risk register may be updated as the budget is finalized and reconciled with funding limits.

Plan Stakeholder Engagement

- The process of developing approaches to involve project stakeholders based on their needs, expectations, interests, and potential impact on the project
- Provides an actionable plan to interact effectively with stakeholders throughout the project life cycle
- Used to identify gaps between current and desired engagement levels (Unaware, Resistant, Neutral, Supportive, Leading)
- Iteratively reviewed and updated as new stakeholders are identified or existing stakeholders change

Plan Stakeholder Engagement

ITTO

Inputs

- Project charter
- Project management plan
 - Resource management plan
 - Communications management plan
 - Risk management plan
- Project documents
 - Assumption log
 - Change log
 - Issue log
 - Project schedule
 - Risk register
 - Stakeholder register
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Benchmarking
- Data analysis
 - Assumption and constraint analysis
 - Root cause analysis
- Decision-making
 - Prioritization/ranking
- Data representation
 - Mind mapping
 - Stakeholder engagement assessment matrix
- Meetings
- Etc.

Outputs

- Project management plan updates
 - Stakeholder engagement plan
- Etc.

Plan Stakeholder Engagement - Inputs

- **Project Charter**
 - Identifies the project sponsor, key stakeholders, and high-level objectives that the engagement plan must support.
- **Project Management Plan**
 - Resource management plan - identifies team members and the resources required to engage stakeholders.
 - Communications management plan - drives how, when, and in what format engagement-related communications occur.
 - Risk management plan - identifies stakeholder-related risks that the engagement plan must mitigate.

Plan Stakeholder Engagement - Inputs

- **Project Documents**
 - Assumption log - assumptions about stakeholders' expectations and engagement preferences.
 - Change log - changes may shift stakeholder engagement needs.
 - Issue log - unresolved issues affecting stakeholder engagement.
 - Project schedule - timing of activities affects when engagement actions are needed.
 - Risk register - stakeholder-related risks to address.
 - Stakeholder register - the master list of stakeholders to be engaged.
- **Agreements**
 - Contracts may dictate engagement approaches with vendors and partner organizations.
- **EEFs and OPAs**
 - Organizational culture, political climate, governance framework, stakeholder engagement policies, templates, lessons learned.

Plan Stakeholder Engagement - Tools

- **Expert Judgment**
 - Input from those experienced in engaging similar stakeholder communities and on similar projects.
- **Data Gathering**
 - Benchmarking - compares stakeholder engagement results from current and prior projects to find effective practices.
- **Data Analysis**
 - Assumption and constraint analysis - tests assumptions about stakeholder behavior to refine the engagement approach.
 - Root cause analysis - identifies underlying reasons for current stakeholder engagement levels.

Plan Stakeholder Engagement - Tools

- **Decision-Making**
 - Prioritization/ranking - prioritizes stakeholders so engagement effort goes where it has the most impact.
- **Data Representation**
 - Mind mapping - visually organizes stakeholders and the relationships among them.
 - Stakeholder engagement assessment matrix - shows current (C) vs desired (D) engagement levels across Unaware, Resistant, Neutral, Supportive, and Leading.
- **Meetings**
 - Workshops with the team and key stakeholders to develop and validate the engagement strategies.

Plan Stakeholder Engagement - Stakeholder Engagement Assessment Matrix

C = Current engagement level, D = Desired engagement level. Where C and D differ, communication and engagement actions must close the gap.

Stakeholder	Unaware	Resistant	Neutral	Supportive	Leading
Mary (Sponsor / CEO)				C	D
Andrew (PM)					C D
Bob (Front-End)				C D	
Bill (Back-End)			C	D	
Christine (UX/Content)				C D	
Marketing Team			C	D	
IT Operations		C	D		
Consultants (End Users)	C			D	

 Current engagement

 Desired engagement

Plan Stakeholder Engagement - Output

- **Stakeholder Engagement Plan**
 - A component of the Project Management Plan that identifies the strategies and actions required to promote productive involvement of stakeholders.
 - Documents desired and current engagement levels of key stakeholders.
 - Identifies the scope and impact of change to stakeholders.
 - Identifies interrelationships and potential overlap between stakeholders.

Plan Communications Management

- The process of developing an appropriate approach and plan for project communications based on stakeholder information needs and project requirements
- Identifies and documents the approach to communicate most effectively and efficiently with stakeholders
- Considers what info is needed, when it is needed, who needs it, who provides it, who receives it, in what format, and at what frequency

Plan Communications Management - ITTO

Inputs

- Project charter
- Project management plan
 - Resource management plan
 - Stakeholder engagement plan
- Project documents
 - Requirements documentation
 - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Communication requirements analysis
- Communication technology
- Communication models
- Communication methods
- Interpersonal and team skills
 - Communication styles assessment
 - Political awareness
 - Cultural awareness
- Data representation
 - Stakeholder engagement assessment matrix
- Meetings
- Etc.

Outputs

- Project management plan updates
 - Communications management plan
 - Stakeholder engagement plan
- Project document updates
 - Project schedule
 - Stakeholder register
- Etc.

Plan Communications Management

- Inputs

- **Project Charter**
 - Identifies key stakeholders, sponsor, and the project purpose, all of which inform communication needs.
- **Project Management Plan**
 - Resource management plan - identifies the team members and resources who will produce or consume project communications.
 - Stakeholder engagement plan - identifies stakeholders and their desired engagement levels which drive what must be communicated to whom.

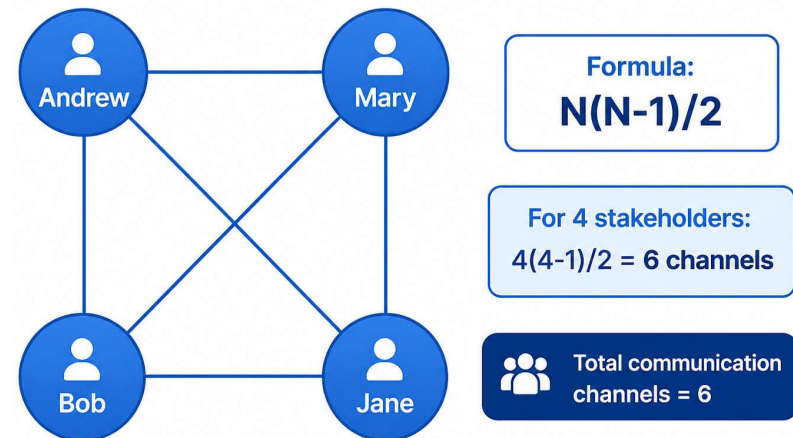
Plan Communications Management - Inputs

- **Project Documents**
 - Requirements documentation - communication requirements (reporting cadence, languages, formats) may be explicitly stated.
 - Stakeholder register - the master list of who must be communicated with.
- **Enterprise Environmental Factors**
 - Organizational culture, political climate, governance, personnel administration, stakeholder risk thresholds, and established communication channels.
- **Organizational Process Assets**
 - Organizational policies and procedures for communication, templates, historical communication artifacts, and lessons learned.

Plan Communications Management - Tools

- **Expert Judgment**
 - Specialists with experience in communications, organizational politics, and similar projects.
- **Communication Requirements Analysis**
 - Determines the information needs of project stakeholders.
 - Number of potential communication channels
 - **FORMULA: $N(N-1)/2$**
 - where N is the number of stakeholders.
 - Adding stakeholders rapidly multiplies channels.

Potential Communication Channels



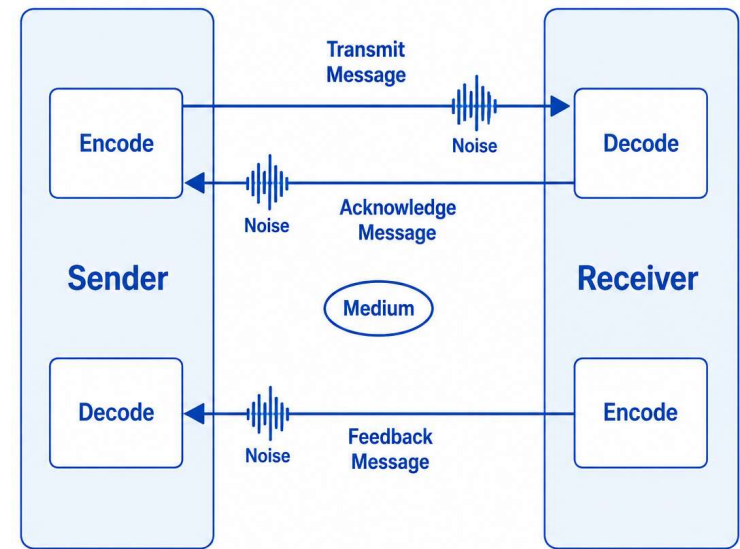
Plan Communications Management - Tools

- **Communication Technology**
 - Includes the tools, systems, and software used to share information with project stakeholders.
 - Common methods include meetings, conversations, written documents, databases, social media, websites, and collaboration tools.
 - Factors that affect the choice of technology include:
 - **Urgency:** how quickly the information is needed.
 - **Availability and reliability:** whether stakeholders can access and use the technology.
 - **Ease of use:** whether the tool is simple enough or requires training.
 - **Project environment:** virtual teams, time zones, languages, and culture.
 - **Sensitivity and confidentiality:** whether security or privacy controls are needed.

Plan Communications Management - Tools

- **Communication Models**
 - A communication model shows how information moves between a sender and receiver.
 - The basic model focuses on delivering the message:
 - **Encode:** sender turns the message into words, symbols, or media.
 - **Transmit:** message is sent through a communication channel.
 - **Decode:** receiver interprets the message.
 - The interactive model adds understanding:
 - **Acknowledge:** receiver confirms the message was received.
 - **Feedback/Response:** receiver responds to confirm understanding.
 - The sender is responsible for making the message clear, complete, and properly understood.
 - The receiver is responsible for receiving, interpreting, and responding appropriately.
 - Noise can interfere with communication, such as distractions, technology issues, culture, language, assumptions, or bias.
 - Cross-cultural communication may be affected by age, nationality, profession, gender, personality, and background.

Plan Communications Management - Tools



Communication Model

Plan Communications Management - Tools

- **Communication Methods**
 - Techniques used to share information with project stakeholders.
 - **Interactive communication:** real-time, two-way or multidirectional communication.
 - Examples: meetings, phone calls, IM
 - **Push communication:** information is sent directly to specific recipients.
 - Examples: emails, reports, memos, voicemails
 - **Pull communication:** recipients access information when needed.
 - Examples: intranet sites, web portals, eLearning.
- Communication can also occur as:
 - **Interpersonal:** one-on-one communication.
 - **Small group:** communication among 3–6 people.
 - **Public:** one speaker or group addressing a large audience.
 - **Mass communication:** message sent to a large or anonymous audience.
 - **Networks/social computing:** many-to-many communication using social tools and media.

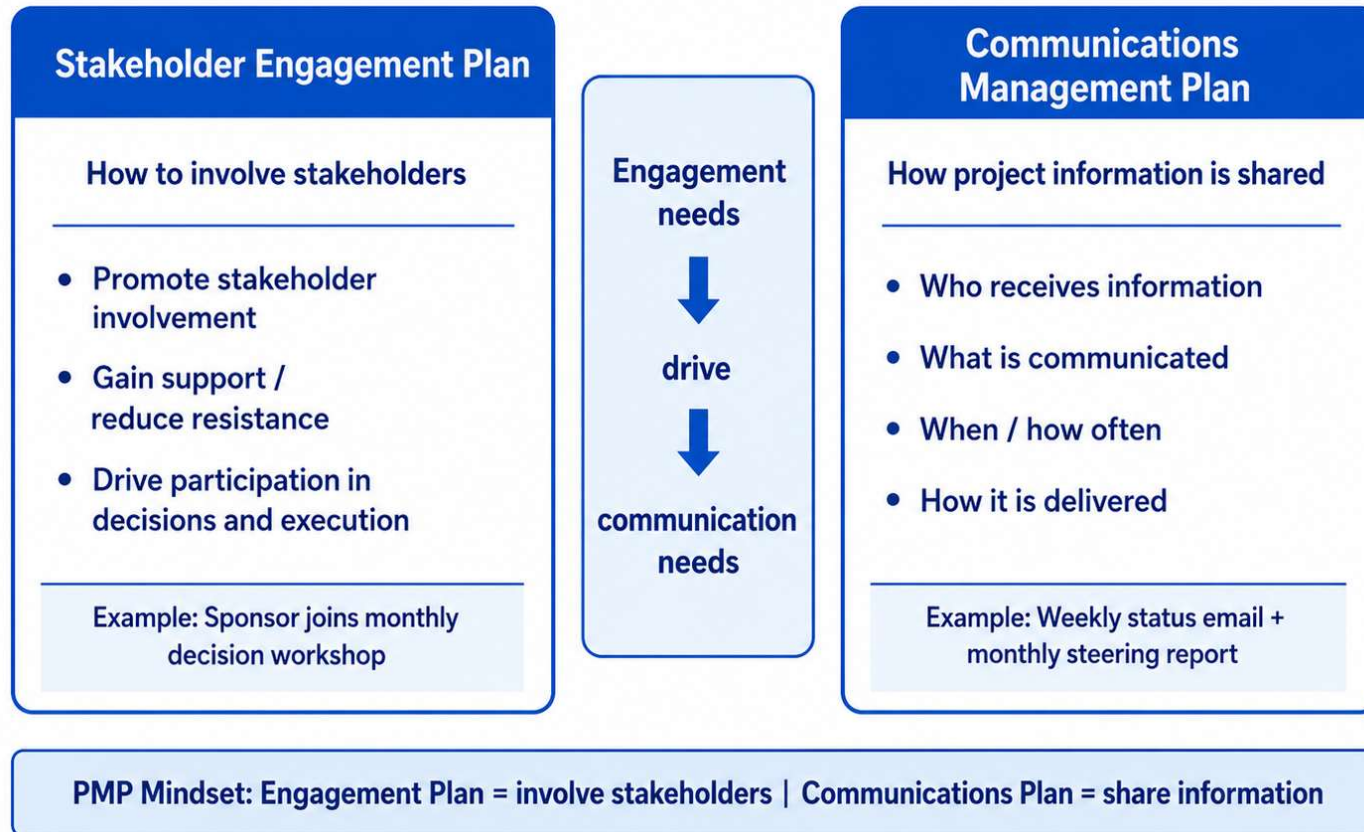
Plan Communications Management - Tools

- **Interpersonal and Team Skills**
 - Communication styles assessment - identifies the preferred method, format, and content of communication for each stakeholder.
 - Political awareness - understands the power relationships in and outside the project.
 - Cultural awareness - understands cultural differences and language barriers that affect communication.
- **Data Representation & Meetings**
 - Stakeholder engagement assessment matrix - drives what info each stakeholder needs.
 - Communication planning meetings with the team and key stakeholders to develop and validate the plan.

Plan Communications Management - Output

- **Communications Management Plan**
 - A component of the Project Management Plan that explains how project information will be created, shared, monitored, and updated.
 - It defines who needs information, what information is needed, when it is needed, and how it will be delivered.
 - It identifies stakeholder communication requirements, message format, language, content, and level of detail.
 - It defines communication methods and technologies, such as meetings, reports, emails, dashboards, websites, or project software.
 - It identifies who is responsible for communicating information and who can approve confidential information.
 - It includes timing, frequency, escalation processes, acknowledgments, and response requirements.
 - It may include templates, meeting guidelines, report formats, terminology, workflows, and communication constraints.
 - The plan should be updated as stakeholders, scope, team structure, or project conditions change.

Stakeholder Engagement Plan vs Communications Management Plan



Plan Communications Management - Output

- **Stakeholder Engagement Plan & Document Updates**
 - Stakeholder engagement plan - updated as communication needs are clarified.
 - Project schedule - communication activities may need to be added.
 - Stakeholder register - updated with communication preferences.

Plan Resource Management

- The process of defining how to estimate, acquire, manage, and use physical and team resources
- Establishes the approach and level of management effort needed to manage project resources based on the type and complexity of the project
- Identifies the various roles needed for the project, the responsibilities of those roles, and the required competencies
- Considers cultural, geographical, and organizational factors in resource planning

Plan Resource Management

Inputs

- Project charter
- Project management plan
 - Quality management plan
 - Scope baseline
- Project documents
 - Project schedule
 - Requirements documentation
 - Risk register
 - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Interviews
- Data analysis
 - SWOT analysis
- Data representation
 - Hierarchical charts
 - Responsibility assignment matrix
 - Text-oriented formats
- Organizational theory
- Meetings
- Green human resource management
- Resource-based view
- Etc.

Outputs

- Project management plan updates
 - Resource management plan
- Team charter
- Project document updates
 - Assumption log
 - Risk register
- Etc.

Plan Resource Management - Inputs

- **Project Charter**
 - Identifies the high-level project description, key milestones, and pre-assigned resources or sponsors that frame resource planning.
- **Project Management Plan**
 - Quality management plan - influences resources needed (skills, certifications) to meet quality standards.
 - Scope baseline - WBS identifies the deliverables that will require resources.

Plan Resource Management - Inputs

- **Project Documents**
 - Project schedule - timing of activities drives when resources are needed.
 - Requirements documentation - dictates the skills and physical resources needed.
 - Risk register - identified resource-related risks (e.g., key personnel loss).
 - Stakeholder register - identifies stakeholders who influence resource planning (HR, finance, vendors).
- **EEFs and OPAs**
 - Organizational culture and structure, geographic distribution, marketplace conditions, organizational policies and procedures, role descriptions, and templates.

Plan Resource Management - Tools

- **Expert Judgment**
 - Specialists with expertise in negotiating for resources, personnel administration, and the discipline of the project.
- **Data Gathering**
 - Interviews - one-on-one discussions with stakeholders to understand resource availability and needs.
- **Data Analysis**
 - SWOT analysis - examines team strengths, weaknesses, opportunities, and threats to inform resource planning.

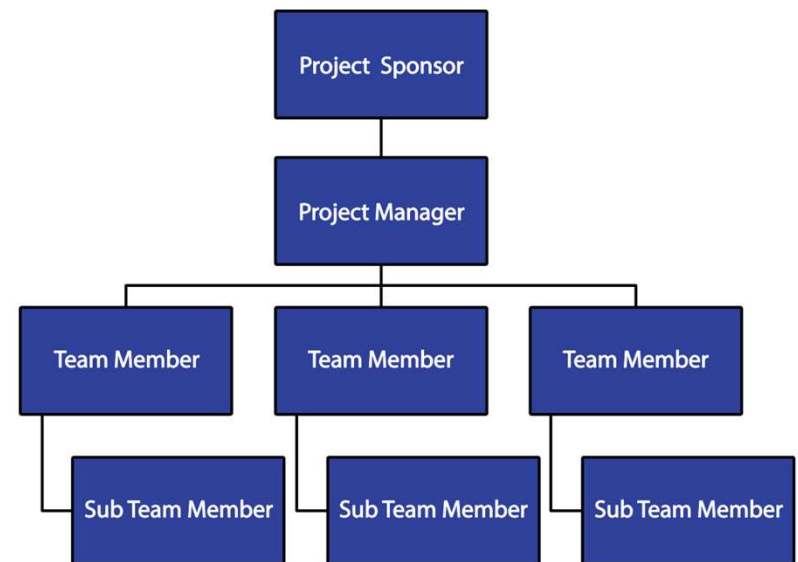
SWOT ANALYSIS



Plan Resource Management - Tools

- **Data Representation**

- Hierarchical charts - org charts showing reporting lines.
- Responsibility Assignment Matrix (RAM) - shows who is responsible for what work; common form is RACI (Responsible, Accountable, Consulted, Informed).
- Text-oriented formats - position descriptions, role-responsibility-authority forms.



Simple project organization chart

Plan Resource Management - RACI Chart Example

RACI Matrix – Web Design Project

Project Activities	Mary (Sponsor)	Andrew (PM)	Bob (Team Member)	Bill (Team Member)	Christine (Team Member)
1. Define website goals	A	R	C	C	C
2. Gather requirements	C	A/R	C	C	C
3. Create project plan	I	A/R	C	C	C
4. Design website layout	I	A	R	C	C
5. Develop website	I	A	R	C	C
6. Content creation	I	A	C	C	R
7. Test website	I	A	R	C	C
8. Review & approve	A	R	C	C	C
9. Launch website	I	A	C	R	C

R – Responsible

The person(s) who do the work

A – Accountable

The person who is ultimately accountable

C – Consulted

The person(s) whose input is required

I – Informed

The person(s) who need to be kept informed



Plan Resource Management - Tools

- **Organizational Theory**
 - Provides information regarding the way in which people, teams, and organizational units behave.
- **Meetings**
 - Resource planning meetings with the team and key stakeholders.
- **Green HRM & Resource-Based View**
 - Green Human resource management - integrates environmental sustainability into HR practices (e.g., remote work to reduce travel).
 - Resource-based view - views people and physical resources as a source of competitive advantage.

Plan Resource Management - Output

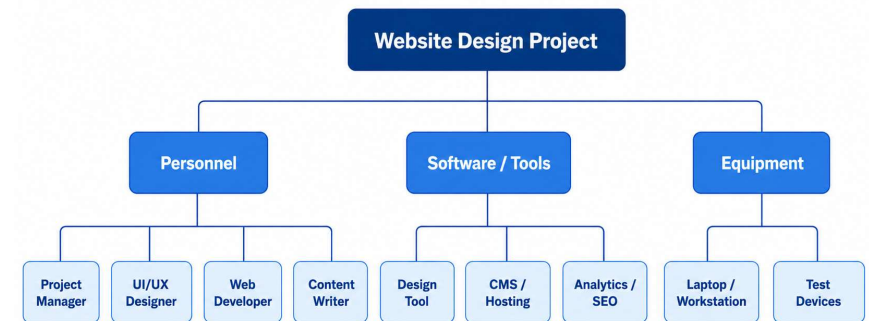
- **Resource Management Plan**
 - The resource management plan explains how project resources will be acquired, allocated, monitored, and controlled.
 - It may cover both team resources and physical resources.
 - It defines how resources will be identified, estimated, acquired, managed, and released.
 - It includes roles and responsibilities:
 - **Role:** the function assigned to a person.
 - **Authority:** the right to make decisions and use resources.
 - **Responsibility:** the work a team member is expected to perform.
 - **Competence:** the skills needed to complete the work.
 - It may include the project organization chart, staffing approach, training, team development, and recognition plan.

Plan Resource Management - Output

- **Team Charter**
 - A team charter defines the team's values, agreements, and operating guidelines.
 - It sets clear expectations for acceptable team behavior.
 - It helps reduce misunderstandings and improve productivity.
 - The team charter may include:
 - Team values
 - Communication guidelines
 - Decision-making processes
 - Conflict resolution processes
 - Meeting guidelines
 - Team agreements
 - It works best when the team helps create it or has input into it.
 - All team members are responsible for following the charter.
 - The charter can be reviewed and updated as the team changes or new members join.

Plan Resource Management - Output

- **Project Document Updates**
 - Assumption log - assumptions about resource availability documented.
 - Risk register - resource-related risks identified.
 - Resource Breakdown Structure - organizes project resources by category and type.



Resource Breakdown Structure

Estimate Resources

- The process of estimating team resources and the type and quantities of materials, equipment, and supplies necessary to perform project work
- This process helps identify possible resource shortages or surpluses early.
- It supports better resource allocation and helps manage resource-related risks.
- Estimate Resources is performed once or at predefined points in the project.
- It is closely connected to the schedule because resource availability can affect activity duration.

Estimate Resources

ITTO

Inputs

- Project management plan
 - Resource management plan
 - Schedule management plan
 - Scope baseline
- Project documents
 - Activity attributes
 - Activity list
 - Assumption log
 - Cost estimates
 - Resource calendars
 - Risk register
- Project schedule
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Bottom-up estimating
- Analogous estimating
- Parametric estimating
- Data analysis
 - Alternative analysis
- Project management information system
- Meetings
- Data gathering
 - Interviews
- Artificial intelligence
- Predictive analytics
- Virtual reality
- Augmented reality
- Branch and bound
- Genetic algorithms
- Constructive cost model
- Etc.

Outputs

- Resource requirements
- Basis of estimates
- Resource breakdown structure
- Project document updates
 - Activity attributes
 - Assumption log
 - Lessons learned register
 - Risk register
- Etc.

Estimate Resources - Inputs

- **Project Management Plan**
 - Resource management plan - guides how resources are estimated, acquired, and managed.
 - Schedule management plan - dictates timing of resource needs.
 - Scope baseline - WBS identifies the work to be resourced.

Estimate Resources - Inputs

- **Project Documents**

- Activity attributes - successor/predecessor relationships, leads/lags, resource requirements.
- Activity list - the activities requiring resources.
- Assumption log - assumptions about resource availability and skill levels.
- Cost estimates - cost may influence resource selection (e.g., contractor vs employee).
- Resource calendars - working days and shifts when resources are available.
- Risk register - resource-related risks (skill gaps, single points of failure).

- **Project Schedule**

- Activities and their dates determine when each resource is needed.

- **EEFs and OPAs**

- Resource location, marketplace conditions, published estimating data, lessons learned, and historical info.

Estimate Resources - Tools

- **Expert Judgment**

- Specialists with experience in resource planning, estimating, and the relevant skill area.

- **Bottom-Up Estimating**

- Resource needs are estimated at the activity or work-package level, then aggregated. Most accurate but most time-consuming.

- **Analogous Estimating**

- Uses values from similar previous projects as the basis for estimating resources for the current project. Quick but less accurate.

Estimate Resources - Tools

- **Parametric Estimating**
 - Uses statistical relationships (e.g., person-hours per page, square feet per worker) between historical data and other variables.
- **Data Analysis**
 - Alternative analysis - evaluates options like make-vs-buy, capability vs commodity resources, or different mixes of skill levels.
- **Project Management Information System**
 - Resource management software, scheduling tools, and estimating systems used to model resource needs.
- **Meetings & Interviews**
 - Discussions with team members, subject matter experts, and prior project teams to refine resource estimates.

Estimate Resources - Tools

- **Artificial Intelligence (AI):** analyzes past project data to recommend the type and quantity of resources needed.
 - Example: suggests the number of developers, designers, and testers for a website project.
- **Predictive Analytics:** uses historical data and trends to forecast future resource needs and possible shortages.
 - Example: predicts when extra developers may be needed to meet a deadline.
- **Virtual Reality (VR):** simulates project environments to help identify needed workers, tools, and equipment.
 - Example: simulates a construction site before work begins.
- **Augmented Reality (AR):** overlays digital information onto the real world to improve resource planning.
 - Example: estimates paint and labor needs by projecting coverage onto walls.

Estimate Resources - Tools

- **Branch and Bound:** finds the best resource allocation under constraints.
 - A way to test different resource choices and eliminate the bad options until you find the best one.
 - Example: assigns developers to tasks to reduce delays.
- **Genetic Algorithms:** test many resource scenarios to find the best mix of cost, time, and workload.
 - The tool may test many possible team plans:
 - Plan 1: two developers on design, one on backend
 - Plan 2: one designer, two developers, one tester
 - Example: balances team assignments on a large website project.
- **COCOMO:** estimates software development effort, time, and resources based on project size and complexity.
 - Example: estimates how many developers are needed for a website project.

Estimate Resources - Output

- **Resource Requirements**
 - Resource requirements identify the types and quantities of resources needed for project work.
 - They may be defined for each activity, work package, WBS branch, or the entire project.
 - Resources may include people, equipment, materials, supplies, facilities, or technology.
 - The level of detail depends on the project and application area.
 - The documentation may include assumptions about resource type, quantity, and availability.
- **Basis of Estimates**
 - Documents how the estimates were derived: the method, assumptions, constraints, range, and confidence level.

Estimate Resources - Output

- **Resource Breakdown Structure (RBS)**
 - Hierarchical representation of resources by category (labor, materials, equipment, facilities) and type (skill level, grade).
 - Used to organize and report project resource data.



- **Project Document Updates**
 - Activity attributes - updated with refined resource requirements per activity.
 - Assumption log - new assumptions documented.
 - Lessons learned register - estimating insights captured.
 - Risk register - resource-related risks identified during estimation.

Plan Risk Management

- The process of defining how to conduct risk management activities for a project
- Defines the approach, tools, data sources, roles and responsibilities, budget, timing, risk categories, definitions of probability and impact, and the probability/impact matrix

Plan Risk Management

Inputs

- Project charter
- Project management plan
 - All components
- Project documents
 - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Interviews
- Data analysis
 - Stakeholder analysis
- Meetings
- Etc.

Outputs

- Project management plan updates
 - Risk management plan
- Etc.

Plan Risk Management - Inputs

- **Project Charter**

- Provides high-level project description, summary milestones, and stakeholder list - which all influence the level of risk management effort.

- **Project Management Plan**

- All components of the PM Plan are inputs because risk management cuts across every aspect of the project.
- Especially the scope, schedule, cost, quality, resource, communications, procurement, and stakeholder plans - each introduces specific categories of risk.

Plan Risk Management - Inputs

- **Project Documents**
 - Stakeholder register - identifies stakeholders whose risk attitudes and tolerances must be reflected in the risk management plan.
- **Enterprise Environmental Factors**
 - Overall risk thresholds set by the organization, regulatory requirements, marketplace conditions, organizational culture, and risk attitude of stakeholders.
- **Organizational Process Assets**
 - Organizational risk policy, risk categories, common definitions of risk concepts and terms, risk statement formats, and standard templates.

Plan Risk Management - Tools

- **Expert Judgment**

- Specialists with knowledge of the project's industry, type of project, risk identification techniques, and risk management practices.
- Often includes input from senior management, sponsors, and other PMs experienced with similar projects.

- **Data Gathering**

- Interviews - one-on-one discussions with key stakeholders to gather perspectives on risk approach, risk attitudes, and any concerns.

Plan Risk Management - Tools

- **Data Analysis**

- Stakeholder analysis - examines the risk attitudes and tolerances of project stakeholders, which inform the definitions and thresholds in the risk management plan.

- **Meetings**

- Risk planning meetings with the team, sponsor, key stakeholders, and risk experts.

Plan Risk Management - Output

- **Risk Management Plan**

- A component of the PM Plan that describes how risk management activities will be structured and performed, can include:
 - Risk strategy - general approach for managing risk on the project.
 - Methodology - specific approaches, tools, and data sources used to perform risk management.
 - Funding - identifies funds needed for risk management and protocols for use of contingency and management reserves.
 - Timing - when and how often risk management will be performed.
 - Risk categories - groupings (e.g., technical, external, organizational, project management) often shown in a Risk Breakdown Structure (RBS).
 - Definitions of risk probability and impact - explicit definitions of each level (very low to very high).
 - Probability and Impact Matrix - the prioritization matrix used to qualitatively assess risks.
 - Reporting formats - how outcomes of risk management are documented, analyzed, and communicated.
 - Tracking - how risk activities will be recorded and audited.

Identify Risks - Risk Breakdown Structure (RBS)



Identify Risks

- Identifying individual project risks as well as sources of overall project risk
 - **Individual project risk:** a specific uncertain event or condition that could affect one or more project objectives.
 - Example: A key developer may become unavailable, causing a website feature to be delayed.
 - **Overall project risk:** the combined effect of all risks and uncertainty on the entire project.
 - Example: The website project may miss the launch date because of multiple risks, such as scope changes, testing defects, and stakeholder approval delays.
- Iterative because new individual risks may emerge as the project progresses through its life cycle

Identify Risks

Inputs

- Project management plan
 - Requirements management plan
 - Schedule management plan
 - Financial management plan
 - Quality management plan
 - Resource management plan
 - Risk management plan
 - Scope baseline
 - Schedule baseline
 - Cost baseline
- Project documents
 - Assumption log
 - Cost estimates
 - Duration estimates
 - Issue log
 - Lessons learned register
 - Requirements documentation
 - Resource requirements
 - Stakeholder register
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Brainstorming
 - Checklists
 - Interviews
- Data analysis
 - Root cause analysis
 - Assumption and constraint analysis
 - SWOT analysis
 - Document analysis
- Interpersonal and team skills
 - Facilitation
- Prompt lists
- Meetings
- Artificial intelligence
- Etc.

Outputs

- Risk register
- Risk report
- Project document updates
 - Assumption log
 - Issue log
 - Lessons learned register
- Etc.

Identify Risks - Inputs

- **Project Management Plan**
 - Requirements management plan - identifies areas of high uncertainty in requirements that may introduce risk.
 - Schedule management plan - identifies time-related risks (e.g., aggressive deadlines).
 - Financial management plan - identifies cost and funding risks.
 - Quality management plan - identifies areas susceptible to quality risk.
 - Resource management plan - identifies resource availability and skill-related risks.
 - Risk management plan - the playbook governing the Identify Risks process: roles, categories, definitions, and tools to use.
 - Scope, schedule, and cost baselines - establish the reference points; risks are identified relative to what could threaten meeting them.

Identify Risks - Inputs

- **Project Documents**
 - Assumption log - assumptions and constraints can become risks if proven wrong.
 - Cost estimates / duration estimates - estimates with wide ranges or low confidence indicate risk.
 - Issue log - existing issues may indicate emerging or underlying risks.
 - Lessons learned register - prior project risks that may apply here.
 - Requirements documentation - unclear, complex, or novel requirements signal risk.
 - Resource requirements - shortages or specialized skill needs introduce risk.
 - Stakeholder register - stakeholder conflict, low engagement, or attitude shifts can be risks.

Identify Risks - Inputs

- **Agreements**
 - Vendor contracts may contain risk-shifting clauses, penalties, or warranties that affect identified risks.
- **EEFs and OPAs**
 - Published material (industry studies, benchmarks), academic studies, risk databases, organizational risk registers, lessons learned, and templates.

Identify Risks - Tools

- **Expert Judgment**

- Specialists with experience on similar projects, in the same business area, or with the relevant technology.

- **Data Gathering**

- Brainstorming - team or workshop session focused on generating risk ideas (often using risk categories from the RBS as prompts).
- Checklists - lists of risks identified on previous similar projects; ensures common risks are not missed.
- Interviews - one-on-one or small-group discussions with stakeholders, SMEs, and team members to surface risks.

Identify Risks - Tools

- **Data Analysis**
 - Root cause analysis - examines a problem to determine underlying causes that may also represent risks.
 - Assumption and constraint analysis - tests the validity of assumptions and constraints; if invalidated, they become risks.
 - Document analysis - structured review of project documents may reveal areas of risk.
 - SWOT analysis - examines Strengths, Weaknesses, Opportunities, and Threats from the project

Strengths -Expert team -Management support	Weaknesses -Little free time -High cost
Opportunities -New Market -New IT Systems	Threats -Regulations -Staff Shortage

SWOT diagram

Identify Risks - Tools

- **Interpersonal and Team Skills**
 - Facilitation - improves the effectiveness of brainstorming, workshops, and interviews used to identify risks.
- **Prompt Lists**
 - Predetermined lists of risk categories that might give rise to risks
- **Meetings**
 - Risk-identification workshops with the team and key stakeholders, often facilitated and time-boxed.
- **Artificial Intelligence**
 - AI-assisted risk identification using historical project data, pattern recognition, and natural language processing of project documents.

Identify Risks - Output

- **Risk Register**
 - Contains the list of identified individual project risks.
 - List of identified risks - each with a unique identifier, category, brief description.
 - Potential risk owners - the person responsible for monitoring and responding to each risk.
 - List of potential responses - any preliminary response ideas surfaced during identification.

Identify Risks - Output

- **Risk Report**
 - Presents information on sources of overall project risk and summary information on identified individual project risks.
- **Project Document Updates**
 - Assumption log - new assumptions documented during risk identification.
 - Issue log - existing issues that turn out to be risks moved or linked.
 - Lessons learned register - effective and ineffective techniques for risk identification captured.

Perform Risk Analysis

- Perform Risk Analysis evaluates project risks to understand their probability, impact, and effect on project objectives.
- This process is iterative and may be repeated throughout the project.
- It includes both qualitative and quantitative risk analysis.
 - **Qualitative risk analysis** assesses individual risks based on probability, impact, urgency, manageability, timing, and relationships to other risks.
 - Qualitative analysis helps prioritize which risks need the most attention.
 - **Quantitative risk analysis** numerically analyzes the combined effect of risks and uncertainty on overall project objectives.
 - Quantitative analysis may not be required for every project.

Perform Risk Analysis

- **Risk Characteristics**
- **Urgency**
 - How quickly we must respond to the risk
 - Example: A key server may fail today, so the team must act immediately
 - High: A response must happen immediately to be effective.
 - Low: The response can wait and still be effective.
- **Proximity**
 - How soon the risk may impact the project
 - Example: A supplier delay could affect the project next week
 - High: The risk may impact the project very soon.
 - Low: The risk may not impact the project until much later.
- **Dormancy**
 - How long it takes to discover the impact after the risk happens
 - Example: A software bug occurs today, but the team does not notice it until customers complain later
 - High: The risk occurs, but the impact may not be discovered for a long time.
 - Low: The risk occurs, and the impact is discovered quickly.
- **Manageability**
 - How easy it is to manage the risk or reduce its impact
 - Example: If one team member is unavailable, the PM can assign another trained person
 - High: The risk owner can easily manage the risk or reduce its impact.
 - Low: The risk owner has limited ability to manage the risk or reduce its impact.

Perform Risk Analysis

- **Controllability**
 - How much control the project team has over the risk outcome
 - Example: The team can control internal testing quality, but cannot control a major storm
 - High: The risk owner has strong control over the risk's outcome.
 - Low: The risk owner has little or no control over the risk's outcome.
- **Detectability**
 - How easy it is to notice the risk before or when it occurs
 - Example: A schedule delay is easy to detect when tasks are not being completed on time
 - High: The risk can be detected and recognized easily.
 - Low: The risk is difficult to detect or recognize.
- **Connectivity**
 - How much the risk is connected to other risks
 - Example: A vendor delay can also cause schedule delays, cost increases, and quality issues
 - High: The risk is connected to many other project risks.
 - Low: The risk is mostly independent and not connected to many other risks.

Perform Risk Analysis

Inputs

- Project management plan
 - Risk management plan
 - Scope baseline
 - Schedule baseline
 - Cost baseline
- Project documents
 - Assumption log
 - Cost estimates
 - Duration estimates
 - Resource requirements
 - Risk register
 - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering and analysis
 - Interviews
- Interpersonal and team skills
 - Facilitation
- Risk categorization
- Data analysis
 - Risk probability and impact assessment
 - Simulations
 - Sensitivity analysis
 - Decision tree analysis
 - Influence diagrams
- Data representation
 - Probability and impact matrix
- Etc.

Outputs

- Project document updates
 - Assumption log
 - Issue log
 - Risk register
 - Risk report
- Etc.



Perform Risk Analysis - Inputs

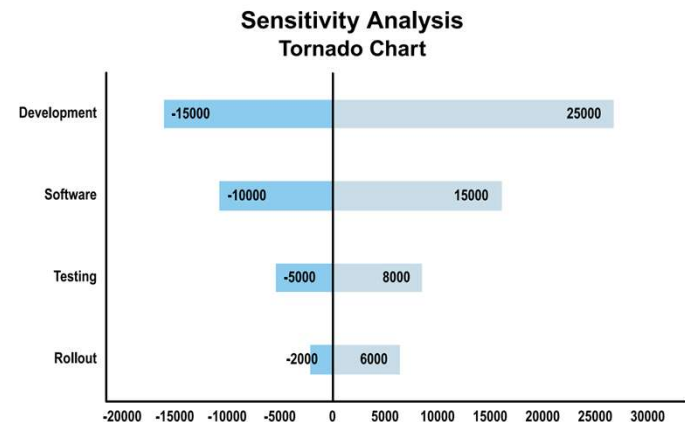
- **Project Management Plan**
 - Risk management plan - defines roles, the methodology, and the probability/impact definitions used during analysis.
 - Scope, schedule, and cost baselines - the reference points against which risk impacts are measured.
- **Project Documents**
 - Assumption log - assumptions still in play that could drive risk.
 - Cost and duration estimates - their ranges feed into quantitative risk analysis (e.g., Monte Carlo).
 - Resource requirements - resource gaps may amplify specific risks.
 - Risk register - the primary input; provides the list of individual risks to be analyzed and prioritized.
 - Stakeholder register - identifies stakeholders whose judgment will be used in probability/impact assessments.
- **EEFs and OPAs**
 - Industry studies, benchmarks, published risk databases, organizational risk policies, and historical risk data from similar projects.

Perform Risk Analysis - Tools

- **Expert Judgment**
 - Specialists with prior experience in similar projects estimate probability, impact, and other risk parameters where data is limited.
- **Data Gathering and Analysis**
 - Interviews - structured discussions with risk owners and SMEs to elicit probability, impact, and other parameters.
- **Interpersonal and Team Skills**
 - Facilitation - improves the effectiveness of risk workshops where probability and impact are assessed.
- **Risk Categorization**
 - Risks are grouped by sources, areas affected, or other useful categories (often by RBS) to identify areas of the project most exposed to risk.

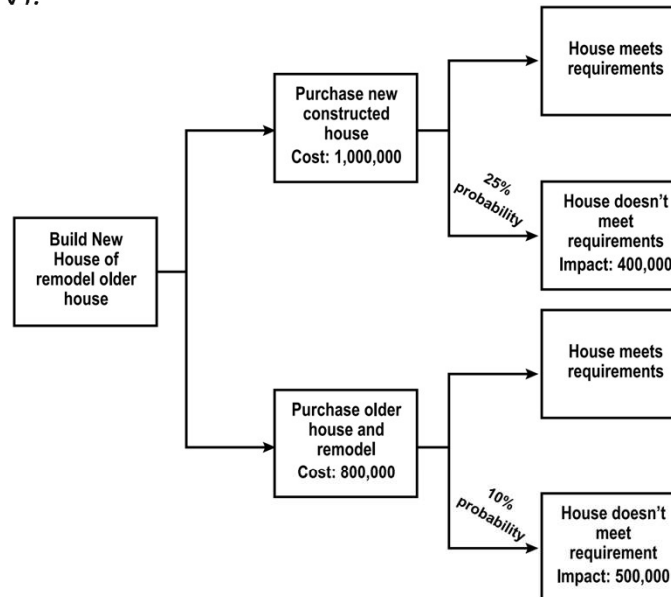
Perform Risk Analysis - Tools

- **Data Analysis - Qualitative**
 - Risk probability and impact assessment - examines the likelihood of each risk and its potential effect on project objectives.
- **Data Analysis - Quantitative**
 - Simulations (typically Monte Carlo) - model the combined effect of risks on project outcomes by running thousands of trials with input variables sampled from probability distributions.
 - Sensitivity analysis - determines which individual risks have the most potential impact on project outcomes (often visualized as a tornado diagram).



Perform Risk Analysis - Tools

- Decision tree analysis - structures decisions under uncertainty by mapping options, probabilities, and outcomes to compute Expected Monetary Value (EMV).



	Initial Cost	Risk Cost	Probability	EMV	Total
New Constructed House	1,000,000	400,000	25%	100,000	1,100,000
Remodel Older House	800,000	500,000	10%	50,000	850,000

To calculate the EMV, multiply the probability by the risk costs. To get the total, add the EMV to the initial cost.

Perform Risk Analysis - Tools

- Influence diagrams - graphical aids for decision making under uncertainty showing causal influences and time ordering of events.
- **Data Representation**
 - Probability and impact matrix - a grid that maps risk probability against impact and assigns an overall risk rating (low/moderate/high/very high).

PROBABILITY AND IMPACT MATRIX

(Website Remaking Project)

PROBABILITY (Chance of Occurrence)	IMPACT (Effect on Project Objectives)			
	LOW (1)	MODERATE (2)	HIGH (3)	VERY HIGH (4)
VERY HIGH (4)	MODERATE Minor image update delay	HIGH Testing defects	VERY HIGH Data migration failure	VERY HIGH Launch delay
HIGH (3)	LOW Small content revision	MODERATE Content migration errors	HIGH Scope changes	VERY HIGH Launch delay
MODERATE (2)	LOW Small content revision	MODERATE Stakeholder approval delay	HIGH SEO ranking drop	HIGH Scope changes
LOW (1)	LOW Minor image update delay	LOW Small content revision	MODERATE Plugin compatibility issue	MODERATE SEO ranking drop

RISK RATING KEY

LOW

=

Low priority risk

MODERATE

=

Medium priority risk

HIGH

=

High priority risk

VERY HIGH

=

Very high priority risk

Perform Risk Analysis - Output

- **Project Document Updates**
 - Assumption log - new assumptions identified during analysis.
 - Issue log - issues identified during the analysis process.
- **Risk Register Updates**
 - Probability and impact assessment for each individual risk.
 - Risk score and risk rating (low/moderate/high/very high).
 - Risk priority - the order in which risks should be addressed.
 - Risk categorization - which categories show the most exposure.
- **Risk Report Updates**
 - Assessment of overall project risk exposure.
 - Prioritized list of individual project risks.
 - Trends in quantitative risk analysis results.
 - Recommended risk responses (input to Plan Risk Responses).

Plan Risk Responses

- The process of developing options, selecting strategies, and agreeing on actions to address overall project risk exposure and to treat individual project risks
- Identifies appropriate ways to address individual risks and the overall project risk
- Risk responses must be appropriate, cost-effective, agreed upon, owned by a responsible person, and timely

Plan Risk Responses

Inputs

- Project management plan
 - Resource management plan
 - Risk management plan
 - Cost baseline
- Project documents
 - Lessons learned register
 - Project schedule
 - Project team assignments
 - Resource calendars
 - Risk register
 - Risk report
 - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data gathering
 - Interviews
- Interpersonal and team skills
 - Facilitation
- Strategies for threats
- Strategies for opportunities
- Contingent response strategies
- Strategies for overall project risk
- Data analysis
 - Alternative analysis
 - Cost-benefit analysis
- Decision-making
 - Multicriteria decision analysis
- Etc.

Outputs

- Change requests
- Project management plan updates
 - Schedule management plan
 - Financial management plan
 - Quality management plan
 - Resource management plan
 - Procurement management plan
 - Scope baseline
 - Schedule baseline
 - Cost baseline
- Project document updates
 - Assumption log
 - Cost forecasts
 - Lessons learned register
 - Project schedule
 - Project team assignments
 - Risk register
 - Risk report
- Etc.

Plan Risk Responses - Inputs

- **Project Management Plan**

- Resource management plan - shows what resources can be allocated to risk responses.
- Risk management plan - the playbook for response planning, including roles and contingency reserve rules.
- Cost baseline - establishes the funding envelope; responses requiring additional funding may need a baseline change.

Plan Risk Responses - Inputs

- **Project Documents**

- Lessons learned register - prior responses that worked or didn't.
- Project schedule - drives when risk responses must be implemented.
- Project team assignments - identifies who is available to own and execute responses.
- Resource calendars - confirms when resources will be available for response actions.
- Risk register - the prioritized list of risks awaiting response strategies.
- Risk report - overall project risk exposure that must be addressed at the project level.
- Stakeholder register - identifies stakeholders who must approve or be involved in responses.

- **EEFs and OPAs**

- Stakeholder risk appetite, organizational risk policies, templates for risk response plans, lessons learned, and historical response strategies.

Plan Risk Responses - Risk Response Strategies

THREATS (Negative Risks)

AVOID	TRANSFER	MITIGATE	ACCEPT	ESCALATE
Eliminate the threat by removing the cause. Example: Change scope, schedule, or strategy to remove the risky work.	Shift the impact to a third party. Example: Insurance, warranties, fixed-price contracts, outsourcing.	Reduce probability and/or impact. Example: Add redundancy, prototyping, testing, or experienced people.	Acknowledge the risk; take no action unless it occurs. Example: Active accept = contingency reserve. Passive accept = deal with it if it happens.	The risk is outside the project's authority; raise it. Example: Escalate to the program, portfolio, sponsor, or senior management level.

OPPORTUNITIES (Positive Risks)

EXPLOIT	SHARE	ENHANCE	ACCEPT	ESCALATE
Ensure the opportunity is realized. Example: Assign best resources, use new technology, eliminate uncertainty.	Allocate ownership to a third party best able to capture it. Example: Joint ventures, partnerships, special-purpose teams.	Increase probability and/or positive impact. Example: Add resources, accelerate schedule, identify triggers that maximize the opportunity.	Acknowledge it; take advantage if it happens. Example: No proactive action; be ready if the opportunity materializes.	The opportunity is beyond the project's scope; raise it. Example: Escalate to program, portfolio, sponsor, or senior management for action.

Plan Risk Responses - Tools

- **Expert Judgment**
 - Specialists with experience handling similar risks (financial, technical, contractual, legal, negotiation).
- **Data Gathering & Interpersonal Skills**
 - Interviews - structured conversations with risk owners and SMEs to develop and refine response options.
 - Facilitation - improves the effectiveness of response-planning workshops.
- **Contingent Response Strategies**
 - Pre-defined responses that are only executed if specific trigger conditions occur.
- **Strategies for Overall Project Risk**
 - Address total project risk exposure rather than individual risks (may include adjusting the project scope, schedule, cost, or quality plan).
- **Data Analysis & Decision-Making**
 - Alternative analysis - compares response options.
 - Cost-benefit analysis - quantifies whether the response is worth the cost.
 - Multicriteria decision analysis - selects among responses by weighting multiple criteria.

Plan Risk Responses - Output

- **Change Requests**
 - Planned responses often require changes to baselines or other PM Plan components; submitted via Integrated Change Control.
- **Project Management Plan Updates**
 - Schedule, financial, quality, resource, and procurement management plans may all be updated to incorporate risk responses.
 - Scope, schedule, and cost baselines may change to reflect approved responses (e.g., adding contingency buffers).

Plan Risk Responses - Output

- **Project Document Updates**
 - Assumption log - assumptions related to chosen responses.
 - Cost forecasts - updated to reflect cost of responses.
 - Lessons learned register - response-planning lessons.
 - Project schedule - response activities added.
 - Project team assignments - owners assigned to risk responses.
 - **Risk register** - updated with chosen response strategies, specific actions, owners, due dates, and any residual or secondary risks.
 - Risk report - updated with the agreed responses for individual and overall project risk.

Executing

- 8 Processes across all 4 domains
- Executes the project management plan to build the deliverables

Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Governance	Initiate Project or Phase	- Integrate and Align Project Plans - Plan Sourcing Strategy	- Manage Project Execution - Manage Quality Assurance - Manage Project Knowledge	- Monitor and Control Project Performance - Assess and Implement Changes	Close Project or Phase
Scope		- Plan Scope Management - Elicit and Analyze Requirements - Define Scope - Develop Scope Structure		- Monitor and Control Scope - Validate Scope	
Schedule		- Plan Schedule Management - Develop Schedule		Monitor and Control Schedule	
Finance		- Plan Financial Management - Estimate Costs - Develop Budget		Monitor and Control Finances	
Stakeholders	Identify Stakeholders	- Plan Stakeholder Engagement - Plan Communications Management	- Manage Stakeholder Engagement - Manage Communications	- Monitor Stakeholder Engagement - Monitor Communications	
Resources		- Plan Resource Management - Estimate Resources	- Acquire Resources - Lead the Team	Monitor and Control Resourcing	
Risk		- Plan Risk Management - Identify Risks - Perform Risk Analysis - Plan Risk Responses	Implement Risk Responses	Monitor Risks	

Manage Project Execution

- Where the team performs the work defined in the project management plan.
- Complete deliverables and meet project objectives.
- The project manager leads the team, coordinates work, and aligns team knowledge and skills.
- Approved changes are implemented
- Deliverables are created as the team performs planned project activities.
- Work performance data is collected and shared for monitoring and controlling.
- Quality is managed by ensuring processes are effective and deliverables meet agreed requirements.
- In adaptive projects, work is executed based on prioritized requirements or the sprint backlog.

Manage Project Execution

Inputs

- Project management plan
 - Any component
- Project documents
 - Change log
 - Lessons learned register
 - Milestone list
 - Project communications
 - Project schedule
 - Requirements traceability matrix
 - Risk register
 - Risk report
- Approved change requests
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Project management information system
- Meetings
 - Daily coordination meetings
- Etc.

Outputs

- Deliverables
- Work performance data
- Issue log
- Change requests
- Project management plan updates
 - Any component
- Project document updates
 - Activity list
 - Assumption log
 - Lessons learned register
 - Requirements documentation
 - Risk register
 - Stakeholder register
- Organizational process asset updates
- Etc.

Manage Project Execution - Inputs

- **Project Management Plan**
 - Any component of the PM Plan may be an input - the plan and its baselines guide what is executed and how.
- **Project Documents**
 - Change log - tracks all change requests and their dispositions.
 - Lessons learned register - captures and applies knowledge gained earlier in the project.
 - Milestone list - shows scheduled milestones the execution work must hit.
 - Project communications - prior reports, presentations, and notes from stakeholders.
 - Project schedule - identifies activities, dates, and resources for execution.
 - Requirements traceability matrix - links requirements to deliverables being produced.
 - Risk register and risk report - inform risk-based execution decisions.

Manage Project Execution - Inputs

- **Approved Change Requests**
 - Change requests already approved by Integrated Change Control are scheduled for implementation as part of execution.
- **EEFs and OPAs**
 - Stakeholder risk thresholds, infrastructure, communication channels, organizational policies, and templates that influence execution.

Manage Project Execution - Tools

- **Expert Judgment**
 - Specialists with expertise in technical knowledge of the industry and the focus of the project guide execution decisions.
- **Project Management Information System (PMIS)**
 - Software tools and infrastructure such as scheduling tools, work authorization systems, configuration management systems, information collection and distribution systems, and dashboards.
 - Provides automated access to information needed to manage the project effectively.
- **Meetings**
 - Used to discuss and address project issues during executing.
 - Daily coordination meetings (stand-ups in adaptive projects) keep the team aligned on what was done, what is being done, and any blockers.
 - Other meetings include kickoff, technical, planning, and status meetings.

Manage Project Execution - Output

- **Deliverables**
 - Any unique and verifiable product, result, or capability to perform a service produced by the project.
- **Work Performance Data**
 - Raw observations and measurements identified during activities (e.g., percent complete, technical performance measures, start/finish dates, costs, defects).
- **Issue Log**
 - Documents project issues - what they are, who is responsible, target resolution dates, and status.

Manage Project Execution - Output

- **Change Requests**
 - Formal proposals to modify any document, deliverable, or baseline. Submitted to Integrated Change Control.
 - May include corrective actions, preventive actions, defect repairs, and updates to PM Plan.
- **Project Management Plan Updates**
 - Any component of the PM Plan may be updated based on changes resulting from execution.
- **Project Document Updates**
 - Activity list, assumption log, lessons learned register, requirements documentation, risk register, and stakeholder register may all be updated during execution.
- **OPA Updates**
 - Updates to standards, guidelines, templates, and lessons learned in the organization's process asset library.

Manage Quality Assurance

- Manage Quality Assurance ensures project processes are performed according to stakeholder expectations.
- The key benefit is increasing the likelihood of meeting project objectives.
- It helps identify ineffective processes and causes of poor project performance.
- **Quality assurance** focuses on whether the project processes are being followed effectively.
 - It includes standards, compliance, regulations, audits, and process improvement.
- **Quality control** focuses on whether the deliverables meet defined specifications and quality thresholds.
 - Quality control includes testing, inspections, defects, and product acceptance.

Manage Quality Assurance

Inputs

- Project management plan
- Project documents
 - All components
- Organizational process assets
 - Policies
 - Procedures
 - Regulations
- Etc.

Tools & Techniques

- Audits
- Checklists
- Data representation
 - Affinity diagrams
 - Cause-and-effect diagrams
 - Flowcharts
- Decision-making
- Problem-solving
- Process improvement
- Etc.

Outputs

- Quality reports
- Change requests
- Project management plan updates
- Project document updates
- Etc.

Manage Quality Assurance - Inputs

- **Project Management Plan**
 - The quality management plan defines acceptable levels of project and product quality, how to ensure they are met, and the metrics used.
 - Other components inform what processes are being executed and how they should be assessed.
- **Project Documents**
 - All relevant project documents - including the lessons learned register, quality control measurements, quality metrics, and risk report - feed quality assurance activities.
- **Organizational Process Assets**
 - Policies - corporate quality policies, audit policies, and compliance policies the project must follow.
 - Procedures - standard processes used by the organization for ensuring quality.
 - Regulations - external rules (industry, government, legal) the project work must comply with.

Manage Quality Assurance - Tools

- **Audits**

- A structured, independent process used to determine if project activities comply with organizational and project policies, processes, and procedures.
- Identifies all good and best practices being implemented; identifies all gaps and shortcomings; shares good practices with similar projects.

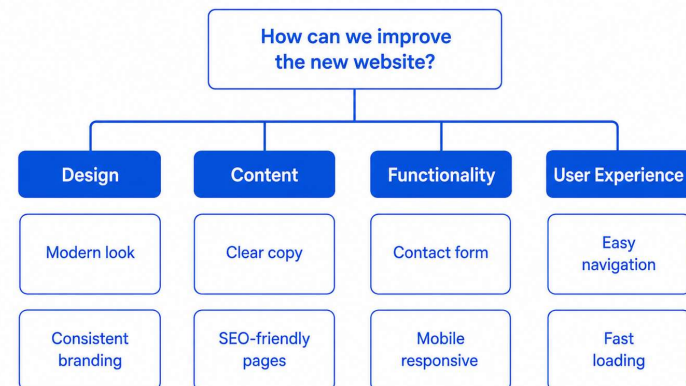
- **Checklists**

- A structured tool used to verify that a set of required steps has been performed.

Manage Quality Assurance - Tools

- **Data Representation**
- Affinity diagrams - organize potential causes of defects into groups showing common areas to focus on.

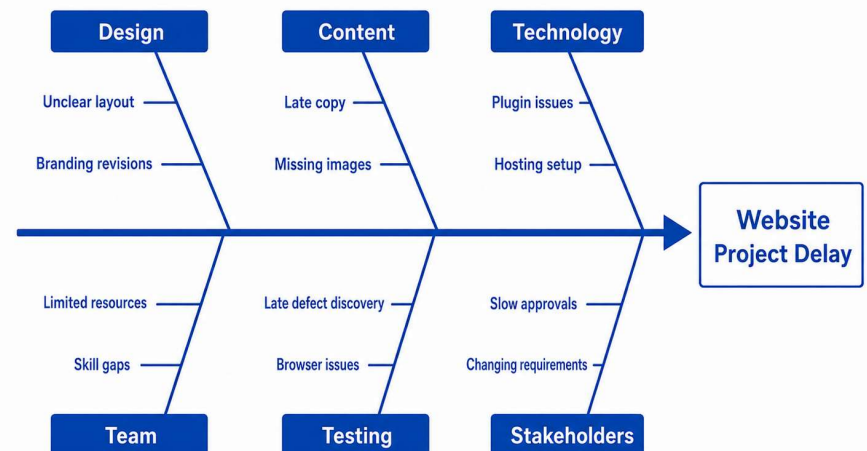
Affinity Diagram – Web Design Project



Manage Quality Assurance - Tools

- **Data Representation**
- Cause-and-effect diagrams (Ishikawa/fishbone) - break down the causes of a problem into discrete branches that help identify the main or root cause.

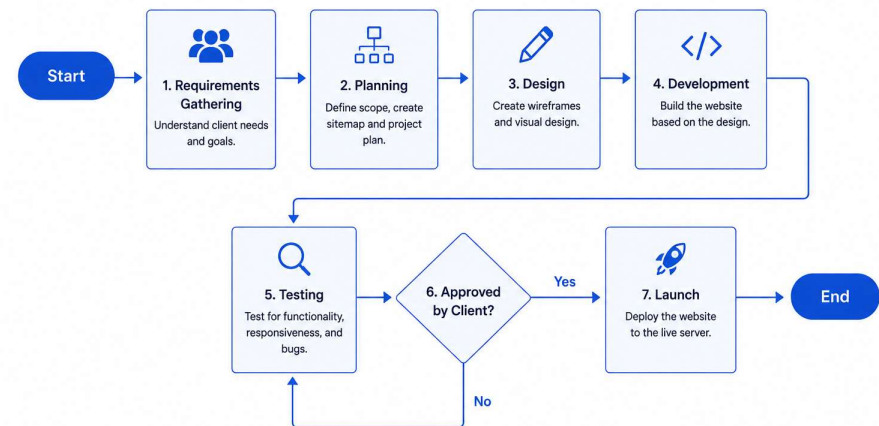
Fishbone Diagram – Website Design Project



Manage Quality Assurance - Tools

- **Data Representation**
- Flowcharts - show the steps of a process and help identify quality issues or process improvement opportunities.

Web Design Project Flowchart



Manage Quality Assurance - Tools

- **Decision-Making, Problem-Solving, Process Improvement**

- Decision-making techniques (multicriteria, voting) help select the best quality improvement option.
- Problem-solving uses methodical, structured techniques to address quality issues.
- Process improvement (Six Sigma, Kaizen) identifies activities that add no value and removes them.

Manage Quality Assurance - Output

- **Quality Reports**

- Documents the results of quality assurance activities, including issues that need to be addressed, recommendations for improvement, and information about quality findings.

- **Change Requests**

- Submitted when quality issues require process or product changes; routed through Integrated Change Control.

- **Project Management Plan Updates**

- The quality, scope, schedule, and cost management plans may be updated based on the results of quality assurance.

- **Project Document Updates**

- Issue log, lessons learned register, and risk register may be updated as a result of quality assurance activities.

Manage Project Knowledge

- Uses existing knowledge and creates new knowledge to help achieve project objectives.
- It also captures new knowledge from the project for future projects and organizational learning.
- Knowledge management includes both:
 - **Explicit knowledge:** documented knowledge such as manuals, procedures, databases, lessons learned, and templates.
 - **Tacit knowledge:** personal experience, insights, skills, and judgment that are harder to document.
 - Tacit knowledge is often shared through conversations, mentoring, collaboration, workshops, and team interactions.
- AI can help capture and organize knowledge, but AI outputs must be validated and sensitive information must be protected.

Manage Project Knowledge

Inputs

- Project management plan
 - All components
- Project documents
 - Lessons learned register
 - Project team assignments
 - Resource breakdown structure
 - Stakeholder register
- Deliverables
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Knowledge management
- Information management
- After-action reviews
- In-progress postmortems
- Storytelling
- Retrospective meetings
- Interpersonal and team skills
 - Active listening
 - Facilitation
 - Leadership
 - Networking
 - Political awareness
- Etc.

Outputs

- Lessons learned register
- Project management plan updates
 - Any component
- Organizational process asset updates
- Etc.

Manage Project Knowledge - Inputs

- **Project Management Plan**
 - All components of the PM Plan are inputs because knowledge must be managed across every aspect of the project.
- **Project Documents**
 - Lessons learned register - captures knowledge gained so it can be applied during the project and shared after closure.
 - Project team assignments - identifies who has what knowledge and skills.
 - Resource breakdown structure - identifies team composition and may help locate experts.
 - Stakeholder register - identifies stakeholders who can provide knowledge or who need to receive it.
- **Deliverables**
 - Each deliverable's creation generates knowledge about how it was produced and what worked.
- **EEFs and OPAs**
 - Organizational culture and structure (impacts knowledge sharing willingness), geographic distribution of facilities, knowledge management systems, and lessons learned databases.

Manage Project Knowledge

- Tools

- **Expert Judgment**
 - Specialists in knowledge management, information management, organizational learning, and knowledge and information management tools.
- **Knowledge Management**
 - Connects people so they can work together to create new knowledge, share tacit knowledge, and integrate knowledge of diverse team members.
 - Tools include networking, communities of practice, meetings, and shadowing.
- **Information Management**
 - Tools and techniques used to create and connect people to explicit knowledge - including methods for codifying explicit knowledge, lessons learned register, and library services.

Manage Project Knowledge

- Tools

- **After-Action Reviews & Postmortems**

- After-action reviews - structured reviews after a phase or milestone to capture what happened, why, and what to improve.
- In-progress postmortems - similar but conducted during the project rather than at the end.

- **Storytelling**

- Sharing experiences, especially failures and successes, in narrative form to transfer tacit knowledge.

- **Retrospective Meetings**

- Regular meetings (often used in adaptive projects) where the team reflects on recent work and identifies improvements.

- **Interpersonal and Team Skills**

- Active listening - genuinely engaging with the speaker to understand their full message.
- Facilitation - guiding group sessions to productive outcomes.
- Leadership - inspiring and motivating others to share knowledge.
- Networking - building relationships across the organization to surface and share knowledge.
- Political awareness - understanding power and influence dynamics that affect knowledge sharing.

Manage Project Knowledge - Output

- **Lessons Learned Register**
 - Created early in the project and updated throughout.
 - At project closure, finalized and added to the organizational lessons learned repository.
- **Project Management Plan Updates**
 - Any component of the PM Plan may be updated as a result of new knowledge gained.
- **Organizational Process Asset Updates**
 - Process documentation, templates, knowledge bases, and lessons learned repositories are updated for use by future projects.

Manage Stakeholder Engagement

- The process of communicating and working with stakeholders to meet their needs and expectations, address issues, and foster appropriate stakeholder engagement
- Allows the PM to increase support and minimize resistance from stakeholders
- Involves activities such as engaging stakeholders at appropriate stages, managing expectations, addressing concerns, and clarifying issues

Manage Stakeholder Engagement

Inputs

- Project management plan
 - Communications management plan
 - Risk management plan
 - Stakeholder engagement plan
 - Change management plan
- Project documents
 - Change log
 - Issue log
 - Lessons learned register
 - Stakeholder register
 - Risk register
 - Status report
 - Project schedule
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Communication skills
 - Feedback
- Interpersonal and team skills
 - Conflict management
 - Cultural awareness
 - Negotiation
 - Observation/conversation
 - Political awareness
- Ground rules
- Meetings
- Etc.

Outputs

- Change requests
- Project management plan updates
 - Communications management plan
 - Stakeholder engagement plan
- Project document updates
 - Change log
 - Issue log
 - Lessons learned register
 - Stakeholder register
- Etc.

Manage Stakeholder Engagement - Inputs

- **Project Management Plan**
 - Communications management plan - dictates how engagement communications are delivered.
 - Risk management plan - guides handling of stakeholder-related risks.
 - Stakeholder engagement plan - identifies engagement strategies and required engagement levels.
 - Change management plan - prescribes how stakeholder concerns leading to changes are processed.
- **Project Documents**
 - Change log - lets stakeholders see what has changed and how their concerns were addressed.
 - Issue log - tracks stakeholder issues and resolutions.
 - Lessons learned register - applies prior engagement lessons.
 - Stakeholder register - the master list of who to engage.
 - Risk register - identifies stakeholder-related risks.
 - Status report - the artifact often delivered during engagement activities.
 - Project schedule - establishes timing for engagement activities.
- **EEFs and OPAs**
 - Organizational culture, political climate, governance, communication policies, change-management procedures, and lessons learned.

Manage Stakeholder Engagement - Tools

- **Expert Judgment**
 - Specialists with knowledge of senior management, the industry, communication and business analysis, and the organization's political dynamics.
- **Communication Skills**
 - Feedback - provides information about reactions to communications, deliverables, and current project status; essential for adjusting engagement.

Manage Stakeholder Engagement - Tools

- **Interpersonal and Team Skills**
 - Conflict management - the PM addresses conflict in a timely and constructive manner to achieve a high-performing team.
 - Cultural awareness - helps the PM and the team to communicate effectively by considering cultural differences.
 - Negotiation - achieving support or agreement that supports the project work or its outcomes.
 - Observation/conversation - keeps the PM in touch with the work and attitudes of the team and stakeholders.
 - Political awareness - understands the strategies of the organization, who wields power and influence, and how to leverage these strategies.
- **Ground Rules**
 - Defined in the team charter; set expected behavior for team members and stakeholders during engagement.
- **Meetings**
 - Used to discuss and address any concerns or issues, report on project performance, and engage stakeholders.

Manage Stakeholder Engagement - Output

- **Change Requests**
 - May result from engagement activities (e.g., a stakeholder raises a concern that requires a change to scope, schedule, or another aspect).
- **Project Management Plan Updates**
 - Communications management plan - revised to fit lessons learned during engagement.
 - Stakeholder engagement plan - updated as engagement experience reveals what works and what does not.
- **Project Document Updates**
 - Change log - documents new change requests resulting from engagement.
 - Issue log - new issues raised by stakeholders are added; resolved issues are closed.
 - Lessons learned register - what worked and what did not in engaging stakeholders.
 - Stakeholder register - updated with new engagement levels or new stakeholder information.

Manage Communications

- The process of ensuring timely and appropriate collection, creation, distribution, storage, retrieval, management, monitoring, and ultimate disposition of project information
- Enables an efficient and effective information flow between the project team and stakeholders

Manage Communications

Inputs

- Project management plan
 - Resource management plan
 - Communications management plan
 - Stakeholder engagement plan
- Project documents
 - Change log
 - Issue log
 - Lessons learned register
 - Quality report
 - Risk report
 - Stakeholder register
- Work performance reports
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Communication technology
- Communication methods
- Communication skills
 - Communication competence
 - Feedback
 - Nonverbal communication
 - Presentations
- Project management information system
- Project reporting
- Interpersonal and team skills
 - Active listening
 - Conflict management
 - Cultural awareness
 - Meeting management
 - Networking
 - Political awareness
- Meetings
- Etc.

Outputs

- Project communications
- Project management plan updates
 - Communications management plan
 - Stakeholder engagement plan
- Project document updates
 - Issue log
 - Lessons learned register
 - Project schedule
 - Risk register
 - Stakeholder register
- Organizational process asset updates
- Etc.

Manage Communications - Inputs

- **Project Management Plan**
 - Resource management plan - identifies the team members responsible for producing and distributing communications.
 - Communications management plan - the playbook driving what is communicated, how, and when.
 - Stakeholder engagement plan - identifies who to communicate with and at what level.
- **Project Documents**
 - Change log - communicated to relevant stakeholders.
 - Issue log - issues and their statuses are communicated.
 - Lessons learned register - useful for shaping future communications.
 - Quality report - quality status communicated to stakeholders.
 - Risk report - overall risk exposure and individual risks communicated.
 - Stakeholder register - identifies recipients of each communication.
- **Work Performance Reports**
 - Performance information formatted for distribution to stakeholders (status reports, dashboards, memos).
- **EEFs and OPAs**
 - Organizational culture, regulatory requirements, communication channels, communication policies, templates, and historical communication artifacts.

Manage Communications - Tools

- **Communication Technology**
 - Email, instant messaging, video conferencing, intranets, portals - the tools that physically transmit communications.
- **Communication Methods**
 - Interactive (face-to-face, calls, video), push (email, reports), pull (portals, repositories).
- **Communication Skills**
 - Communication competence - tailored communications that consider stakeholder attitudes, knowledge level, and expertise.
 - Feedback - confirms the message was received and understood; enables two-way communication.
 - Nonverbal communication - tone, body language, and appearance influence how messages are received.
 - Presentations - formal delivery of information to stakeholders, often with visual aids.

Manage Communications

- Tools

- **PMIS & Project Reporting**

- PMIS provides standard tools for distributing reports, dashboards, and notifications to all stakeholders.
- Project reporting - the act of collecting and distributing project information.

- **Interpersonal and Team Skills**

- Active listening - acknowledges and confirms understanding.
- Conflict management - resolves communication conflicts.
- Cultural awareness - tailors messages to cultural norms.
- Meeting management - keeps meetings productive.
- Networking - builds informal communication channels.
- Political awareness - navigates organizational dynamics.

- **Meetings**

- Daily, weekly, milestone, and ad-hoc meetings to deliver and discuss project information.

Manage Communications - Output

- **Project Communications**
 - Performance reports, deliverable status, schedule progress, and cost incurred - the actual artifacts delivered to stakeholders.
 - Format and frequency depends on the urgency and complexity of the message and the audience.
- **Project Management Plan Updates**
 - Communications management plan - updated when current approach is shown to be ineffective.
 - Stakeholder engagement plan - updated as engagement evolves through communication.
- **Project Document Updates**
 - Issue log, lessons learned register, project schedule, risk register, and stakeholder register may all be updated based on communication activities.
- **Organizational Process Asset Updates**
 - Communication artifacts (presentations, reports, templates) may be added to the OPA repository for use by future projects.

Acquire Resources

- The process of obtaining team members, facilities, equipment, materials, supplies, and other resources necessary to complete project work
- Outlines and guides the selection of resources and assigns them to their respective activities

Acquire Resources

Inputs

- Project management plan
 - Resource management plan
 - Procurement management plan
 - Cost baseline
- Project documents
 - Project schedule
 - Resource calendars
 - Resource requirements
 - Stakeholder register
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Decision-making
 - Multicriteria decision analysis
- Interpersonal and team skills
 - Negotiation
 - Problem-solving
- Preassignment
- Virtual teams
- Etc.

Outputs

- Physical or virtual resource assignments
- Project team assignments
- Resource calendars
- Change requests
- Project management plan updates
 - Resource management plan
 - Cost baseline
- Project document updates
 - Lessons learned register
 - Project schedule
 - Resource breakdown structure
 - Resource requirements
 - Risk register
 - Stakeholder register
- Enterprise environmental factor updates
- Organizational process asset updates
- Etc.

Acquire Resources - Inputs

- **Project Management Plan**

- Resource management plan - identifies how resources are to be acquired, the rules of preassignment, and the training plan.
- Procurement management plan - guides external resource acquisition through procurement.
- Cost baseline - provides the budget envelope for resource acquisition.

- **Project Documents**

- Project schedule - identifies when each resource is needed; drives acquisition timing.
- Resource calendars - existing or assumed availability windows.
- Resource requirements - what skills, materials, and equipment must be acquired.
- Stakeholder register - identifies stakeholders who can influence or provide resources (functional managers, vendors).

- **EEFs and OPAs**

- Existing information on organizational resources, personnel administration policies, marketplace conditions, organizational structure, geographic locations, and templates for staff hiring/contracting.

Acquire Resources - Tools

- **Decision-Making**
 - Multicriteria decision analysis - selects among candidate resources by weighting criteria such as availability, cost, experience, ability, knowledge, skills, attitude, and international factors.
- **Interpersonal and Team Skills**
 - Negotiation - used with functional managers (for internal resources), vendors (for external), and other PMs competing for the same resources.
 - Problem-solving - addresses resource conflicts and shortages as they arise.
- **Preassignment**
 - When physical or team resources for a project are determined in advance (e.g., named in the charter, promised in a proposal, or required by contract).
- **Virtual Teams**
 - Groups of people with a shared goal who fulfill their roles with little or no time spent meeting face-to-face.
 - Allows acquisition of resources from anywhere geographically; requires good communication tools.

Acquire Resources - Output

- **Physical or Virtual Resource Assignments**
 - Documents the actual physical resources (materials, equipment, facilities) and virtual resources assigned to the project.
- **Project Team Assignments**
 - Documents the team members and their roles and responsibilities for the project.
- **Resource Calendars**
 - Identifies the working days, shifts, and start/end of normal business hours when each specific resource is available.

Acquire Resources - Output

- **Change Requests**
 - May result when resource acquisition leads to changes to the project plan (e.g., a different skill mix requires schedule or cost adjustments).
- **Project Management Plan Updates**
 - Resource management plan - reflects actual experience.
 - Cost baseline - may change due to actual resource costs.
- **Project Document Updates**
 - Lessons learned register, project schedule, resource breakdown structure, resource requirements, risk register, and stakeholder register may all be updated.
- **EEF & OPA Updates**
 - Resource availability information updated in the organizational systems for use by future projects.

Lead the Team

- The process of tracking team member performance, providing feedback, resolving issues, and managing team changes to optimize project performance
- Encompasses developing the team and managing team conflicts to improve interactions and overall team performance
- Provides leadership, motivation, and support that increases team performance
- Influences team behavior, manages conflict, and resolves issues

Lead the Team

Inputs

- Project management plan
 - Resource management plan
- Project documents
 - Issue log
 - Lessons learned register
 - Project schedule
 - Resource calendars
 - Team charter
 - Work performance reports
 - Project team assignments
- Team performance assessments
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Colocation
- Virtual teams
- Communication technology
- Interpersonal and team skills
 - Conflict management
 - Influencing
 - Motivation
 - Negotiation
 - Team building
 - Decision-making
 - Critical thinking
 - Coaching and mentoring
 - Training
- Problem-solving
 - Six Thinking Hats®
- Retrospectives
- Recognition and rewards
- Individual and team assessments
- Data analysis
 - SWOT analysis
- Meetings
- Emotional intelligence
- Organizational cultural intelligence
- Leadership
 - Distributed management and leadership
 - Centralized management and leadership
 - Servant leadership
- Tuckman ladder
- Project management information system
- Virtual collaboration tools
- Etc.

Outputs

- Team performance assessments
- Change requests
- Project management plan updates
 - Resource management plan
 - Schedule baseline
 - Cost baseline
- Project document updates
 - Issue log
 - Lessons learned register
 - Project schedule
 - Resource calendars
 - Team charter
 - Project team assignments
- Enterprise environmental factor updates
- Organizational process asset updates
- Etc.

Lead the Team - Inputs

- **Project Management Plan**
 - Resource management plan - guidance on how to manage and release team members.
- **Project Documents**
 - Issue log - documents issues that the leader must address.
 - Lessons learned register - prior leadership lessons applied.
 - Project schedule - drives when team activities occur and when team development opportunities exist.
 - Resource calendars - identify when team members are available.
 - Team charter - documents team values and operating guidelines.
 - Work performance reports - communicate status to the team and identify areas needing leadership intervention.
 - Project team assignments - identify team composition.
- **Team Performance Assessments**
 - Formal or informal evaluations of team effectiveness used as the basis for actions to improve performance.
- **EEFs and OPAs**
 - HR management policies, performance management systems, rewards programs, employee development, training records, and security clearance procedures.

Lead the Team - Tools

- **Colocation & Virtual Teams**

- Colocation - placing many or all of the team in the same physical location to enhance ability to perform.
- Virtual teams - distributed teams enabled by communication technology; expand the talent pool but require deliberate engagement.

- **Communication Technology & Virtual Collaboration**

- Email, chat, video conferencing, shared portals, and collaboration platforms keep the team aligned across geography and time zones.

Lead the Team - Tools

- **Interpersonal and Team Skills**

- Influencing - shaping team behavior without formal authority through persuasion, building trust, and demonstrating expertise.
- Motivation - providing reasons for team members to act; intrinsic (autonomy, mastery, purpose) and extrinsic (rewards, recognition).
- Negotiation - resolving competing demands and agreements between team members.
- Team building - activities that enhance the team's social relations and build a collaborative working environment.
- Decision-making - selecting between alternatives.
- Critical thinking - evaluating evidence and reasoning.
- Coaching and mentoring - developing team members' skills and confidence.
- Training - formal skill development for team members.

Lead the Team - Tools

- **Problem-Solving**

- Six Thinking Hats: This is a problem-solving and decision-making technique that encourages teams to look at a situation from multiple perspectives.
- Each “hat” represents a different way of thinking, helping teams avoid bias and consider all angles before making decisions.
- The blue hat manages the process and focuses on the big picture
- The white hat looks at facts and data
- The red hat expresses emotions and intuition
- The black hat identifies risks and potential problems
- The yellow hat focuses on benefits and positive outcomes
- The green hat encourages creativity and new ideas

- **Retrospectives**

- Periodic team reflections to identify what is working and what should change.

Lead the Team - Tools

Emotional Intelligence and Cultural Intelligence

- Emotional intelligence is the ability to understand and manage your own emotions and the emotions of others.
- Project managers need emotional intelligence to stay calm, lead effectively, manage conflict, and build trust.
- Daniel Goleman's five elements of emotional intelligence are:
 - Self-awareness: understanding your emotions and how they affect others.
 - Self-regulation: controlling reactions and staying calm under pressure.
 - Motivation: having the drive, persistence, and optimism to achieve goals.
 - Empathy: understanding other people's feelings and perspectives.
 - Social skills: communicating, influencing, and building strong relationships.
- Organizational cultural intelligence is the ability to work effectively with people from different cultures and backgrounds.
- It means recognizing communication and work-style differences without making assumptions or stereotypes.

Lead the Team - Tools

- **Leadership Approaches**
 - **Distributed Management and Leadership**
 - Decision-making and authority are shared across the team.
 - Team members take ownership of work in their areas.
 - Useful for collaborative, Agile, and self-organizing teams.
 - Example: designers, developers, and testers make decisions within their expertise.
 - **Centralized Management and Leadership**
 - Decision-making authority stays with one person or a small leadership group.
 - Work is directed from the top down.
 - Useful when control, consistency, or fast direction is needed.
 - Example: the project manager assigns tasks and makes key decisions.
 - **Servant Leadership**
 - The leader supports the team instead of controlling the team.
 - Focuses on removing blockers, providing resources, and helping people succeed.
 - Useful for empowering teams and improving collaboration.
 - Example: the project manager helps the website team solve issues so they can deliver quality work.

Lead the Team - Tools

Individual and Team Assessments

- Individual and team assessments help the project manager understand how the team is performing.
- The project manager assesses both:
 - Individual team members: strengths, weaknesses, skills, behavior, and development needs.
 - The team as a whole: collaboration, trust, communication, conflict resolution, and decision-making.
- These assessments help identify performance gaps and improvement opportunities.
- They can also help the project manager assign work, provide coaching, build trust, and improve teamwork.
- **Data Analysis & Meetings**
 - SWOT analysis applied to the team identifies strengths to build on and weaknesses to address.
 - Meetings - regular team meetings, one-on-ones, and all-hands
- **PMIS & Virtual Tools**
 - Resource management, scheduling, performance reporting tools, and virtual collaboration platforms.

Lead the Team - Tools

- Interpersonal and Team Skills
- Conflict management:

CONFLICT RESOLUTION TECHNIQUES

TECHNIQUE	MEANING	TYPICAL OUTCOME
1. COLLABORATE / PROBLEM-SOLVE	Work together to understand the issue and find the best solution for everyone.	WIN-WIN Best approach.
2. COMPROMISE / RECONCILE	Each side gives up something to reach a middle-ground solution.	LOSE-LOSE Nobody gets exactly what they want.
3. SMOOTH / ACCOMMODATE	Keep the peace by giving in and focusing on agreement.	YOU LOSE Harmony matters more than your position.
4. FORCE / DIRECT	Use authority to push your solution through.	WIN-LOSE My way or the highway.
5. WITHDRAW / AVOID	Step away, delay the issue, or avoid the conflict.	NO REAL RESULT The problem usually stays there.

Lead the Team - Tools

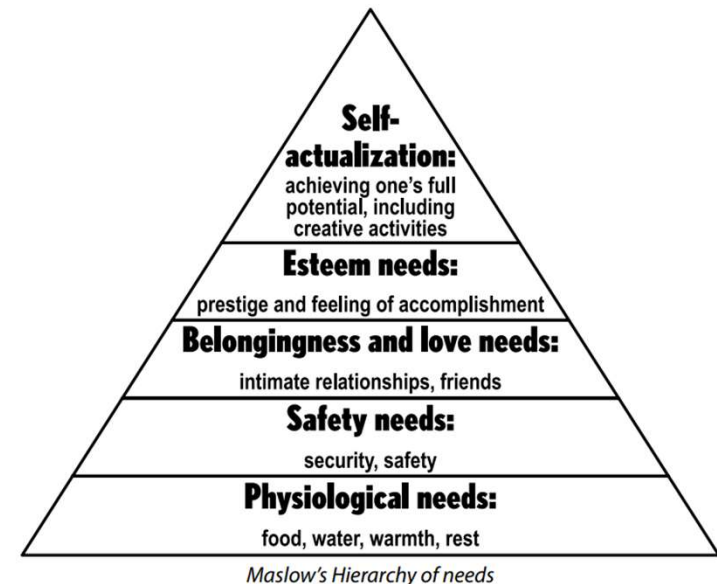
- **Conflict management steps:**
- Identify the root cause of the problem (not just the symptoms).
- Analyze the problem (cause-and-effect diagram).
- Identify solutions.
- Implement the selected solution.
- Review the solution.
- Confirm that the solution solved the problem.

Lead the Team - Tools

- **Recognition and Rewards**

- **Maslow's Hierarchy of Needs**

- Maslow's theory explains that people are motivated by different levels of needs.
 - Lower-level needs must usually be satisfied before higher-level needs become strong motivators.
 - The needs move from basic physical needs to higher psychological and personal growth needs.



Lead the Team - Tools

- **Recognition and Rewards**
- Herzberg's Hygiene and Motivational Factors
 - Hygiene factors prevent dissatisfaction, such as salary, policies, and work environment.
 - Motivational factors create satisfaction, such as achievement, growth, recognition, and advancement.
- Intrinsic vs. Extrinsic Motivation
 - Extrinsic motivation comes from outside rewards, such as pay or bonuses.
 - Intrinsic motivation comes from inside the person, such as autonomy, mastery, and purpose.
- McClelland's Theory of Needs
 - Achievement: motivated by challenging goals and accomplishment.
 - Power: motivated by leadership, influence, and responsibility.
 - Affiliation: motivated by belonging, teamwork, and relationships.
- Theory X, Theory Y, and Theory Z
 - Theory X: assumes people dislike work and need close supervision.
 - Theory Y: assumes people are self-motivated and want to do good work.
 - Theory Z: focuses on meaning, values, loyalty, well-being, and long-term commitment.

Lead the Team - Tools

(Myers-Briggs Type Indicator)

MBTI helps project managers understand different personality types and work preferences.

- It can improve communication, teamwork, conflict management, and leadership.
- MBTI groups people into 16 personality types based on four areas:
 - **Extroversion vs. Introversion:** how people focus their energy.
 - **Sensing vs. Intuition:** how people take in information.
 - **Thinking vs. Feeling:** how people make decisions.
 - **Judging vs. Perceiving:** how people approach structure and flexibility.
- In the Lead the Team process, MBTI can help the project manager adapt their leadership style.
- Diagram:
 - https://en.wikipedia.org/wiki/Myers%E2%80%93Briggs_Type_Indicator

What's Your Personality Type?

Use the questions on the outside of the chart to determine the four letters of your Myers-Briggs type.
For each pair of letters, choose the side that seems most natural to you, even if you don't agree with every description.

1. Are you outwardly or inwardly focused? If you:

- Could be described as talkative, outgoing
- Like to be in a fast-paced environment
- Tend to work out ideas with others, think out loud
- Enjoy being the center of attention

then you prefer

E

Extraversion

- Could be described as reserved, private
- Prefer a slower pace with time for contemplation
- Tend to think things through inside your head
- Would rather observe than be the centre of attention

then you prefer

I

Introversion

2. How do you prefer to take in information? If you:

- Focus on the reality of how things are
- Pay attention to concrete facts and details
- Prefer ideas that have practical applications
- Like to describe things in a specific, literal way

then you prefer

S

Sensing

- Imagine the possibilities of how things could be
- Notice the big picture, see how everything connects
- Enjoy ideas and concepts for their own sake
- Like to describe things in a figurative, poetic way

then you prefer

N

Intuition

ISTJ
Responsible, sincere, analytical, reserved, realistic, systematic. Hardworking and trustworthy with sound practical judgment.

ISTP
Action-oriented, logical, analytical, spontaneous, reserved, independent. Enjoy adventure, skilled at understanding how mechanical things work.

ESTP
Outgoing, realistic, action-oriented, curious, versatile, spontaneous. Pragmatic problem solvers and skillful negotiators.

ESTJ
Efficient, outgoing, analytical, systematic, dependable, realistic. Like to run the show and get things done in an orderly fashion.

ISFJ
Warm, considerate, gentle, responsible, pragmatic, thorough. Devoted caretakers who enjoy being helpful to others.

ISFP
Gentle, sensitive, nurturing, helpful, flexible, realistic. Seek to create a personal environment that is both beautiful and practical.

ESFP
Playful, enthusiastic, friendly, spontaneous, tactful, flexible. Have strong common sense, enjoy helping people in tangible ways.

ESFJ
Friendly, outgoing, reliable, conscientious, organised, practical. Seek to be helpful and please others, enjoy being active and productive.

INFJ
Idealistic, organised, insightful, dependable, compassionate, gentle. Seek harmony and cooperation, enjoy intellectual stimulation.

INFP
Sensitive, creative, idealistic, perceptive, caring, loyal. Value inner harmony and personal growth, focus on dreams and possibilities.

ENFP
Enthusiastic, creative, spontaneous, optimistic, supportive, playful. Value inspiration, enjoy starting new projects, see potential in others.

ENFJ
Caring, enthusiastic, idealistic, organised, diplomatic, responsible. Skilled communicators who value connection with people.

INTJ
Innovative, independent, strategic, logical, reserved, insightful. Driven by their own original ideas to achieve improvements.

INTP
Intellectual, logical, precise, reserved, flexible, imaginative. Original thinkers who enjoy speculation and creative problem solving.

ENTP
Inventive, enthusiastic, strategic, enterprising, inquisitive, versatile. Enjoy new ideas and challenges, value inspiration.

ENTJ
Strategic, logical, efficient, outgoing, ambitious, independent. Effective organisers of people and long-range planners.

3. How do you prefer to make decisions? If you:

- Make decisions in an impersonal way, using logical reasoning
- Value justice, fairness
- Enjoy finding the flaws in an argument
- Could be described as reasonable, level-headed

then you prefer

T

Thinking

- Base your decisions on personal values and how your actions affect others
- Value harmony, forgiveness
- Like to please others and point out the best in people
- Could be described as warm, empathetic

then you prefer

F

Feeling

4. How do you prefer to live your outer life? If you:

- Prefer to have matters settled
- Think rules and deadlines should be respected
- Prefer to have detailed, step-by-step instructions
- Make plans, want to know what you're getting into

then you prefer

J

Judging

- Prefer to leave your options open
- See rules and deadlines as flexible
- Like to improvise and make things up as you go
- Are spontaneous, enjoy surprises and new situations

then you prefer

P

Perceiving

• Diagram

- https://en.wikipedia.org/wiki/Myers%E2%80%93Briggs_Type_Indicator



Lead the Team - Tools

TUCKMAN'S LADDER

How teams develop over time

Stage	Description
1. Forming	The team comes together and gets to know each other. Roles are still unclear and uncertainty is high.
2. Storming	Conflicts begin as team members share ideas, challenge each other, and disagree on how work should be done.
3. Norming	The team begins to align, agree on working methods, and build trust.
4. Performing	The team works well together, solves problems, and performs effectively with less supervision.
5. Adjourning	The work is completed, and the team is released or reassigned.

Note: Teams may move through these stages again when new members join or the team changes.

Lead the Team - Tools

Common Schedule Behaviors

- **Student Syndrome:** delaying work until the last possible moment.
 - Example: A developer has 5 days to finish a task but waits until day 4 to start.
- **Parkinson's Law:** work expands to fill the time available.
 - Example: A task that could take 2 days stretches to 5 days because 5 days were scheduled.
- **Self-Protection:** hiding delays or problems to avoid blame.
 - Example: A team member knows a website feature is behind but does not report it until the last minute.
- **Sandbagging:** intentionally overestimating time or effort to look better later.
 - Example: A developer says a task will take 10 days even though it can be done in 5.
- **Dropped Baton:** work is handed off poorly and information is missed.
 - Example: A designer does not clearly explain requirements to the developer, causing rework.

Lead the Team - Output

- **Team Performance Assessments**
 - The PM evaluates and addresses team performance regularly.
 - Indicators include improvements in skills, improvements in competencies, reduced staff turnover rate, and increased team cohesiveness.
- **Change Requests**
 - Resource changes (additions, removals) and other leadership-driven adjustments processed through Integrated Change Control.
- **Project Management Plan Updates**
 - Resource management plan, schedule baseline, and cost baseline may be updated.
- **Project Document Updates**
 - Issue log, lessons learned register, project schedule, resource calendars, team charter, and project team assignments may all be updated.
- **EEF & OPA Updates**
 - Input to organizational performance appraisals, personnel skill updates, lessons-learned templates, and training needs.

Implement Risk Responses

- The process of implementing agreed-upon risk response plans
- Ensures that planned risk responses are executed as planned to address overall project risk exposure, minimize threats, and maximize opportunities
- Without implementation, risk responses are merely good intentions on paper - this is where the actual risk-reduction work happens

Implement Risk Responses

Inputs

- Project management plan
 - Risk management plan
- Project documents
 - Lessons learned register
 - Risk register
 - Risk report
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Interpersonal and team skills
 - Influencing
- Project management information system
- Etc.

Outputs

- Change requests
- Project document updates
 - Issue log
 - Lessons learned register
 - Project team assignments
 - Risk register
 - Risk report
- Etc.

Implement Risk Responses - Inputs

- **Project Management Plan**
 - Risk management plan - documents the roles, responsibilities, and authority of each team member when implementing risk responses.
- **Project Documents**
 - Lessons learned register - prior implementation lessons applied (what worked, what failed during execution).
 - Risk register - documents the agreed response strategies for each risk along with the assigned owner and any specified actions to take.
 - Risk report - documents the current overall project risk exposure and the agreed response strategy.
- **Organizational Process Assets**
 - Lessons learned repository from similar completed projects with details of risk implementation.

Implement Risk Responses - Tools

- **Expert Judgment**
 - Specialists with expertise in validating or modifying risk responses if needed and deciding how to implement them most efficiently and effectively.
- **Interpersonal and Team Skills**
 - Influencing - the PM may need to influence the risk owner (or others involved) to take the necessary action on time. Especially important when the PM lacks formal authority over the owner.
- **Project Management Information System**
 - PMIS includes scheduling, resource, and cost software to ensure agreed risk response plans and their associated activities are integrated into the project alongside other project activities.

Implement Risk Responses - Output

- **Change Requests**
 - Implementing risk responses may require changes to the cost or schedule baselines, or other components of the PM Plan; processed through Integrated Change Control.
- **Project Document Updates**
 - Issue log - new issues raised during implementation (e.g., a planned response failed) are recorded.
 - Lessons learned register - updated with effective and ineffective responses to inform future projects.
 - Project team assignments - team member assignments may be updated as risk responses are executed.
 - Risk register - updated with the results of executing the responses (including any secondary or residual risks).
 - Risk report - updated to reflect the current status of significant agreed-upon risk responses and current overall project risk exposure.

Monitoring and Controlling

- 10 Processes
- Ensures the work follows the project management plan
- Reports on the progress of the project

Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Governance	Initiate Project or Phase	- Integrate and Align Project Plans - Plan Sourcing Strategy	- Manage Project Execution - Manage Quality Assurance - Manage Project Knowledge	- Monitor and Control Project Performance - Assess and Implement Changes	Close Project or Phase
Scope		- Plan Scope Management - Elicit and Analyze Requirements - Define Scope - Develop Scope Structure		- Monitor and Control Scope - Validate Scope	
Schedule		- Plan Schedule Management - Develop Schedule		Monitor and Control Schedule	
Finance		- Plan Financial Management - Estimate Costs - Develop Budget		Monitor and Control Finances	
Stakeholders	Identify Stakeholders	- Plan Stakeholder Engagement - Plan Communications Management	- Manage Stakeholder Engagement - Manage Communications	- Monitor Stakeholder Engagement - Monitor Communications	
Resources		- Plan Resource Management - Estimate Resources	- Acquire Resources - Lead the Team	Monitor and Control Resourcing	
Risk		- Plan Risk Management - Identify Risks - Perform Risk Analysis - Plan Risk Responses	Implement Risk Responses	Monitor Risks	

Monitor and Control Project Performance

- Monitoring is ongoing throughout the project and focuses on collecting, measuring, and assessing project data.
- Controlling means taking corrective or preventive action, replanning when needed, and following up to resolve issues.
- This process compares actual performance against the project plan.
- It tracks work completed, resources used, budget spent, risks, changes, and deliverable status.
- It provides stakeholders with accurate information about project health and progress.
- It helps determine whether deliverables are on track to meet planned benefits and acceptance criteria.

Monitor and Control Project Performance

- Measurement should focus only on useful, relevant, and actionable metrics.
- Metrics should be SMART: specific, measurable, achievable, relevant, and time-bound.
- Common measurement pitfalls include:
 - **Hawthorne effect:** people change behavior because they know they are being measured.
 - **Vanity metrics:** data that looks good but does not help decision-making.
 - **Demoralization:** unrealistic goals can hurt morale.
 - **Misusing metrics:** focusing on the wrong measures or short-term numbers.
 - **Confirmation bias:** interpreting data to support what you already believe.
 - **Correlation vs. causation:** assuming one thing caused another without proof.

Monitor and Control Project Performance

Inputs

- Project management plan
 - Any component
- Project documents
 - Assumption log
 - Basis of estimates
 - Cost forecasts
 - Issue log
 - Lessons learned register
 - Milestone list
 - Quality reports
 - Risk register
 - Risk report
 - Schedule forecasts
- Work performance information
- Agreements
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data analysis
 - Alternative analysis
 - Cost-benefit analysis
 - Earned value analysis
 - Root cause analysis
 - Trend analysis
 - Variance analysis
- Decision-making
 - Voting
- Meetings
- Project dashboards
- Visual controls
- Information radiators
- Etc.

Outputs

- Work performance reports
- Change requests
- Project management plan updates
 - Any component
- Project document updates
 - Cost forecasts
 - Issue log
 - Lessons learned register
 - Risk register
 - Schedule forecasts
- Etc.

Monitor and Control Project Performance - Inputs

- **Project Management Plan**
 - Any component of the PM Plan, especially the baselines (scope, schedule, cost, performance measurement) used as the comparison reference.
- **Project Documents**
 - Assumption log - assumptions and constraints to be monitored.
 - Basis of estimates - reference for evaluating cost and schedule variances.
 - Cost forecasts - projections of future costs based on current performance.
 - Issue log - tracks open project issues that may affect performance.
 - Lessons learned register - captures and applies learning during monitoring.
 - Milestone list - milestones being tracked for schedule status.

Monitor and Control Project Performance - Inputs

- **Project Documents (cont.)**
 - Quality reports - quality status that may affect overall performance.
 - Risk register and risk report - identified risks and overall risk exposure.
 - Schedule forecasts - projected future schedule performance.
- **Work Performance Information**
 - Performance data analyzed and integrated based on comparisons across knowledge areas (e.g., status of deliverables, status of change requests, forecasted estimates to complete).
- **Agreements**
 - Contracts that govern work and contain performance metrics that must be monitored.
- **EEFs and OPAs**
 - Project management information systems, stakeholder expectations and risk thresholds, monitoring policies and templates.

Monitor and Control Project Performance - Tools

- **Expert Judgment**
 - Specialists in interpretation of information, techniques to identify project performance trends, and statistical analysis.
- **Data Analysis**
 - Alternative analysis - selects corrective or preventive actions when variances occur.
 - Cost-benefit analysis - determines best corrective action in terms of cost and benefit.
 - Earned value analysis - integrates scope, schedule, and cost measures (PV, EV, AC) to assess performance and progress.
 - Root cause analysis - identifies the main reason for variances.
 - Trend analysis - examines past results to predict future performance.
 - Variance analysis - reviews differences between planned and actual performance.

Monitor and Control Project Performance - Tools

- **Decision-Making**
 - Voting and other techniques used to converge on corrective or preventive actions.
- **Meetings, Dashboards, Visual Controls**
 - Status meetings, project review meetings, and inspection meetings used to communicate performance.
 - Project dashboards consolidate KPIs in real time.
 - Visual controls and information radiators (kanban boards, burn-down charts) make project status visible to all stakeholders.

Monitor and Control Project Performance - Output

- **Work Performance Reports**
 - Physical or electronic representation of work performance information intended to generate decisions, actions, or awareness.
 - Examples: status reports, memos, justifications, information notes, electronic dashboards, recommendations, and updates.
- **Change Requests**
 - Generated when actual performance deviates from the plan beyond thresholds.
 - Includes corrective actions, preventive actions, defect repairs, and updates to plan/baselines.
- **Project Management Plan Updates**
 - Any component of the PM Plan may be updated based on performance findings (often after change-request approval).
- **Project Document Updates**
 - Cost forecasts, issue log, lessons learned register, risk register, and schedule forecasts are updated to reflect current performance and projections.

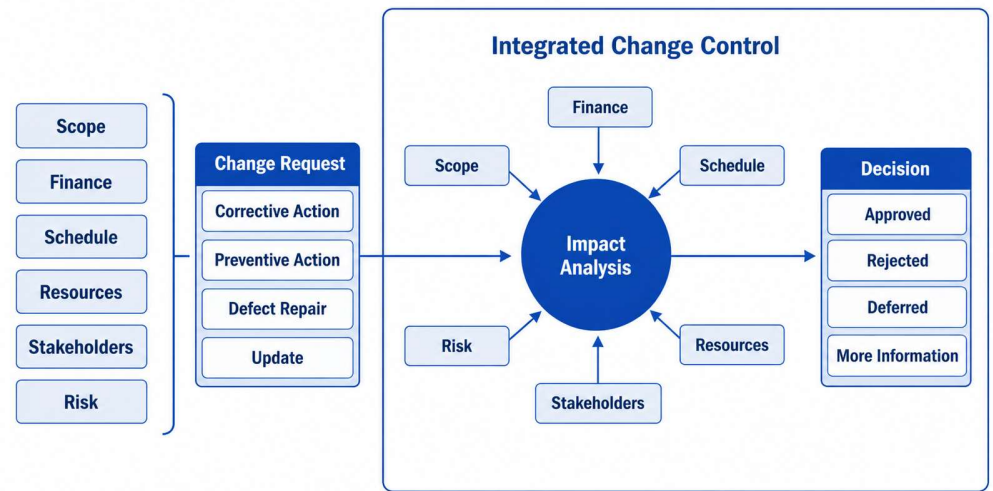
Assess and Implement Changes

- The process of reviewing all change requests, approving and managing changes to deliverables, project documents, and the project management plan, and communicating the decisions
- Also known as Perform Integrated change control
- Reviews requests on the impact on scope, schedule, cost, quality, resources, and risks before approval
- Approved changes are implemented through the executing processes; rejected changes are recorded in the change log

Assess and Implement Changes

- **Predictive Approach**
- In predictive projects, changes are formally controlled after baselines are established.
- Once a baseline exists, changes to scope, schedule, cost, resources, risk, or other controlled artifacts require a formal change request.
- Change requests should be documented in writing and entered into the change control system.
- Each change request should include impact analysis, such as cost, schedule, scope, resource, stakeholder, and risk impacts.
- A change request may involve:
 - Corrective action
 - Preventive action
 - Defect repair
 - Updates
- The change may be approved, rejected, deferred, or sent back for more information.
- A change control board may review and decide on changes when required.
- While waiting for a decision, the project team should continue approved project work.
- Approved changes may require updates to the project management plan, project documents, cost estimates, schedule, resources, and risk responses.

Assess and Implement Changes



Assess and Implement Changes

- **Adaptive Approach**
 - In adaptive projects, changes are usually managed through the **product backlog**, not a formal change request process.
 - Stakeholder change ideas are captured as backlog items.
 - Changes are continuously reviewed, assessed, and prioritized during the project.
 - Impact analysis is still performed to understand effects on value, scope, time, cost, risk, and priorities.
 - There is usually no formal approval process like in predictive projects.
 - If the change has value, it can be added to the backlog and prioritized.
 - If the change is low priority, it is effectively deferred.
 - If the change is not added or is removed from the backlog, it is effectively rejected.

Assess and Implement Changes

Inputs

- Project management plan
 - Change management plan
 - Configuration management plan
 - Scope baseline
 - Schedule baseline
 - Cost baseline
- Project documents
 - Basis of estimates
 - Change log
 - Requirements traceability matrix
 - Risk report
- Work performance reports
- Change requests
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Change control tools
- Data analysis
 - Alternative analysis
 - Cost-benefit analysis
- Decision-making
 - Voting
 - Autocratic decision-making
 - Multicriteria decision analysis
- Meetings
- Integrated change control
- Backlog management
- Etc.

Outputs

- Approved change requests
- Project management plan updates
 - Any component
- Project document updates
 - Change log
- Etc.

Assess and Implement Changes

- Inputs

- **Project Management Plan**
 - Change management plan - defines how changes will be managed and controlled.
 - Configuration management plan - defines those items that are configurable, those that require formal change control, and the process for controlling changes.
 - Scope, schedule, and cost baselines - the references against which changes are evaluated.
- **Project Documents**
 - Basis of estimates - shows how the cost, duration, and resource estimates were derived; used to evaluate the impact of changes.
 - Change log - records all change requests and their current dispositions.
 - Requirements traceability matrix - tracks impact of changes on requirements.
 - Risk report - identifies overall project risk and impacts of proposed changes on risk exposure.
- **Work Performance Reports**
 - Used by the CCB to assess change requests in the context of current project performance.
- **Change Requests**
 - Formal proposals to modify any document, deliverable, or baseline - the main work item for this process.
- **EEFs and OPAs**
 - Government and industry regulations, change control procedures, configuration management knowledge base, and templates.



Assess and Implement Changes - Tools

- **Expert Judgment**

- Specialists in technical knowledge of the industry, legal/regulatory matters, change control, configuration management, risk, and quality assess change requests.

- **Change Control Tools**

- Change control tools can be manual or automated and should be selected based on stakeholder needs, organizational rules, and project constraints.
- Configuration control focuses on identifying, tracking, verifying, and auditing project deliverables and processes.
- Change control focuses on documenting, reviewing, approving, rejecting, or deferring changes to project documents, deliverables, or baselines.
- These tools help ensure changes are registered, assessed, communicated, and controlled so the project maintains accountability and alignment.

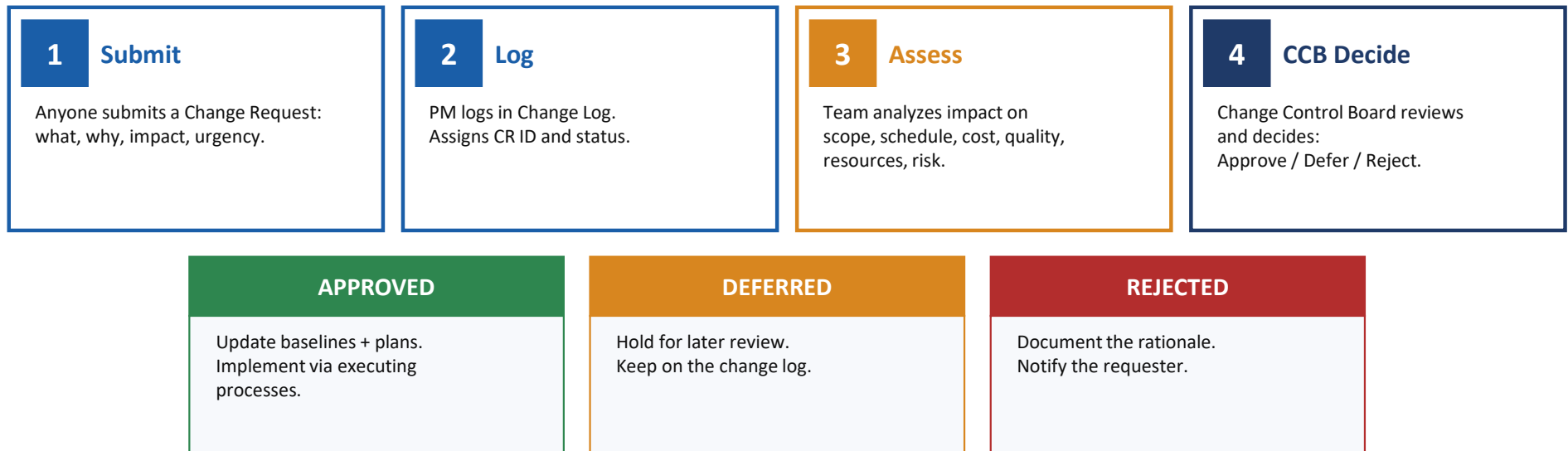
- **Data Analysis**

- Alternative analysis - evaluates options for implementing the change.
- Cost-benefit analysis - determines if the change is worthwhile based on cost vs benefit.

Assess and Implement Changes - Tools

- **Decision-Making**
 - Voting - reaching a decision through unanimity, majority, or plurality.
 - Autocratic decision-making - one person decides for the group (used when speed is critical).
 - Multicriteria decision analysis - evaluates change requests against multiple weighted criteria.
- **Meetings,**
 - Change control meetings - convened by the CCB to review change requests and decide on approval, deferral, or rejection.
- **Backlog management**
 - in adaptive projects, ongoing reprioritization and refinement of the backlog handles many change requests.
- **Integrated change control**
 - Integrated change control ensures project changes are managed in a coordinated and controlled way.
 - Change requests are reviewed and analyzed before decisions are made.
 - Impact analysis helps evaluate effects on scope, schedule, cost, resources, risk, and stakeholders.
 - Decisions are documented, communicated, and implemented only after approval.

Assess and Implement Changes - Integrated Change Control Process



Every change request follows the same flow: Submit -> Log -> Assess -> CCB Decide -> Communicate. Decisions branch to Approved, Deferred, or Rejected.

Assess and Implement Changes - Output

- **Approved Change Requests**
 - Change requests that have been processed and approved by the responsible party (PM, sponsor, or CCB).
 - Become inputs to the executing processes for implementation.
- **Project Management Plan Updates**
 - Any component of the PM Plan may be updated as a result of approved changes - including baselines, which can only be changed via formal change control.
- **Project Document Updates**
 - Change log - updated to reflect dispositions of all change requests (approved, deferred, rejected, withdrawn).
 - Notification of disposition is sent to the requestor and all affected stakeholders.

Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Governance	Initiate Project or Phase	- Integrate and Align Project Plans - Plan Sourcing Strategy	- Manage Project Execution - Manage Quality Assurance - Manage Project Knowledge	- Monitor and Control Project Performance - Assess and Implement Changes	Close Project or Phase
Scope		- Plan Scope Management - Elicit and Analyze Requirements - Define Scope - Develop Scope Structure		- Monitor and Control Scope - Validate Scope	
Schedule		- Plan Schedule Management - Develop Schedule		Monitor and Control Schedule	
Finance		- Plan Financial Management - Estimate Costs - Develop Budget		Monitor and Control Finances	
Stakeholders	Identify Stakeholders	- Plan Stakeholder Engagement - Plan Communications Management	- Manage Stakeholder Engagement - Manage Communications	- Monitor Stakeholder Engagement - Monitor Communications	
Resources		- Plan Resource Management - Estimate Resources	- Acquire Resources - Lead the Team	Monitor and Control Resourcing	
Risk		- Plan Risk Management - Identify Risks - Perform Risk Analysis - Plan Risk Responses	Implement Risk Responses	Monitor Risks	

Monitor and Control Scope

- Monitor and Control Scope is the ongoing process of tracking and managing project scope.
- It ensures changes to scope are reviewed, controlled, and processed properly.
- It checks that deliverables remain aligned with the scope baseline and quality requirements.
- The process monitors results to determine whether deliverables meet stakeholder expectations.
- It helps prevent uncontrolled scope changes and scope creep.

Monitor and Control Scope

Inputs

- Project management plan
 - Scope management plan
 - Quality management plan
- Project documents
 - Requirements documentation
 - Performance measurement baseline
 - Quality test and evaluation documents
- Quality metrics
- Approved change requests
- Deliverables
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Data analysis
 - Variance analysis
 - Trend analysis
 - Root cause analysis
- Performance reviews
- Audits and inspections
- Testing/product evaluations
- Data representations
- Process automations
- Etc.

Outputs

- Quality reports
- Verified deliverables
- Change requests
- Quality control measurements
- Project management plan updates
 - Scope management plan
 - Quality management plan
- Project document updates
 - Requirements documentation
 - Lessons learned register
- Work performance information
- Etc.

Monitor and Control Scope - Inputs

- **Project Management Plan**
 - Scope management plan - defines how scope changes will be processed.
 - Quality management plan - defines how product quality will be verified, supporting scope monitoring.
- **Project Documents**
 - Requirements documentation - basis for confirming actual deliverables match what was requested.
 - Performance measurement baseline - the integrated scope, schedule, and cost baseline used as the comparison reference.
 - Quality test and evaluation documents - test plans, test results, and inspection records.
- **Quality Metrics**
 - Specific measurable attributes a deliverable must satisfy.
- **Approved Change Requests**
 - Requests already approved through Integrated Change Control that affect scope.
- **Deliverables**
 - Output of project work that must be measured against the scope baseline.
- **EEFs and OPAs**
 - Quality control standards, scope-related policies, checklists, lessons learned, and historical information.

Monitor and Control Scope - Tools

- **Data Analysis**
 - Variance analysis - compares baselines to actual results to determine if variances are within acceptable thresholds.
 - Trend analysis - examines project performance over time to determine whether performance is improving or degrading.
 - Root cause analysis - identifies the underlying cause of any scope variance so corrective action can address the source.
- **Performance Reviews**
 - Comparison and analysis of work performed against the scope baseline at scheduled intervals.
- **Audits and Inspections**
 - Examination of work products and processes to confirm scope is being followed and quality standards are being met.

Monitor and Control Scope - Tools

- **Testing/Product Evaluations**
 - Structured tests and evaluations to confirm deliverables meet documented requirements before acceptance.
- **Data Representations**
 - Charts and graphs (control charts, scatter diagrams, histograms) used to visualize scope status and detect patterns.
- **Process Automations**
 - Automated tools (CI/CD, automated testing, monitoring systems) that continuously verify scope compliance.

Monitor and Control Scope - Output

- **Quality Reports**
 - Document the status of quality activities and any scope-related quality issues.
- **Verified Deliverables**
 - Deliverables that have passed quality control and are ready to be sent to Validate Scope for customer acceptance.
- **Change Requests**
 - Submitted to Integrated Change Control for any required corrective actions, preventive actions, or defect repairs.
- **Quality Control Measurements**
 - Documented results of QC activities used to evaluate the quality processes.

Validate Scope

- Validate Scope is the process of formally accepting completed deliverables.
- It confirms that deliverables meet the agreed requirements and quality standards.
- Stakeholders review the deliverables and provide formal acceptance.
- This process helps increase the chance that the final product, service, or result will be accepted.
- It also helps ensure the delivered work provides value and meets expectations.

Validate Scope

Inputs

- Project documents
 - Requirements documentation
 - Scope baseline
 - Quality reports
 - Work performance data
 - Quality control measurements
- Verified deliverables
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Data gathering
- Data analysis
- Inspection
- Decision-making
- Customer talks and tests
- Process analysis
- Review meetings
- Etc.

Outputs

- Accepted deliverables
- Change requests
- Project document updates
 - Quality reports
 - Work performance information
 - Requirements documentation
- Lessons learned updates
- Etc.



Validate Scope - Inputs

- **Project Documents**

- Requirements documentation - the detailed requirements against which deliverables are validated.
- Scope baseline - the approved scope statement, WBS, and WBS dictionary used to confirm what was supposed to be delivered.
- Quality reports - results of QA activities that may highlight issues affecting acceptance.
- Work performance data - actual progress data on each deliverable.
- Quality control measurements - inspection results that support objective acceptance.

- **Verified Deliverables**

- Project deliverables that have been verified by Control Quality and are ready for formal acceptance by the customer or sponsor.

- **EEFs and OPAs**

- Customer review processes, project management information systems, organizational acceptance procedures, and templates.

Validate Scope - Tools

- **Data Gathering**
 - Techniques like interviews and questionnaires used to collect customer feedback on deliverables.
- **Data Analysis**
 - Comparison of deliverables against acceptance criteria documented in the scope statement and requirements.
- **Inspection**
 - Examining a work product to determine whether it conforms to documented standards (also called walkthroughs, audits, reviews).
- **Decision-Making**
 - Voting or other techniques used to reach a conclusion on whether to accept, reject, or request changes to deliverables.
- **Customer Talks and Tests**
 - Discussions with the customer and structured tests (including UAT) to confirm deliverables meet expectations.
- **Process Analysis**
 - Examination of the validation process itself to identify improvements.
- **Review Meetings**
 - Formal sessions where deliverables are walked through with the customer or sponsor for sign-off.

Validate Scope - Output

- **Accepted Deliverables**
 - Deliverables that meet acceptance criteria and are formally signed off by the customer or sponsor.
 - Documentation of acceptance is sent to the Close Project or Phase process.
- **Change Requests**
 - Deliverables that are not accepted may have a change request submitted, which is then processed via integrated change control.
- **Project Document Updates**
 - Quality reports - updated with validation outcomes.
 - Work performance information - documents which deliverables were accepted, rejected, or returned with feedback.
 - Requirements documentation - updated to reflect any approved changes.
- **Lessons Learned Updates**
 - Captures techniques that were effective at validating scope, common reasons deliverables were rejected, and improvements for future projects.

Monitor and Control Schedule

- Tracks project progress and updates the schedule based on actual or forecasted performance.
- The key benefit is keeping the project schedule realistic throughout the project.
- In predictive projects, changes to the schedule baseline must go through integrated change control.
- In adaptive projects, the focus is on velocity, backlog reprioritization, retrospectives, and accepted work per iteration.
- For contracted work, supplier status reports, milestone updates, and walkthroughs help verify schedule progress.

Monitor and Control Schedule

Inputs

- Project management plan
 - Schedule management plan
 - Scope baseline
 - Performance measurement baseline
- Product backlog
- Project documents
 - Lessons learned register
 - Project calendars
 - Project schedule
 - Resource calendars
 - Risk register
 - Schedule data
- Work performance data
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Data analysis
 - Earned value analysis
 - Burnup/burndown chart
 - Performance reviews
 - Trend analysis
 - Variance analysis
 - What-if scenario analysis
- Critical path method
- Critical chain method
- Project management information system
- Resource optimization
- Leads and lags
- Schedule compression
- Branch and bound
- Velocity
- Daily coordination meetings
- Sprint reviews
- Backlog refinement
- Etc.

Outputs

- Work performance information
- Schedule forecasts
- Change requests
- Project management plan updates
 - Schedule management plan
 - Schedule baseline
 - Cost baseline
 - Performance measurement baseline
- Project document updates
 - Assumption log
 - Basis of estimates
 - Lessons learned register
 - Project schedule
 - Resource calendars
 - Risk register
 - Schedule data
- Etc.

Monitor and Control Schedule - Inputs

- **Project Management Plan**
 - Schedule management plan - defines how schedule will be managed and controlled.
 - Scope baseline - reference for assessing scope-related schedule impacts.
 - Performance measurement baseline - integrated scope, schedule, and cost baseline used for variance analysis.
- **Product Backlog**
 - In adaptive projects, the prioritized list of features that drives sprint planning and release scheduling.
- **Project Documents**
 - Lessons learned register - applies prior schedule-control lessons.
 - Project calendars / Resource calendars - identify working days and resource availability.
 - Project schedule - current approved schedule against which actuals are compared.
 - Risk register - identifies schedule risks and their responses.
 - Schedule data - supplementary data used to model the schedule.
- **Work Performance Data**
 - Raw observations: which activities have started, percent complete, finish dates, and duration of completed activities.
- **EEFs and OPAs**
 - Schedule control standards, project management information systems, schedule monitoring policies, templates, and lessons learned.

Monitor and Control Schedule - Tools

- **Data Analysis**
 - Earned value analysis - schedule variance (SV) and schedule performance index (SPI) measure schedule performance.
 - Burnup/burndown chart - visualizes work remaining over time vs the ideal trajectory in adaptive projects.
 - Performance reviews - measure, compare, and analyze schedule performance against the baseline.
 - Trend analysis - examines project performance over time to determine if schedule performance is improving or deteriorating.
 - Variance analysis - investigates differences between baseline and actual performance.
 - What-if scenario analysis - assess feasibility of the schedule under adverse conditions.
- **Critical Path & Critical Chain Methods**
 - Used to update the schedule and confirm whether the critical path has shifted.

Monitor and Control Schedule - Tools

- **PMIS & Resource Optimization**
 - PMIS provides schedule reporting and tracking. Resource optimization adjusts schedule based on availability.
- **Schedule Compression**
 - Crashing or fast tracking applied to recover slipped schedule (with cost or risk impact).
- **Adaptive Techniques**
 - Velocity - average story points completed per sprint, used to forecast remaining work.
 - Daily coordination meetings (stand-ups), sprint reviews, backlog refinement keep the team aligned and the schedule current.

Monitor and Control Schedule - Output

- **Work Performance Information**
 - Performance data analyzed in context: SV, SPI, schedule forecasts, completed activities, and current critical path status.
- **Schedule Forecasts**
 - Estimates or predictions of conditions and events in the future based on current performance information.
- **Change Requests**
 - Variance analysis, performance reviews, and progress meetings can lead to change requests for corrective or preventive actions.
- **Project Management Plan Updates**
 - Schedule management plan, schedule baseline, cost baseline, and performance measurement baseline may all be updated through change control.
- **Project Document Updates**
 - Assumption log, basis of estimates, lessons learned register, project schedule, resource calendars, risk register, and schedule data are updated to reflect current state.

Monitor and Control Finances

- Monitor and Control Finances tracks and manages the project's financial health throughout the project.
- It compares actual costs against planned costs to identify variances.
- It includes tracking expenditures, updating financial records, and monitoring the cost baseline.
- It may also involve updating revenue forecasts, cost savings projections, or financial benefits.
- Corrective actions are taken when costs, forecasts, or financial risks move off track.

Monitor and Control Finances

Inputs

- Project management plan
 - Financial management plan
 - Cost baseline
 - Performance measurement baseline
- Project documents
 - Lessons learned register
 - Project funding requirements
- Work performance data
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data analysis
 - Earned value analysis
 - Trend analysis
 - Reserve analysis
- To-complete performance index
- Project management information system
- Etc.

Outputs

- Work performance information
- Revenue and cost forecasts
- Change requests
- Funding proposals
- Project management plan updates
 - Financial management plan
 - Cost baseline
 - Performance measurement baseline
- Project document updates
 - Assumption log
 - Basis of estimates
 - Cost estimates
 - Lessons learned register
 - Risk register
- Etc.

Monitor and Control Finances - Inputs

- **Project Management Plan**
 - Financial management plan - the rules under which monitoring and controlling are performed.
 - Cost baseline - the reference against which actual costs are compared.
 - Performance measurement baseline - the integrated scope, schedule, and cost baseline used in EVA.
- **Project Documents**
 - Lessons learned register - applies prior cost-control lessons.
 - Project funding requirements - includes projected and actual funding consumed.
- **Work Performance Data**
 - Raw observations on costs incurred, work completed, and schedule progress used to calculate EVA metrics.
- **Organizational Process Assets**
 - Existing financial controls, cost variance thresholds, monitoring policies, financial reporting templates, and lessons learned.

Monitor and Control Finances - Tools

- **Expert Judgment**
 - Specialists in financial analysis, EVA, forecasting, and the relevant industry interpret cost performance.
- **Data Analysis**
 - Earned value analysis (EVA) - integrates PV (Planned Value), EV (Earned Value), and AC (Actual Cost) to compute CV, SV, CPI, and SPI.
 - Trend analysis - examines project performance over time to determine if performance is improving or deteriorating; feeds forecasting.
 - Reserve analysis - monitors the use of contingency and management reserves; reserves may be reduced or returned as risks pass.
- **To-Complete Performance Index (TCPI)**
 - Measures the cost performance required to be achieved on the remaining work to meet a specified management goal.
 - $TCPI = (BAC - EV) / (BAC - AC)$ - the efficiency that must be maintained to complete on the original budget.
- **Project Management Information System (PMIS)**
 - Software tools used for monitoring PV, EV, and AC; graphical displays; trending; and reporting forecasts.

Monitor and Control Finances - Output

- **Work Performance Information**
 - Performance data analyzed in context: cost variance (CV), schedule variance (SV), CPI, SPI, EAC, ETC, VAC, and TCPI calculations and analyses.
- **Revenue and Cost Forecasts**
 - Estimate at Completion (EAC) - what the project will cost at completion based on current performance.
 - Estimate to Complete (ETC) - the cost expected to be incurred to complete the remaining work.
- **Change Requests**
 - Variance analyses or trend analyses lead to change requests for corrective or preventive actions.

Monitor and Control Finances - Output

- **Funding Proposals**
 - When the project requires additional funding (e.g., scope increase, risk realization), formal proposals are submitted to the funding authority.
- **Project Management Plan Updates**
 - Financial management plan, cost baseline, and performance measurement baseline may be updated through change control.
- **Project Document Updates**
 - Assumption log, basis of estimates, cost estimates, lessons learned register, and risk register may all be updated based on cost performance and forecasts.

Monitor Stakeholder Engagement

- Monitor Stakeholder Engagement checks whether stakeholder engagement efforts are working effectively.
- It helps identify when stakeholder relationships, communication, or participation need improvement.
- The process is performed throughout the project as stakeholders and project conditions change.
- It may require updating engagement strategies, communication approaches, or project plans.
- The key benefit is maintaining or improving stakeholder support and involvement.

Monitor Stakeholder Engagement

Inputs

- Project management plan
 - Resource management plan
 - Communications management plan
 - Stakeholder engagement plan
- Project documents
 - Issue log
 - Lessons learned register
 - Project communications
 - Risk register
 - Stakeholder register
- Work performance data
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Data analysis
 - Alternative analysis
 - Root cause analysis
 - Stakeholder analysis
- Decision-making
 - Multicriteria decision analysis
 - Voting
- Data representation
 - Stakeholder engagement assessment matrix
- Communication skills
 - Feedback
 - Presentations
- Interpersonal and team skills
 - Active listening
 - Cultural awareness
 - Leadership
 - Networking
 - Political awareness
- Meetings
- Etc.

Outputs

- Work performance information
- Change requests
- Project management plan updates
 - Resource management plan
 - Communications management plan
 - Stakeholder engagement plan
- Project document updates
 - Issue log
 - Lessons learned register
 - Risk register
 - Stakeholder register
- Etc.

Monitor Stakeholder Engagement - Inputs

- **Project Management Plan**
 - Resource management plan - identifies team members and their roles in monitoring engagement.
 - Communications management plan - frames the communications used to monitor engagement.
 - Stakeholder engagement plan - documents the desired engagement levels against which actuals are compared.
- **Project Documents**
 - Issue log - issues raised by or about stakeholders indicate engagement quality.
 - Lessons learned register - prior monitoring approaches that worked or did not.
 - Project communications - actual communications delivered to stakeholders.
 - Risk register - identifies stakeholder-related risks that may have changed.
 - Stakeholder register - the list against which engagement is monitored.
- **Work Performance Data**
 - Raw observations: which stakeholders attended meetings, responded to communications, raised issues, or are actively supporting/resisting the project.
- **EEFs and OPAs**
 - Organizational culture, communications expectations, stakeholder feedback channels, monitoring tools, and lessons learned.

Monitor Stakeholder Engagement - Tools

- **Data Analysis**

- Alternative analysis - evaluates options for adjusting engagement when the current approach is not working.
- Root cause analysis - identifies underlying reasons for engagement gaps.
- Stakeholder analysis - reassesses stakeholder positions, interests, and influence as the project evolves.

- **Decision-Making**

- Multicriteria decision analysis - prioritizes engagement interventions across multiple criteria.
- Voting - converges on the best response to an engagement issue.

- **Data Representation**

- Stakeholder engagement assessment matrix - updated to show current vs desired engagement and any gaps.

Monitor Stakeholder Engagement - Tools

- **Communication Skills**

- Feedback - actively solicited from stakeholders to gauge satisfaction and understanding.
- Presentations - used to share status and elicit engagement.

- **Interpersonal and Team Skills**

- Active listening - reduces misunderstandings and improves engagement.
- Cultural awareness - tailors monitoring approach to cultural norms.
- Leadership - inspires and motivates stakeholder engagement.
- Networking - builds informal channels to gauge engagement quality.
- Political awareness - navigates organizational dynamics.

- **Meetings**

- Status meetings, retrospectives, and one-on-ones used to monitor and adjust engagement.

Monitor Stakeholder Engagement - Output

- **Work Performance Information**
 - Status of stakeholder engagement: which stakeholders are supportive, which are resistant, and where engagement gaps exist.
- **Change Requests**
 - May include corrective actions to engagement strategies or preventive actions to reduce stakeholder-related risks.
- **Project Management Plan Updates**
 - Resource management plan - may need additional resources to engage stakeholders.
 - Communications management plan - revised based on what is and is not working.
 - Stakeholder engagement plan - updated with revised engagement strategies.
- **Project Document Updates**
 - Issue log - new stakeholder issues added.
 - Lessons learned register - updated with monitoring lessons.
 - Risk register - updated as stakeholder-related risks change.
 - Stakeholder register - updated with new engagement levels and any new stakeholders identified.

Monitor Communications

- The process of ensuring the information needs of the project and its stakeholders are met
- Determines whether the planned communication artifacts and activities are having the desired effect of increasing or maintaining stakeholder support for the project

Monitor Communications

Inputs

- Project management plan
 - Resource management plan
 - Communications management plan
 - Stakeholder engagement plan
- Project documents
 - Issue log
 - Lessons learned register
 - Project communications
- Work performance data
- Enterprise environmental factors
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Project management information system
- Data representation
 - Stakeholder engagement assessment matrix
- Interpersonal and team skills
 - Observation/conversation
- Meetings
- Etc.

Outputs

- Work performance information
- Change requests
- Project management plan updates
 - Communications management plan
 - Stakeholder engagement plan
- Project document updates
 - Issue log
 - Lessons learned register
 - Stakeholder register
- Etc.

Monitor Communications - Inputs

- **Project Management Plan**
 - Resource management plan - identifies the resources available to monitor communications.
 - Communications management plan - the reference for what should be happening.
 - Stakeholder engagement plan - documents stakeholder engagement and communication strategies.
- **Project Documents**
 - Issue log - communications-related issues are tracked here.
 - Lessons learned register - prior monitoring lessons applied.
 - Project communications - actual communications delivered to stakeholders, used as evidence of effectiveness.
- **Work Performance Data**
 - Raw observations on what was communicated, when, to whom, and the responses received.
- **EEFs and OPAs**
 - Organizational culture, project management body of knowledge for similar projects, established communication channels, monitoring policies, and templates.

Monitor Communications - Tools

- **Expert Judgment**
 - Specialists in communications, public relations, and the relevant subject matter help interpret communication effectiveness.
- **Project Management Information System**
 - Provides a set of standard tools for the PM to capture, store, and distribute information; metrics from the PMIS show whether communications are being read and acted on.
- **Data Representation**
 - Stakeholder engagement assessment matrix - shows current engagement levels which can be tracked against desired.
- **Interpersonal and Team Skills**
 - Observation/conversation - direct dialogue with team members and stakeholders to identify communication issues.
- **Meetings**
 - Discussions with stakeholders to confirm communications are reaching them and being understood.

Monitor Communications - Output

- **Work Performance Information**
 - Information on how project communication is performing: stakeholder feedback, response rates, channel effectiveness, gaps in coverage.
- **Change Requests**
 - Change requests submitted via Integrated Change Control to revise communication processes, technology, or content.
- **Project Management Plan Updates**
 - Communications management plan - updated to reflect what was learned about the effectiveness of various approaches.
 - Stakeholder engagement plan - updated as engagement patterns are observed.
- **Project Document Updates**
 - Issue log - communications issues recorded and tracked.
 - Lessons learned register - what worked and what didn't in monitoring communications.
 - Stakeholder register - updated with new communication preferences uncovered.

Monitor and Control Resourcing

- Monitor and Control Resourcing ensures project resources are available as planned.
- It focuses on physical and virtual resources, not team member performance.
- Resources may include equipment, materials, supplies, facilities, software, licenses, infrastructure, testing environments, and services.
- The process compares planned resource use against actual resource use.
- Corrective action is taken when resources are unavailable, overused, underused, or causing delays.

Monitor and Control Resourcing

Inputs

- Project management plan
 - Resource management plan
- Project documents
 - Issue log
 - Lessons learned register
 - Physical or virtual resource assignments
 - Project schedule
 - Resource breakdown structure
 - Risk register
- Work performance data
- Agreements
- Organizational process assets
- Etc.

Tools & Techniques

- Data analysis
 - Alternative analysis
 - Cost-benefit analysis
 - Performance reviews
 - Trend analysis
- Problem-solving
- Interpersonal and team skills
 - Negotiation
 - Influencing
- Project management information system
- Value stream mapping
- Continuous improvement
- Theory of constraints
- Control charts
- Branch and bound
- Etc.

Outputs

- Work performance information
- Change requests
- Project management plan updates
 - Resource management plan
 - Schedule baseline
 - Cost baseline
- Project document updates
 - Assumption log
 - Issue log
 - Lessons learned register
 - Physical resource assignments
 - Resource breakdown structure
 - Risk register
- Etc.

Monitor and Control Resourcing

- Inputs

- **Project Management Plan**
 - Resource management plan - the reference for monitoring and controlling resource use.
- **Project Documents**
 - Issue log - resource-related issues raised and tracked.
 - Lessons learned register - prior monitoring lessons applied.
 - Physical or virtual resource assignments - shows what should be in place.
 - Project schedule - drives expected timing of resource use.
 - Resource breakdown structure - the organized view of resources to monitor.
 - Risk register - resource-related risks (key person, vendor failure) being monitored.
- **Work Performance Data**
 - Raw data on resource use: hours worked, materials consumed, equipment usage, idle time.
- **Agreements**
 - Vendor contracts that specify resource availability and performance requirements.
- **Organizational Process Assets**
 - Resource policies and procedures, monitoring templates, control thresholds, and historical resource utilization data.

Monitor and Control Resourcing - Tools

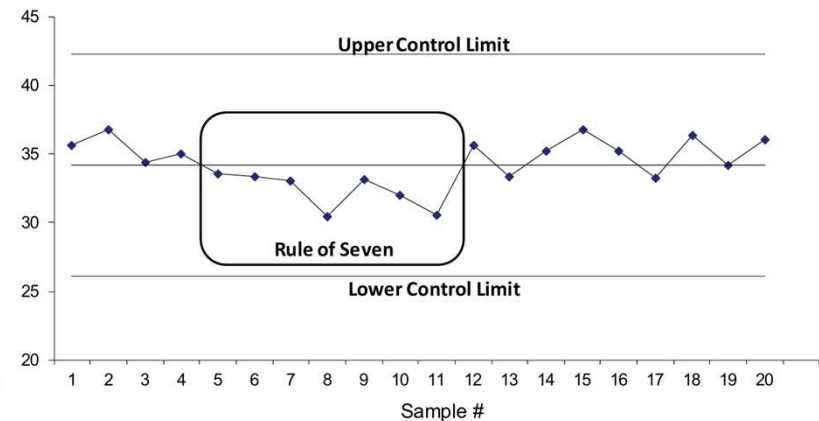
- **Data Analysis**
 - Alternative analysis - evaluates options for adjusting resource allocation when shortages or surpluses occur.
 - Cost-benefit analysis - determines best corrective action based on cost vs benefit.
 - Performance reviews - measures, compares, and analyzes planned vs actual resource utilization.
 - Trend analysis - examines resource utilization over time to predict future needs.
- **Problem-Solving**
 - Structured techniques (5 Whys, root cause analysis) to address resource issues.

Monitor and Control Resourcing - Tools

- **Interpersonal and Team Skills**
 - Negotiation - used to renegotiate resource availability with functional managers and vendors.
 - Influencing - used when the PM lacks formal authority over resources.
- **PMIS & Optimization Techniques**
 - PMIS - resource management dashboards and utilization reports.
 - Value stream mapping - identifies waste in the resource-to-deliverable workflow.
 - Continuous improvement - ongoing optimization of resource use (Lean, Kaizen).
 - Theory of constraints - identifies and improves the bottleneck resource.
 - Branch and bound - mathematical optimization of resource allocation.

Monitor and Control Resourcing - Tools

- **Control Charts**
 - **Used to determine if a process is stable and predictable**
 - Shows whether performance is normal or out of control
 - **Tracks results over time**
 - Examples: defects, cost variance, schedule variance, or scope changes
 - **Control limits**
 - Based on statistical calculations
 - Show the normal range of process performance
 - **Specification limits**
 - Based on customer or project requirements
 - Show the maximum and minimum values allowed
 - **Corrective action**
 - If results go outside the control limits, investigate and correct the process
 - **Rule of Seven**
 - If **7 data points in a row** are on the same side of the mean, investigate
 - The points do **not** have to be outside the control limits
 - This may show a pattern, trend, or process problem



Monitor and Control

Resourcing - Output

- **Work Performance Information**
 - Resource utilization status: actual vs planned, variances, future resource needs, and recommendations.
- **Change Requests**
 - Submitted via Integrated Change Control to revise resource assignments, baselines, or processes.
- **Project Management Plan Updates**
 - Resource management plan - updated to reflect lessons from monitoring.
 - Schedule and cost baselines - may be updated through approved change requests.
- **Project Document Updates**
 - Assumption log - assumptions about resources updated.
 - Issue log - resource-related issues recorded and tracked.
 - Lessons learned register - what worked and what didn't in resource monitoring.
 - Physical resource assignments - updated as resources are reallocated.
 - Resource breakdown structure - updated as resource categories or types change.
 - Risk register - resource-related risks updated.

Monitor Risks

- Monitor Risks tracks identified risks and watches for new risks throughout the project.
- It ensures risk response plans are being implemented as planned.
- It evaluates whether risk responses are effective or need to be changed.
- This process is performed throughout the project.

Monitor Risks

Inputs

- Project management plan
 - Risk management plan
- Project documents
 - Issue log
 - Lessons learned register
 - Risk register
 - Risk report
- Work performance data
- Work performance reports
- Etc.

Tools & Techniques

- Data analysis
 - Technical performance analysis
 - Reserve analysis
- Audits
- Meetings
- Etc.

Outputs

- Work performance information
- Change requests
- Project management plan updates
 - Any component
- Project document updates
 - Assumption log
 - Issue log
 - Lessons learned register
 - Risk register
 - Risk report
- Organizational process asset updates
- Etc.

Monitor Risks - Inputs

- **Project Management Plan**
 - Risk management plan - provides guidance on monitoring and controlling risk: when to use reserves, how to verify assumptions, how often to monitor.
- **Project Documents**
 - Issue log - tracks issues that may have evolved from risks; helps identify whether responses have been effective.
 - Lessons learned register - applies prior monitoring lessons.
 - Risk register - the master list being monitored.
 - Risk report - current overall project risk exposure being tracked.
- **Work Performance Data**
 - Raw observations on the status of risks: which have occurred, which are no longer applicable, the status of agreed responses, and any new risks identified.
- **Work Performance Reports**
 - Performance information on cost, schedule, scope, and quality - the basis for assessing whether the risk responses and overall risk approach are working.

Monitor Risks - Tools

- **Data Analysis**
 - Technical performance analysis - compares technical accomplishments during execution to the schedule of technical achievement.
 - Reserve analysis - throughout execution, monitors the amount of contingency reserve remaining versus the amount of risk remaining.
- **Audits**
 - Risk audits - examine and document the effectiveness of risk responses in dealing with identified risks.
- **Meetings**
 - Risk reviews - held periodically to examine the status of risk responses, identify new risks, reassess current risks, close out outdated risks, and discuss risk-related issues.

Monitor Risks - Output

- **Work Performance Information**
 - Information on how risk management is performing: how the actual risks differ from those that were predicted; and how effective the responses have been.
- **Change Requests**
 - Recommended corrective actions - bringing the project back into compliance with the project management plan (e.g., implementing contingency plans).
- **Project Management Plan Updates**
 - Any component of the PM Plan may be updated as a result of monitoring.
- **Project Document Updates**
 - Assumption log - updated with new assumptions or invalidated ones.
 - Issue log - new issues uncovered during monitoring.
 - Lessons learned register - what worked or didn't in monitoring risks.
 - Risk register - new risks added; existing risks reassessed; outdated risks closed; outdated assumptions and impacts updated.
 - Risk report - updated to reflect current overall project risk exposure.
- **Organizational Process Asset Updates**
 - Templates for risk management plans, risk registers, risk reports, and risk breakdown structures may be updated based on lessons learned.

Closing

- 1 Process
- Bring the phase or project to a official end

Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Governance	Initiate Project or Phase	- Integrate and Align Project Plans - Plan Sourcing Strategy	- Manage Project Execution - Manage Quality Assurance - Manage Project Knowledge	- Monitor and Control Project Performance - Assess and Implement Changes	Close Project or Phase
Scope		- Plan Scope Management - Elicit and Analyze Requirements - Define Scope - Develop Scope Structure		- Monitor and Control Scope - Validate Scope	
Schedule		- Plan Schedule Management - Develop Schedule		Monitor and Control Schedule	
Finance		- Plan Financial Management - Estimate Costs - Develop Budget		Monitor and Control Finances	
Stakeholders	Identify Stakeholders	- Plan Stakeholder Engagement - Plan Communications Management	- Manage Stakeholder Engagement - Manage Communications	- Monitor Stakeholder Engagement - Monitor Communications	
Resources		- Plan Resource Management - Estimate Resources	- Acquire Resources - Lead the Team	Monitor and Control Resourcing	
Risk		- Plan Risk Management - Identify Risks - Perform Risk Analysis - Plan Risk Responses	Implement Risk Responses	Monitor Risks	

Close Project or Phase

- Close Project or Phase finalizes all activities for a project, phase, release, iteration, or contract.
- It applies to both successful and unsuccessful work.
- The process confirms that planned work has been completed or properly ended.

Close Project or Phase

Inputs

- Project charter
- Project management plan
 - All components
- Project documents
 - Assumption log
 - Basis of estimates
 - Change log
 - Issue log
 - Lessons learned register
 - Milestone list
 - Project communications
 - Quality control measurements
 - Quality reports
 - Requirements documentation
 - Risk register
 - Risk report
- Accepted deliverables
- Business documents
 - Business case
 - Benefits management plan
- Agreements
- Procurement documentation
- Organizational process assets
- Etc.

Tools & Techniques

- Expert judgment
- Data analysis
 - Document analysis
 - Regression analysis
 - Trend analysis
 - Variance analysis
- Meetings
- Etc.

Outputs

- Project document updates
 - Lessons learned register
- Final product, service, or result transition
- Final report
- Organizational process asset updates
- Etc.

Close Project or Phase - Inputs

- **Project Charter**
 - Documents the project's success criteria, exit criteria, approval requirements, and the assigned project manager - all of which are confirmed at closure.
- **Project Management Plan**
 - All components of the PM Plan are inputs because closure must verify that every aspect of the plan has been completed.
- **Project Documents**
 - Assumption log - documented assumptions confirmed or invalidated during the project.
 - Basis of estimates - used in performance review and lessons learned.
 - Change log - records all changes made during the project.
 - Issue log - records of all issues raised; all should be resolved or transferred at closure.
 - Lessons learned register - finalized at closure for transfer to organizational repositories.
 - Milestone list - basis for confirming all major milestones were achieved.

Close Project or Phase - Inputs

- **Project Documents (cont.)**
 - Project communications - record of communications during the project; archived at closure.
 - Quality control measurements - documented results of QC activities.
 - Quality reports - status of quality activities.
 - Requirements documentation - confirms all requirements were met or formally addressed.
 - Risk register and risk report - documents risks and their outcomes.
- **Accepted Deliverables**
 - Deliverables formally signed off by the customer or sponsor through Validate Scope.
- **Business Documents**
 - Business case - reviewed at closure to confirm the business need was met.
 - Benefits management plan - basis for handing off benefit realization tracking to operations.
- **Agreements & Procurement Documentation**
 - Contracts must be formally closed; procurement records (statements of work, change requests, deliverables, payments, inspection results) are archived.
- **OPAs**
 - Organizational closure procedures and templates, lessons learned repositories, configuration management knowledge bases.

Close Project or Phase - Tools

- **Expert Judgment**
 - Specialists in management control, audit, legal and procurement, and regulations help ensure all closure activities are performed correctly.
- **Data Analysis**
 - Document analysis - assessing project documentation to identify lessons learned and knowledge sharing for future projects.
 - Regression analysis - examining the relationships between different project variables that contributed to project outcomes.
 - Trend analysis - validating project models and adjusting for future projects.
 - Variance analysis - improves the metrics of the organization by comparing what was originally planned and the end result.
- **Meetings**
 - Closeout meetings, lessons learned meetings, and final stakeholder meetings used to confirm closure activities, document lessons, and celebrate success.

Close Project or Phase - Output

- **Project Document Updates**
 - Lessons learned register - finalized to include final information on closure activities, used as input for the organizational lessons learned repository.
- **Final Product, Service, or Result Transition**
 - Formal handoff of the final product, service, or result to the operating group or customer who will support, use, or sustain it.
 - Includes operations and maintenance documentation, training, and warranty terms.

Close Project or Phase - Output

- **Final Report**
 - Provides a summary of project performance: scope, quality, cost, schedule, summary of validation against objectives, summary of risks, and summary of issues encountered and how they were resolved.
 - Confirms the project's business need has been met and that benefits will be tracked beyond project closure.
- **Organizational Process Asset Updates**
 - Project documents, operational documents, project or phase closure documents, and historical information all added to the organization's process asset repository.
 - Lessons learned, templates, and process improvements available for future projects.

Close Project or Phase - Project Closure Checklist

Deliverables & Acceptance

- ☒ All deliverables formally accepted by sponsor/customer
- ☒ Final product, service, or result transitioned to operations
- ☒ All requirements traced to deliverables and confirmed

Documentation

- ☒ Final project report completed and shared
- ☒ Lessons learned register finalized
- ☒ All project documents archived per policy
- ☒ OPA repository updated with templates and artifacts

Financial

- ☒ Final cost reconciliation completed
- ☒ Outstanding invoices paid; budget closed
- ☒ Any unused contingency reserves released

Contracts & Procurement

- ☒ All vendor contracts formally closed
- ☒ Final payments processed; warranties documented
- ☒ Procurement audit completed

People & Resources

- ☒ Team members released to other assignments
- ☒ Performance assessments completed
- ☒ Recognition and rewards delivered
- ☒ Equipment / facilities released



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Standard closure activities organized by category. All items must be completed before the project can be formally closed.

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